



SASTRA

ENGINEERING · MANAGEMENT · LAW · SCIENCES · HUMANITIES · EDUCATION

DEEMED TO BE UNIVERSITY

(U/S 3 of the UGC Act, 1956)



THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

T H A N J A V U R | K U M B A K O N A M | C H E N N A I

**Course Directory for
B. Tech. in Computer Science & Engineering
(Students admitted from 2021-22)
(Total Credits: 180)**

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

- PEO1.** Empower graduates to model computing systems by applying mathematical and engineering principles
- PEO2.** Enable graduates to create and implement solutions with advanced computing skills addressing industrial and scientific challenges
- PEO3.** Provide broad knowledge base covering different areas – from Digital Electronics to Advanced Computer Architecture, from Algorithms to Machine Intelligence, from Database to Data analytics, from System Software to Advanced Operating System, from Computer Networks to Internet of Things
- PEO4.** Equip graduates with capability to design, implement and test computational approaches to develop solutions
- PEO5.** Impart independent learning skills to successfully pursue higher studies and engage in innovative research
- PEO6.** Inculcate professional ethics and work with commitment for progress of society

PROGRAMME OUTCOMES (POs)

Engineering graduates will be able to:

- PO1.** Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2.** Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems to reach substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3.** Design/development of solutions: Design solutions for complex engineering problems design systems, components or processes that meet the specified needs with appropriate consideration towards public health and safety, and the cultural, societal, and environmental considerations.
- PO4.** Conduct investigations of complex problems: Use research-based knowledge and research methods including statistical tools, analysis, data interpretation, and synthesis of the information to provide valid conclusions.
- PO5.** Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- PO6.** The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7.** Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8.** Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9.** Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10.** Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend

and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAMME SPECIFIC OUTCOMES (PSOs)

Upon completion of the B.Tech. Computer Science and Engineering, the graduates will be able to:

PSO1. Ability to comprehend the problem and apply various programming skills to develop innovative and quality software product for the changing business needs.

PSO2. Able to provide acceptable technical solutions for multidisciplinary areas of engineering by utilizing the skills of Data Analytics, Cloud Computing and Machine Learning.

Scheme of Study**I Semester (25 Credits)**

Course Code	Course Title	Periods of Instruction			Credits
		L	T	P	
ENG101	Technical Communication	1	0	2	2
MAT101R01	Engineering Mathematics – I	3	1	0	4
CSE101	Problem Solving & Programming in C	3	0	2	4
PHY101	Engineering Physics	3	0	2	4
EEE101	Basic Electrical Engineering	2	0	2	3
CIV101	Basic Civil Engineering	2	0	2	3
CIV102	Engineering Mechanics	2	1	0	3
MEC102	Introduction to Engineering Design	2	0	0	2
	Total	18	2	10	25

II SEMESTER (23 Credits)

Course Code	Course Title	Periods of Instruction			Credits
		L	T	P	
BIT101	Biology For Engineers	2	0	0	2
MAT102R01	Engineering Mathematics – II	3	1	0	4
CSE201	Object Oriented Programming in C++	3	0	2	4
CHY101	Engineering Chemistry	3	0	2	4
EIE101	Basic Electronics Engineering	2	0	2	3
MEC101	Basic Mechanical Engineering	2	0	2	3
CIV103	Engineering Graphics	1	0	4	3
	Total	16	1	12	23

III Semester (22 Credits)

Course Code	Course Name	No. of Contact Hours / Week			Credits
		L	T	P	
MAT201R01	Engineering Mathematics – III	3	1	0	4
CSE208	Java Programming	3	0	2	4
CSE105	Computer Organization	4	0	0	4
CSE103	Data Structures	4	0	0	4
ECE101	Digital System Design	4	0	0	4
CSE104	Data Structures Laboratory	0	0	2	1
ECE103	Digital System Design Laboratory	0	0	2	1
TOTAL		18	1	6	22

IV Semester (21 Credits)

Course Code	Course Name	No. of Contact Hours / Week			Credits
		L	T	P	
MAT302	Discrete Structures	3	1	0	4
CSE305	Design & Analysis of Algorithms	4	0	0	4
INT102R01	Fundamentals of Database Management Systems	3	0	0	3
CSE205R01	Computer Architecture	4	0	0	4
	Departmental Elective – I	3	1	0	4
CSE206R01	Computer System Design Laboratory	0	0	2	1
CSE306	Design & Analysis of Algorithms Laboratory	0	0	2	1
TOTAL		17	2	4	21

V Semester (22 Credits)

Course Code	Course Name	No. of Contact Hours / Week			Credits
		L	T	P	
CSE301	Theory of Computation	3	1	0	4
CSE308	Operating Systems	3	0	0	3
CSE302	Computer Networks	4	0	0	4
	Departmental Elective – II	3	1	0	4
	Departmental Elective – III	3	1	0	4
CSE309	Operating Systems Laboratory	0	0	2	1
CSE303	Computer Networks Laboratory	0	0	2	1
TNP101	Soft Skills – I	0	0	2	1
TOTAL		16	3	6	22

VI Semester (24 Credits)

Course Code	Course Name	No. of Contact Hours / Week			Credits
		L	T	P	
CSE313	Cryptography & Network Security	3	1	0	4
CSE314R01	Software Engineering Practices	3	0	0	3
CSE402	Compiler Engineering	4	0	0	4
	Departmental Elective – IV	3	1	0	4
	Departmental Elective – V	3	1	0	4
CSE315R01	Software Engineering Practices Laboratory	0	0	2	1
CSE403	Compiler Engineering Laboratory	0	0	2	1
TNP102	Soft Skills – II	0	0	2	1
CSE300	Mini Project	0	0	4	2
TOTAL		16	3	10	24

VII Semester (22 Credits)

Course Code	Course Name	No. of Contact Hours / Week			Credits
		L	T	P	
INT404R01	Big Data Analytics	3	0	0	3
CSE407R01	Cloud Computing	3	1	0	4
CSE409	Parallel & Distributed Systems	4	0	0	4
CSE415R01	Internet of Things	4	0	0	4
	Departmental Elective – VI	3	1	0	4
MAN105	Professional Ethics	1	0	0	1
CSE411	Parallel & Distributed Systems Laboratory	0	0	2	1
CSE431	Internet of Things Laboratory	0	0	2	1
TOTAL		18	2	4	22

VIII Semester (21 Credits)

Course Code	Course Name	No. of Contact Hours / Week			Credits
		L	T	P	
OEXXXX	Open Elective	3	0	0	3
OEXXXX	Open Elective	3	0	0	3
OEXXXX	Open Elective	3	0	0	3
CSE400	Project & Viva Voce	0	0	24	12
TOTAL		9	0	24	21

Additional Course for Lateral Entry Students – Proposed – for III Semester (Mandatory)

Course Code	Course Title	No. of Contact Hours / Week			Credits
		L	T	P	
CSE101	Problem Solving & Programming in C	3	0	2	4

Additional Course for Lateral Entry Students – Proposed – for IV Semester (Mandatory)

Course Code	Course Title	No. of Contact Hours / Week			Credits
		L	T	P	
CSE201	Object Oriented Programming in C++	3	0	2	4

List of Electives – Proposed – for IV Semester (Any one to be selected)

Course Code	Course Title	No. of Contact Hours / Week			Credits
		L	T	P	
MAT301R01	Engineering Mathematics – IV	3	1	0	4
MAT304R01	Mathematics for Cyber Security	3	1	0	4
CSE202	Computer Graphics using OpenGL	3	1	0	4
CSE307	Object Oriented Analysis & Design	3	1	0	4

List of Electives – Proposed – for V Semester (Any two to be selected)

Course Code	Course Title	No. of Contact Hours / Week			Credits
		L	T	P	
MAT305	Statistical Foundations for Computer Science	3	1	0	4
INT301	Network Tools & Techniques	3	0	2	4
INT310	Artificial Intelligence	3	1	0	4
CSE304	Python Programming with Web Frameworks	3	0	2	4
CSE203	Linux Programming	3	0	2	4
CSE401	Digital Image Processing	3	1	0	4
CSE310	System Software	3	1	0	4
CSE311	System Modelling & Simulation	3	1	0	4

Course Code	Course Title	No. of Contact Hours / Week			Credits
		L	T	P	
CSE312	Script Programming	3	0	2	4

List of Electives – Proposed – for VI Semester (Any two to be selected)

Course Code	Course Title	No. of Contact Hours / Week			Credits
		L	T	P	
INT302	Green Computing	4	0	0	4
INT303	Data Warehousing & Data Mining	3	1	0	4
INT309	Domain-centric Security	3	0	2	4
ICT307	Network Administration & Management Tools	3	1	0	4
ICT403	Wireless Sensor Networks	3	1	0	4
CSE404	Evolutionary Computing	3	1	0	4
CSE405R01	Natural Language Processing	3	1	0	4
CSE406	Ethical Hacking	3	1	0	4
CSE408	Graph Theory & its Applications	3	1	0	4
INT430	Information Security	3	1	0	4

List of Electives – Proposed – for VII Semester (Any one to be selected)

Course Code	Course Title	No. of Contact Hours / Week			Credits
		L	T	P	
MAT328	Optimization Techniques for Computing	3	1	0	4
INT405R01	Machine Learning Techniques	3	1	0	4
INT406	High Performance Computing	3	1	0	4
INT409	Embedded Security	3	0	2	4
ICT401	Mobile Computing	4	0	0	4
CSE413R01	Computer Vision & Pattern Recognition	3	1	0	4
CSE414	Software Testing & Quality Assurance	3	1	0	4

Course Code	Course Title	No. of Contact Hours / Week			Credits
		L	T	P	
CSE410	Embedded Systems	4	0	0	4
CSE416	Virtualization Techniques	3	1	0	4
CSE417	Robotics	3	1	0	4
CSE418	Brain Computing	3	1	0	4
INT410	Cyber Physical Systems	3	1	0	4
CSE423	Cognitive Computing	3	1	0	4
INT327	Low Code Programming with Oracle Apex	3	0	2	4
INT436	Full Stack Web Application Development	3	0	2	4

List of Mandatory Non-Credited Courses

Course Code	Course Title	No. of Contact Hours / Week			Credits
		L	T	P	
MAN101	Constitution of India	1	0	0	0
MAN102	Indian Traditional Knowledge	1	0	0	0
MAN103	Environmental Studies	1	0	0	0
MAN104	NSS Activities	1	0	0	0

<u>Course Distribution</u>	<u>Credits</u>	<u>Percentage</u>
Humanities & Social Sciences	6	3
Basic Sciences (MPC)	24	13
Engineering Sciences	28	16
Professional Subjects – Core	75	42
Professional Subjects – Electives	24	13
Open Electives	9	5
Project Work	12+2 =14	8
TOTAL	180	100

L	T	P	C
3	1	0	4

Course Code: MAT201R01

Semester: III

ENGINEERING MATHEMATICS – III
(Common to all Branches)

Course Objective:

The course aims to train the learner to understand Laplace transforms techniques and thereby solve problems arising in engineering disciplines, understand Fourier series expansions and their application, acquire knowledge on complex differentiation and integration and evaluate typical integrations arising in many engineering disciplines using complex differentiation and integration.

UNIT - I: Laplace Transforms

15 Periods

Introduction to Laplace transforms – Sufficient conditions for existence - Properties of the Laplace transform - Transforms of derivatives and derivatives of transforms - Shifting theorems - Change of scale property - Convolution theorem - Periodic function theorem - Inverse Laplace transforms – Solution of first and second order ordinary differential equations and simultaneous differential equations using Laplace Transforms – Problems in Electrical circuits, Mechanical vibrations.

UNIT - II : Fourier Series

15 Periods

Introduction to Fourier series- Dirichlet's conditions - Fourier series of odd and even functions - Half-Range Fourier series and Parseval's theorem - Root-mean square value of a function, Complex form of Fourier series - Harmonic analysis - Frequency response and oscillating systems.

UNIT - III : Complex Differentiation

15 Periods

Analytic functions - Cauchy Riemann Equations and other properties - Harmonic functions – Construction of an Analytic function by Milne-Thomson method only - Conformal mappings - Mappings $\sin z$, e^z , $1/z$ only – Mobius transformation - Analysing AC circuits – Heat transfer problems.

UNIT - IV : Complex Integration

15 Periods

Cauchy's integral theorem and integral formula – Taylor and Laurent's series - Types of Singularities – Calculus of residues – Cauchy's residue theorem - Evaluation of Real definite integrals (Type I and II only) by contour integration - Problems in power spectrum involving contour integrals.

TEXTBOOK

1. Glyn James, *Advanced Modern Engineering Mathematics*, Fifth Edition, Pearson Education Limited, England, 2018.

REFERENCES

1. H. K. Dass, R. Verma, *Higher Engineering Mathematics* (Revised Third Edition) 2014, S. Chand and Company Limited, New Delhi.
2. Erwin Kreyszig, *Advanced Engineering Mathematics*, Twelfth Edition, John Wiley & Sons, Reprint 2018.

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/117101055/19>
2. <http://farside.ph.utexas.edu/teaching/336L/Fluidhtml/node91.html>
3. <https://nptel.ac.in/courses/111/103/111103070/>
4. <http://93.174.95.29/main/167B8B5341AD80C50444D54F19A4A24B>
5. <http://93.174.95.29/main/E1F91221B00F4E2F8E4F978539808F5B>

UNIT-WISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to:

Unit I	Find the Laplace Transform of standard functions, apply appropriate properties and solve differential equations arising in many fields of Engineering
Unit II	Express a given function in terms of Fourier series and apply them in some related problems arising in engineering
Unit III	Apply standard techniques of complex variable theory in application areas such as heat conduction, elasticity, fluid dynamics and flow of electric current
Unit IV	Evaluate complicated integrals using residue calculus and also solve related application problems in engineering

COURSE LEARNING OUTCOMES

Upon the successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
1	Apply the concepts of Laplace transform to solve related problems arising in different Engineering fields	K3
2	Find the Fourier series expansion of the given continuous or piece-wise continuous function	K3
3	Apply the Fourier series technique to identify the solutions of related application problems.	K3
4	Understand the concept of continuity, differentiability and analyticity of complex functions	K3
5	Find and apply the suitable transformation to the study the given geometrical structure	K3
6	Use the concept of contour integration to evaluate different typical real definite integrals arising in power spectrum	K3

L	T	P	C
3	0	2	4

Course Code: CSE208
Semester: III

JAVA PROGRAMMING

Course Objectives

This course will help the learner to develop Java console applications using packages, interfaces, classes, exception handling mechanism, and Collections framework. The learner can also develop multi-threaded applications using Java thread model. The learner will also be able to design GUI using Swing controls and event handling mechanism.

UNIT - I

12 Periods

Introduction: History and Evolution of Java - Overview of Java - Data types, Variables and Arrays - Operators - Control Statements - Introducing Classes - A closer look at Methods and Classes - Inheritance. Packages and Interfaces - Exception Handling

UNIT - II

11 Periods

Multithreaded Programming: Java Thread Model - The Main Thread - Creating a Thread - Creating Multiple Threads - Thread Priorities - Synchronization. **I/O:** I/O Basics - Reading Console Input - Writing Console Output. **String Handling:** String - String Buffer - String Builder

UNIT - III

11 Periods

Collections Framework: Collection Interfaces - Collection, List, Set, Sorted Set - Collection Classes - Array List, Hash Set, Tree Set - Accessing a collection via an Iterator - Working with Maps - Comparators. **Event Handling:** Introduction - Event Delegation - Model - Event classes - Sources of Events - Event Listener - Adapter Classes - Inner Classes in Event Handling

UNIT - IV

11 Periods

Introducing GUI Programming with Swing: Introducing Swing - Exploring Swing - Introducing Swing Menus

TEXTBOOK

1. Herbert Schildt, *The Complete Reference JAVA*, Tata McGraw Hill, Eleventh Edition, 2019.

REFERENCES

1. Harvey M. Deitel and Paul J. Deitel. *JAVA: How to Program*, Prentice Hall, Ninth Edition, 2012.
2. Kathy Sierra and Bert Bates. *Head First Java*, O'Reilly, Second Edition, 2009.

ONLINE MATERIAL

1. <http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-092-introduction-to-programming-in-java-january-iap-2010/lecture-notes/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Recall the basic syntax and structure of the Java program • Create programs for the given problem using classes • Design programs using Packages and Interfaces
Unit II	• Develop applications using multiple threads • Create program to solve problems using Strings and Files
Unit III	• Develop programs for event handling • Develop applications using collection interfaces • Design applications employing collection classes
Unit IV	• Design GUI using Swings

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Design Classes to model real world objects using inheritance
- Create Classes using Packages, Interfaces and Exception Handling
- Develop applications using Threads and Files
- Create programs for event handling
- Design applications using collection framework
- Create applications Using Swing

LIST OF LABORATORY EXERCISES

1. Program to demonstrate Polymorphism.
2. Program to demonstrate Inheritance.
3. Program to demonstrate Interfaces and Packages.
4. Program to demonstrate Exception handling.
5. Program to demonstrate Multithreading.
6. Program using I/O classes.
7. Program to demonstrate String Handling
8. Program using Set and Comparable interface
9. Program to Handle Mouse Event and Key Event.
10. Program using swing.

L	T	P	C
4	0	0	4

Course Code: CSE105

Semester: III

COMPUTER ORGANIZATION

Course Objectives

This course will enable the learner to understand the basic building blocks of computer, arithmetic operation, pipelined data path, memory hierarchy, storage and interfacing of I/O

UNIT- I

17 Periods

Instructions: Language of the Computer: Introduction - Operations of the Computer Hardware - Operands of the Computer Hardware - Signed and Unsigned Numbers - Representing Instructions in the Computer - Logical Operations - Instruction for Making Decisions - Supporting Procedures in Computer Hardware - Communicating with people - ARM Addressing for 32-Bit Immediate and More Complex Addressing Modes - Parallelism and Instructions: Synchronization. **Arithmetic for Computers:** Introduction - Addition and Subtraction - Multiplication - Division - Floating Point - Parallelism and Computer Arithmetic: Associativity

UNIT - II

15 Periods

The Processor: Introduction - Logic Design Conventions - Building a Datapath - A Simple Implementation Scheme - An Overview of Pipelining - Pipelined Datapath and Control - Data Hazards: Forwarding versus Stalling - Control Hazards - Exceptions - Parallelism and Advanced Instruction-Level Parallelism

UNIT - III

15 Periods

Exploiting Memory Hierarchy: Introduction - The Basics of Caches - Measuring and Improving Cache Performance - Virtual Memory - A Common Framework for Memory Hierarchies - Virtual Machines - Using a Finite-State Machine to Control a Simple Cache - Parallelism and Memory Hierarchies: Cache Coherence

UNIT - IV

13 Periods

Storage and Other I/O Topics: Introduction - Dependability, Reliability, and Availability - Disk Storage - Flash Storage - Connecting Processors, Memory, and I/O Devices - Interfacing I/O Devices to the Processors, Memory and Operating System - I/O Performance Measures: Examples from Disk and File Systems - Designing an I/O System - Parallelism and I/O: Redundant Arrays of Inexpensive Disks

TEXT BOOKS

1. David A. Patterson, John L. Hennessy. *Computer Organization and Design*, Pearson Education, Fourth edition, 2014.
2. David A. Patterson, John L. Hennessy. *Computer Organization and Design*, Pearson Education, Fifth edition, 2016.

REFERENCES

1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky. *Computer Organization*, McGraw Hill, Fifth edition, 2011.
2. William Stallings, *Computer Organization and Architecture - Designing for Performance*, Pearson Education, Tenth edition, 2006.

ONLINE MATERIAL

1. <https://nptel.ac.in/courses/106106092/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Recognise Instructions for making decisions • Outline the basic arithmetic operations
Unit II	• Illustrate pipelined data path and distinguish hazards • Classify parallelism and extend advanced instruction level parallelism
Unit III	• Evaluate the performance of cache memory • Appraise memory hierarchy and need for virtual memory
Unit IV	• Explore storage alternatives and interfacing of I/O devices • Define levels of RAID

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Recognise Instructions for making decisions and outline the basic arithmetic operations
- Illustrate pipelined data path and distinguish hazards
- Classify parallelism and extend advanced instruction level parallelism
- Appraise memory hierarchy, evaluate performance of cache memory and need for virtual memory
- Explore storage alternatives, levels of RAID and interfacing of I/O devices

L	T	P	C
4	0	0	4

Course Code: CSE103
Semester: III

DATA STRUCTURES

Course Objectives

This course will help the learner to manipulate linear and non-linear data structures and use the knowledge to select appropriate data structure for a given application.

UNIT - I

15 Periods

Pseudo code & Recursion: Introduction - Pseudo code - ADT - ADT model - implementations;
Recursion - Designing recursive algorithms. **Searching:** Sequential search - binary search and hashed search - Collision resolution techniques. **Sorting:** Introduction to sorting techniques - Bubble sort and selection sort implementation

Applications: Recursion: GCD - factorial - Fibonacci - Prefix to Postfix conversion - Towers of Hanoi

UNIT - II

15 Periods

General linear lists: Operations - implementation - algorithms. **Linear Lists:** Stacks - Basic Operations - Linked stack - Algorithms - Implementation. Queues - Operations - Linked list design - algorithms. **Complex implementations:** Circularly linked lists - doubly linked lists

Applications: Stack - Infix to Postfix - Evaluation of Postfix. Queue - Categorizing data - Student information system using general linear linked list

UNIT - III

15 Periods

Non-Linear Lists: Trees: Basic concepts - binary trees - traversals. **General Trees:** insertion - deletion - conversion of general to binary tree. **Binary Search Tree:** Basics - operations - threaded trees. **AVL Trees:** Basic concepts and implementation

Applications of Trees: Expression trees and traversal - Student information processing using BST

UNIT - IV

15 Periods

Heaps: Heaps - basics - implementation - algorithms. **Multi-way Trees:** B-tree implementation - algorithms - 2-3 tree - Trie structure. **Graphs:** Elementary operations - Storage Structures - Graph algorithms - BFT - DFT

Applications: Heap Application - Priority Queues - Graph Application - Minimum Spanning Tree Prim's and Kruskal's algorithm

TEXT BOOKS

1. Sanjay Pahuja. *A Practical Approach to Data Structures and Algorithms*, New Age International Publishers, 2010.
2. Richard F. Gilberg, Behrouz A. Forouzan. *Data Structures: A Pseudocode Approach with C*, Thomson Learning, Second Edition, 2007.

REFERENCES

1. Ellis Horowitz, Sartaj Sahni, Susan Anderson Freed. *Fundamentals of Data Structures in C*. University Press, Second Edition, 2008.
2. J.P.Trembley, P.G. Sorenson. *An Introduction to Data Structures with Applications*, Tata McGraw Hill, Second Edition, 2002.

ONLINE MATERIALS

1. https://onlinecourses.nptel.ac.in/noc18_cs25/preview

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Formulate pseudo code for problems • Solve problems using recursion method • Develop programs to process records using general linked list and apply searching and sorting techniques
Unit II	• Differentiate stack and queue data structures • Apply stack and queue data structures to solve problems • Summarize the implementations of linked list
Unit III	• Illustrate different tree data structures • Choose an appropriate tree data structure for the given application • Formulate algorithms to process tree data structures
Unit IV	• Distinguish Multi-way trees and binary trees • Apply graph data structure to solve real life problems • Identify minimum weight spanning tree for weighted graphs

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Write pseudo-code for recursive and non-recursive applications
- Choose appropriate searching and solving strategy in problem solving
- Define and use linear data structures like stack, queue and linked lists for the given problems
- Define non-linear tree data structure and use it to solve the given problems
- Choose the appropriate representation of graph
- Select the appropriate data structure and algorithm to solve the given problem

L	T	P	C
4	0	0	4

Course Code: ECE101

Semester: III

DIGITAL SYSTEM DESIGN

Course Objective

This course enables the learners to understand the design and the basic building blocks of processors through digital logic circuits.

UNIT - I

15 Periods

Boolean Simplification & Logic Families

Review of Number Systems and Codes - Boolean Functions: Boolean Laws- Simplification using the Laws- Minterms - Maxterms - Sum of Product and Product of Sum Forms - Karnaugh Map(up to 5 variables) -Introduction to Quine McClusky Method - NAND, NOR Implementation.

Digital Logic Families: Introduction to Digital Logic Families.

UNIT - II

15 Periods

Combinational Circuits

Combinational Circuits: Half and Full Adder, Ripple Carry Adder, Carry Look Ahead Adder, BCD Adder, Half and Full Subtractor, Parallel Adder/Subtractor, BCD Subtractor-Magnitude Comparators - Multiplexer - Demultiplexer - **Encoder:** Decimal to BCD, Octal to Binary, Priority Encoder - **Decoder:** 2-4 Line, 3-8 Line, 4-16 Line, BCD to Decimal, BCD to Seven Segment decoder - **Code Converters** : Gray to Binary, Binary to Gray, Gray to BCD, Parity Generator/Checker - Implementation of Boolean Functions using Multiplexer and Demultiplexer.

UNIT - III

15 Periods

Sequential Circuits

Flip Flops: RS, D, JK and T, Characteristics Equation and Excitation Table-Master Slave Flip Flops-Realization of One Flip Flop using Other Flip Flop.

Shift Registers: SISO, SIPO, PISO, PIPO, Bi-Directional Shift Register and Universal Shift Register using Multiplexer.

Counters: Design of Synchronous Counters and Asynchronous Counters, Ring Counter, Johnson Counter, Sequence Generator using Counters and Shift Register.

Finite State Machine (FSM): Basic Design Procedure-Mealy & Moore Machine.

UNIT - IV

15 Periods

Introduction to Verilog HDL - Data Types - Value Set, Wire, Wand, Wor, Tri, Register, Input, Output, Inout, Integer, Parameter. Operators - Arithmetic, Relational, Bit-wise, Logical, Conditional, Concatenation, Shift, Reduction Operators. Module - Continuous Assignment, Module instantiation - Gate Primitives - Procedural Assignment - Always and Initial blocks, Conditional statement, Looping statement, Event control. HDL models for simple combinational and sequential circuits.

TEXT BOOKS

1. M.Morris Mano. *Digital Design*, Fourth Edition, Pearson Education, 2011.
2. T.L.Floyd. *Digital Fundamentals*, Tenth Edition, Pearson Education, 2011.

3. Albert Malvino and Leach. *Digital principles and Applications*, Fifth Edition, Tata McGraw Hill, 2005.
4. Michael D.Ciletti. *Advanced Digital Design with the Verilog HDL*, Second Edition, Prentice Hall of India, 2010.

REFERENCES

1. Ronald J.Tocci. Neal S.Widmer. *Digital systems principles and applications*, Eleventh Edition, Prentice Hall of India, 2010.
2. Charles H.Roth, Jr. *Fundamentals of Logic Design*. Fifth Edition, Thomson. 2007.
3. Donald D. Givone. *Digital Principles and Designs*, Tata McGraw Hill, 2003.
4. Bhasker. J. *A Verilog HDL Primer*, Third Edition, B.S.Publications, 2015.

ONLINE MATERIALS

1. NPTEL - <https://nptel.ac.in/courses/117105080>
2. NPTEL- <https://nptel.ac.in/courses/106108099>
3. NPTEL - http://nptel.ac.in/courses/Webcoursecontents/IIT20Guwahati/digital_circuit/frame/index.html

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to:

Unit - I	• Comprehend the number systems and its minimization logic in digital logic circuits • Categorize the various digital logic families based on their characteristics
Unit - II	• Apply and design various combinational circuit for the given logic
Unit - III	• Analyze and implement various synchronous and asynchronous sequential circuits
Unit - IV	• Design the appropriate combinational, sequential circuits for the given application using verilog.

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Comprehend the number systems and its minimization logic in digital logic circuits
- Categorize the various digital logic families based on their characteristics
- Apply and design various combinational circuit for the given logic
- Analyze and implement various synchronous and asynchronous sequential circuits
- Design the appropriate combinational, sequential circuits for the given application using verilog.
- Design a processor using combinational and sequential logic circuits.

L	T	P	C
0	0	2	1

Course Code: CSE104
Semester: III

DATA STRUCTURES LABORATORY

Course Objectives

This course will help the learner to implement the concepts and techniques of data structures using C / C++ / Java.

1. Implementation of sorting techniques (Bubble Sort and Insertion Sort)
2. Implementation of searching techniques (Sequential search, Binary search and Hashed search)
3. Demonstration of Stack and Queue operations using arrays
4. Implementation of Singly Linked List
5. Demonstration of Stack and Queue operations using linked linear list
6. Demonstration of Infix to Postfix conversion using stack
7. Implementation of Doubly Linked List
8. Demonstration of Circular DLL operations
9. Demonstration of BST operations
10. Graph traversal using BFS
11. Graph traversal using DFS
12. Finding Minimum Weight Spanning Tree employing Prim's algorithm

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Evaluate and choose appropriate searching and sorting techniques
- Comprehend the applications of linear data structures like stack and queue
- Implement information processing systems using linked list data structures
- Demonstrate operations on Binary Search Tree
- Employ stack and queue operations to perform Graph traversals
- Construct minimum weight spanning tree of weighted graphs

L	T	P	C
0	0	2	1

Course Code: ECE103

DIGITAL SYSTEM DESIGN LABORATORY

Course Objective

To provide hands on experience to design any combinational / sequential logic circuits for the given application.

LIST OF EXPERIMENTS

Combinational Logic Circuit Design:

1. Design of Adder, Subtractors and Four bit (Modulo 16) Adder / Subtractor
2. Construction of 1- bit Comparator-using 74xx and verification of 4 bit Comparator IC 7485.
3. Design and Implementation of Multiplexer, Demultiplexer and code converter (Gray to Binary)
4. Design of Verification of Decoders using IC 74147 and encoder using IC 7445.

Sequential Logic Circuit Design:

5. Verification of various flip-flops using gates
6. Verification of Shift Register - SIPO/SISO & PIPO/PISO
7. Design of 4 bit sequence generator
8. Design of Mod N Asynchronous and Synchronous counter using IC 7490 and IC 74192
9. Design of 1 bit ALU using logic gates and study of 4 bit ALU using IC 74181
10. Design of digital clock to display minutes and seconds using 7-segment (IC 7447)

Simulation using Verilog HDL:

11. Design of combinational circuits
12. Design of sequential circuits

COURSE LEARNING OUTCOMES

Upon successful completion of each experiment, the learner will be able to

- Construct various arithmetic circuits using ICs
- Implement 1-bit comparator using 74xx IC and extend the design to verify 8 bit comparator using IC 7485.
- Select and design suitable MUX/DEMUX for the given Boolean expression
- Build the appropriate code converters for the given number system
- Analyze and verify the given encoder and decoder circuits using ICs
- Construct different flip flops using logic gates and verify their outputs
- Design various shift registers for the given applications
- Realize the operation of the given 4 bit sequence generator

- Develop N bit synchronous and asynchronous counters using ICs
- Assemble N bit Arithmetic Logic Unit using appropriate logic ICs
- Construct a real time clock using digital logic circuits
- Design the combinational and sequential circuits using Verilog

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Design and implement digital logic systems for various electronic applications

L	T	P	C
3	0	2	4

Course Code: CSE101

Semester: III

PROBLEM SOLVING & PROGRAMMING IN C

Course Objectives

This course will help the learner to formulate simple algorithms for arithmetic & logical problems and translate the algorithms to C programs using constructs like arrays, functions, pointers & structures

UNIT - I

11 Periods

Introduction: Flowchart and algorithms - swapping of two numbers - simple arithmetic problems - even or odd number - greatest of three numbers - finding roots of a quadratic equation - factorial - sum of n numbers - reversing the digits of a number - gcd, base conversion - generation of fibonacci series. **Introduction to C:** Character set - identifiers and keywords - data types - constants - variables and arrays - declarations - expressions - statements - symbolic constant. **Operators and Expressions:** Arithmetic operators - unary operators - relational and logical operators - assignment operators

UNIT - II

11 Periods

Library Functions and Data I/O: getchar - putchar - scanf - printf - gets - puts - interactive programming. **Control Statements:** if else statement - conditional operator - while - do-while and for statements - nested control structures - switch statement - break and continue statements - comma operator. **Functions:** Defining and accessing a function - function prototypes - passing arguments to a function - recursion. **Applications:** Operations on a set of numbers, finding factorial, sum of series, operations on digits of an integer, base conversion, character to number conversion

UNIT - III

11 Periods

Storage Classes: Automatic - External and static variables - multi file programs. **Arrays:** Definition - processing - passing arrays to functions - multi dimensional arrays. **Strings:** Definition - NULL character - initialization of strings - reading and writing a string - processing the strings - character arithmetic. **Applications:** Insertion and deletion in an array, bubble sort, linear and binary search, removal of duplicates from an ordered array, partitioning an array, prime factorization, keyword searching in a text, searching & sorting of strings

UNIT - IV

12 Periods

Pointers: Fundamentals - pointer declarations - passing pointers to a function - pointers and one dimensional arrays - dynamic memory allocation - operations on pointers - pointers and multi dimensional arrays - arrays of pointers - more about pointer declarations. **Structures and Unions:** Defining and processing a structure - user defined data types (typedef) - structures and pointers - passing structures to functions - self referential structures - unions. **Applications:** Programs using array of structures and array operations using pointer arithmetic

TEXTBOOKS

1. Byron, Gottfried. *Programming with C*, McGraw Hill Education, Third Edition, 2017.
2. Dromey, R.G. *How to solve it by computer*, Pearson Education Inc., 2012.

REFERENCES

1. Anil Bikas Chaudhuri. *The art of programming through flowcharts & algorithms*, Firewall Media, First Edition, 2018.
2. Brian W Kernighan. and Dennis M Ritchie. *The C programming language*, Pearson Education, Second Edition, 2017.
3. Pradip Dey. and Manas Ghosh. *Computer fundamentals and programming in C*, Oxford University Press, Second Edition, 2013.

ONLINE MATERIALS

1. NPTEL - <http://nptel.ac.in/courses/106104128/>
2. NPTEL - <http://nptel.ac.in/courses/106102066/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Write simple algorithms for arithmetic and logical problems
Unit II	• Develop programs using input and output statements • Recall the syntax of branching and looping constructs • Select an appropriate construct to solve the given problem • Develop programs using functions
Unit III	• Illustrate the operations on arrays • Demonstrate string operations
Unit IV	• Employ structures and pointers for developing a given application

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Write a simple algorithm for a given problem
- Illustrate declarations and operations in C language
- Develop programs using input and output statements
- Select an appropriate C control structure to solve the given problem
- Illustrate the operations on arrays
- Develop applications using composite data types, functions and pointers

LIST OF LABORATORY EXERCISES

1. Programs using Input, output and assignment statements
2. Programs using Branching statements
3. Programs using Looping statements
4. Programs using Functions
5. Programs using Recursion
6. Programs using Single and Multidimensional Arrays
7. Programs using strings
8. Programs using Pointers
9. Programs using dynamic memory allocation
10. Programs using Structures

L	T	P	C
3	1	0	4

Course Code: MAT302
Semester: IV

DISCRETE STRUCTURES

Course Objectives

This course will help the learner to develop logical thinking and its application to computer science, optimize the Boolean expression, derive closed form and asymptotic expressions from series and recurrences for growth rates of processes, model problems in computer science using graphs and trees, understand the basic concepts of group and its properties.

UNIT - I

15 Periods

Set Theory: Basic Set Operations-Cartesian Product and Power Sets. **Combinatorics:** Basic Counting techniques-Permutations-Partitions of Sets and Laws of addition-Combinations and Binomial theorem **Logic:** Propositions and logical operations-Truth tables and propositions generated by a set-Equivalence and implication-Laws of logic-Mathematical systems-Propositions over a Universe-Mathematical induction-Quantifiers-Review of methods of proof. **More on sets:** Minsets - Duality Principle

UNIT - II

12 Periods

Relations and Graphs: Basic Definitions-Graphs of relations-Properties of relations-Matrices of relations-Closure operations on relations. **Functions:** Definition and notation-Injective, Surjective and Bijective functions-Composition, Identity and Inverses. **Boolean Algebra:** Lattices-Boolean Algebras-Axioms of a Boolean Algebra- Boolean identities - Karnaugh Map method of simplifying Boolean functions Quine-Mc-Clusky tabulation method of simplifying Boolean functions-Boolean expression-Application of Boolean algebra to switching theory

UNIT - III

18 Periods

Recursion and Recurrence Relations: Faces of recursion-Sequences-Recurrence relations-Common recurrence relations-Generating functions-Recursion and Computer Algebra Systems. **Graph Theory:** Introduction-Connectivity- Transversals: Eulerian and Hamiltonian-Graph Optimization-Planarity and Colorings. Introduction to Trees

UNIT - IV

15 Periods

Algebraic Systems: Operations-Algebraic Systems-Some general properties of groups- Z_n the integers modulo n -Subsystems-Direct Products-Isomorphisms. **Group Theory and Applications:** Cyclic groups-Cosets and Factor groups-Permutation groups-Normal Subgroups and Group Homomorphisms-Coding Theory-Group codes. **Introduction to Rings and Fields:** Rings-Basic definitions and Concepts-Fields

TEXTBOOK

1. Alan Doerr and Kenneth Levasseur. *Applied Discrete Structures for Computer Science*, Pearson Education, Second Edition, 2007. (A Free electronic version under Creative Common Licence is available for free download under the title “Applied Discrete Structures”, Version 2, Year 2013).

REFERENCES

1. Kenneth H. Rosen. *Discrete Mathematics and its Applications*, McGraw Hill, Seventh Edition, 2012.
2. Jean Paul Tremblay, Rampurkar Manohar. *Discrete Mathematical Structures With Applications to Computer Science*, McGraw Hill, 2005.
3. NarsinghDeo. *Graph Theory with Applications to Engineering and Computer Science*, PHI, 2004.

ONLINE RESOURCE

1. <https://nptel.ac.in/courses/106105192/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Skill development for application of various methods for solving problems in Combinatorics and for identifying problems that necessitate solution by Mathematical induction and to write algorithms of their own
Unit II	• Skill development for handling various Boolean algebraic operations in problems in switching theory and to study Logic gates from the perspective of Boolean algebraic operators
Unit III	• Acquiring skills and understanding various Graph theoretical concepts and use them in problems from Algebraic Graph theory and to provide graph theoretic approach to solve a wide variety of Engineering problems that need topological and geometrical description of data
Unit IV	• The learner will have thorough knowledge in Group Theory and Coding Theory fundamentals to write algorithms in Machine Learning Environment

COURSE LEARNING OUTCOMES

Upon the completion of this Course, Learner will be able to

- Comprehend formal logical arguments
- Specify and manipulate basic mathematical objects such as sets, functions, and relations and will also be able to verify mathematical properties of these objects possess
- Represent graphs in machines using matrices and apply algorithms to problems using graphical theoretical concepts
- Understand the basic concepts of group, rings and their applications in both algebraic and geometric contexts

Course Code: CSE305
Semester: IV

L	T	P	C
4	0	0	4

DESIGN & ANALYSIS OF ALGORITHMS

Course Objectives

This course will help the learner to acquire knowledge to develop efficient algorithm for a given application by selecting appropriate design technique and analyze its computational complexity

UNIT - I

15 Periods

Introduction: Role of algorithms in computing - Analyzing algorithms - Designing algorithms - Growth of Functions. **Divide and conquer:** maximum subarray problem - Strassen's algorithm for matrix multiplication. **Solving Recurrences:** Substitution method - Recursion tree method - Master method

UNIT - II

15 Periods

Sorting: Heapsort: Heaps - Maintaining the heap property - Building a heap - heap sort algorithm. **Quicksort:** Description of Quicksort - Performance of Quicksort - A randomized version of Quicksort - Analysis of Quicksort. **Sorting in Linear Time:** Lower bounds for sorting - Counting sort - Radix sort - Bucket sort

UNIT - III

15 Periods

Dynamic Programming: Rod cutting - Matrix Chain Multiplication - Longest common subsequence. **Greedy Algorithms:** An activity selection problem - Elements of the greedy strategy - Huffman codes

UNIT - IV

15 Periods

Graph Algorithms: Breadth first search - Depth first search - Topological sort - The algorithms of Kruskal and Prim - Bellman-Ford algorithm - Dijkstra's algorithm - Floyd-Warshall algorithm. **String Matching:** Notation and Terminology - The naïve string matching - Rabin Karp algorithm - Knuth-Morris-Pratt algorithm

TEXTBOOK

1. T.H. Cormen, C.E. Leiserson, R.L. Rivest, and C. Stein. *Introduction to Algorithms*, Prentice Hall of India, Third Edition, 2009. (Paperback-2011)

REFERENCES

1. Anany Levitin. *Introduction to the Design and Analysis of Algorithm*, Pearson Education, Third Edition, 2012.
2. Sara Baase and Allen Van Gelder. *Computer Algorithms - Introduction to Design and Analysis*, Pearson Education, Third Edition, 2008.
3. A.V. Aho, J.E. Hopcroft, and J.D. Ullman. *The Design and Analysis of Computer Algorithms*, Pearson Education, 2003.
4. Jon Kleinberg and Éva Tardos. *Algorithm Design*, Pearson Education, First Edition, 2013.

ONLINE MATERIALS

1. <http://nptel.ac.in/courses/106101060/>
2. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-046j-design-and-analysis-of-algorithms-spring-2015/index.htm#>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Define asymptotic notations for time complexity analysis • Develop algorithms using divide and conquer strategy • Employ the techniques for solving recurrences to find the computational complexity of recursive algorithms
Unit II	• Compare the performance of sorting algorithms, • Identify appropriate sorting algorithm for a given application
Unit III	• Design algorithms for a given application by employing dynamic programming • Write algorithms for a given application using greedy strategy
Unit IV	• Judge and select appropriate graph traversal algorithms for a given application • Judge and select appropriate string matching algorithm based on application requirement.

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Define asymptotic notations for time complexity analysis
- Design algorithms for a given problem by applying strategies like divide and conquer, dynamic programming, and greedy and evaluate their performance
- Select appropriate methodology to develop an algorithm for a given problem
- Employ the techniques for solving recurrences to find the computational complexity of recursive algorithms
- Select appropriate sorting algorithm based on time complexity analysis for a given application
- Choose appropriate graph traversal and string matching algorithms for a given problem

L	T	P	C
3	0	0	3

Course Code: INT102R01
Semester: IV

FUNDAMENTALS OF DATABASE MANAGEMENT SYSTEMS

Course Objectives:

This course will help the learner to infer the fundamentals of data models and to conceptualize and depict a database system using ER diagram. Construct tables and write effective queries to design and implement a database for real-world applications.

UNIT - I

11 Periods

Introduction: Data-Database - Characteristics of the Database Approach - Actors on the scene - Advantages - Data models, Schemas and Instances - Three Schema Architecture and Data Independence - Database Languages and Interfaces - Database System Environment. **Data Modeling:** Data modeling using the ER model

UNIT - II

11 Periods

SQL: Basic SQL - Data Definition and Data Types, Specifying Constraints, Basic Retrieval Queries, Insert, Delete and Update statements - Additional features of SQL - More Complex SQL Retrieval Queries - Specifying Constraints as Assertions and Actions as Triggers - Views - Schema Change Statements in SQL - JDBC - SQL Class Library for Java Programming

UNIT - III

12 Periods

Relational Database Design: Functional Dependency - First, Second and Third Normal Forms - BCNF - Multi valued Dependency and Fourth Normal Form - Join Dependency and Fifth Normal Form. **Transaction Processing Concepts:** Introduction to Transaction Processing - Transaction and System Concepts - Serializability - Concurrency Control Techniques - Two-Phase Locking Techniques, Timestamp Ordering

UNIT - IV

11 Periods

Introduction to Non-Relational Databases (MongoDB): Getting Started - Documents, collections, databases, data types. **Creating, Updating and Deleting documents:** Inserting and saving documents - removing documents - updating documents - Querying - Introduction to find querying - Type-specific Queries

TEXT BOOKS

1. R.Elmasri, S.B.Navathe. *Fundamentals of Database Systems*, Addison Wesley, Seventh Edition, 2016.
2. Kristina Chodorow and Michel Dirolf. *MongoDB: The Definitive Guide*, O'Reilly, Publications, Second Edition, 2013.

REFERENCES

1. Henry F.Kort, Abraham Silberschatz, Sudarshan. *Database System Concepts*, McGraw Hill, Seventh Edition, 2019.
2. S.K.Singh. *Database Systems Concepts, Design and Applications*, Prentice Hall, First Edition, 2009.
3. Raghu Ramakrishnan, Johannes Gehrke. *Database Management Systems*, McGraw Hill, Third Edition, 2014.

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/106105175/1>
2. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-830-database-systems-fall-2010/lecture-notes/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Compare and contrast database systems from file systems and describe the fundamental elements of relational database management systems • Design ER-models to represent simple database application scenarios
Unit II	• Develop SQL queries for the given scenario using DDL and DML commands • Construct connectivity with database using JDBC
Unit III	• Improve the database design by normalization • Design and implement various concurrency control techniques in transaction processing system
Unit IV	• Analyze the recent advancements in MongoDB • Develop and run the application using MongoDB

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Illustrate the fundamental elements of relational database management systems
- Design ER,EER-models to represent simple database application scenarios
- Experiment with SQL to construct queries for the given problem
- Improve the database design using normalization techniques
- Design and implement non relational databases for real world applications
- Develop and run the application using MongoDB

L	T	P	C
4	0	0	4

Course Code: CSE205R01

Semester: IV

COMPUTER ARCHITECTURE

UNIT - I

15 Periods

ARM Architecture and Instruction Set: ARM Design Philosophy - Embedded System Hardware - Embedded System Software - Registers - Current Program Status Register - Pipeline - Exceptions - Interrupts and the Vector Table - ARM Instruction Set - Thumb Instruction Set.

UNIT - II

15 Periods

ARM processor LPC214x: Introduction – Features - Architectural overview –Vectored Interrupt Controller - General Purpose Input / Output – UART - I2C – RTC: Pin & Register Description –Watchdog Timer – ADC & DAC.

UNIT - III

15 Periods

Advanced Architecture Features: Classes of Computers-Defining Computer Architecture - Quantitative Principles of Computer Design. **Instruction-Level Parallelism:** Concepts and Challenges-Basic Compiler Techniques for Exposing ILP-Reducing Branch Costs with Advanced Branch Prediction-Overcoming Data Hazards with Dynamic Scheduling-Dynamic Scheduling: Examples and the Algorithm-Hardware-Based Speculation

UNIT - IV

15 Periods

Vector Architecture and Parallelism: Data level parallelism: Introduction-Vector Architecture-SIMD Instruction Set Extensions for Multimedia-Graphics Processing Units-Detecting and Enhancing Loop-Level Parallelism- **Thread-Level Parallelism:** Introduction-Centralized Shared-Memory Architectures-Performance of Symmetric Shared-Memory Multiprocessors-Distributed Shared-Memory and Directory-Based Coherence-Synchronization: The Basics-Models of Memory Consistency: An Introduction

TEXT BOOKS

1. Andrew N Sloss, Dominic Symes and Chris Wright, "ARM System Developers Guide: Designing and optimizing system software", Morgan Kaufmann, 2004.
2. UM10139LPC214x User manual Rev 4 — 23 April 2012.
3. John L. Hennessy, David A. Patterson. *Computer Architecture - A Quantitative Approach*, Morgan Kaufmann is an imprint of Elsevier, Fifth Edition, 2012.

REFERENCES

1. William Stallings. *Computer Organization and Architecture - Designing for Performance*, Pearson Education, Tenth edition, 2006.

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/106/108/106108100/>
2. <https://nptel.ac.in/courses/106/105/106105193/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	Describe the architecture, instruction set and advanced features of ARM7 processor
Unit II	Identify the significance of ARM processor along with its Pin and register description with its features for designing an embedded system
Unit III	Select techniques for extending basic pipelining to increase parallelism explored at instruction level
Unit IV	- Illustrate vector architecture and summarize SIMD instruction set - Extend enhancement of loop level parallelism and Thread Level Parallelism

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Describe the architecture, instruction set and advanced features of ARM7 processor.
- Identify the significance of ARM based Microcontroller along with its Pin and register description with its features for designing an IoT system
- Select techniques for extending basic pipelining to increase parallelism explored at instruction level
- Illustrate vector architecture and extend enhancement of loop level parallelism and Thread Level Parallelism

Course Code: CSE306
Semester: IV

L	T	P	C
0	0	2	1

DESIGN & ANALYSIS OF ALGORITHMS LABORATORY

Course Objectives

This course will help the learner to select appropriate algorithm and design technique for a given application

1. Programs on Comparison Sorts - Merge Sort, Quick Sort and Heap Sort - performance analysis by counting the number of swaps performed
2. Programs on Linear Sorting techniques - Counting Sort, Radix Sort and Bucket Sort
3. Applications of Divide and Conquer approach - Maximum Subarray problem - with comparison of computation time with Brute Force approach for different input sizes
4. Applications of Dynamic Programming - Matrix Chain Multiplication - comparison of computation time with brute force approach for different input sizes
5. Applications of Dynamic Programming - Longest Common Subsequence - comparison of computation time with brute force approach for different input sizes
6. Applications of Greedy Technique - Text data compression using Huffman Codes - finding compression percentage for different kinds of inputs
7. Programs on Graphs - Topological Sort of Directed Acyclic Graph
8. Programs on Graphs - Minimum Spanning Tree using Prim's and Kruskal's algorithms
9. Programs on Graphs - Single Source Shortest paths using Bellman-Ford algorithm
10. Programs on Graphs - All-Pairs Shortest Paths using Floyd-Warshall algorithm
11. String Matching - with Rabin-Karp algorithm
12. String Matching - Knuth-Morris-Pratt (KMP) algorithm

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Choose a sorting technique for any application
- Examine the performances of algorithms
- Employ Divide and Conquer strategy for solving problems having (optimal) substructure property
- Employ Dynamic Programming strategy for solving optimization problems having optimal substructure property and overlapping subproblem property
- Employ graph algorithms for routing and other applications
- Use string matching algorithms for text processing applications

L	T	P	C
0	0	2	1

Course Code: CSE206R01
Semester: IV

COMPUTER SYSTEM DESIGN LABORATORY

Course Objectives

This course will help the learner to understand assembly language, I/O operations, Timer, ADC operations and Interrupts of ARM architecture.

LIST OF EXPERIMENTS

Assembly Language Programming

1. Programs on Data transfer Instructions
2. Programs on ALU instructions
3. Programs on Stack

Interfacing On and Off Chip Peripherals with ARM

4. Controlling the on-board LEDs
5. Controlling the on-board Switch
6. Designing Program flow control using Internal Timers
7. Implementation of Hardware Interrupt
8. Data transfer using Serial Communication
9. Interfacing ADC
10. Interfacing Stepper Motor

Using GEM5

11. Program on Task context maintenance
12. Performance analysis of cache memory

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Develop Assembly language program for any given problem using ARM processors
- Demonstrate I/O operations for Advanced Microcontroller
- Demonstrate Timer operations
- Illustrate the ADC and UART functions in ARM board
- Demonstrate interrupt routines
- Analysis of system architecture using GEM5

L	T	P	C
3	0	2	4

Course Code: CSE201
Semester: IV

OBJECT ORIENTED PROGRAMMING IN C++

Course Objective

This course will help the learner to employ object oriented concepts for developing programs catering to different applications

UNIT - I

11 Periods

Introduction: Fundamentals of object oriented programming - procedure oriented programming vs. Object Oriented Programming (OOP), Object Oriented Programming concepts - classes, reusability, encapsulation, inheritance, polymorphism, dynamic binding, and message passing. **C++ Programming Basics:** Output using cout - directives - input with cin - type bool - Manipulators - type conversions. **Functions:** Returning values from functions - reference arguments - overloaded functions - inline functions - default arguments - returning by reference

UNIT - II

11 Periods

Objects and Classes: Implementation of classes in C++ - C++ objects as physical objects - C++ objects as data types - constructors - objects as function arguments - returning object from function - default copy constructor - structures and classes - objects and memory - static class data - const data and classes. **Arrays and String Arrays Fundamentals:** Arrays as class member data - arrays of objects - strings - standard C++ string class

UNIT - III

11 Periods

Operator Overloading: Overloading unary operators - overloading of binary operators - data conversion - pitfalls of operators overloading and conversion - keywords explicit and mutable. **Inheritance:** Concept of inheritance - derived class and base class - derived class constructors - overriding member function - class hierarchies - public and private inheritance - levels of inheritance - multiple inheritance - ambiguity in multiple inheritance. **Aggregation:** classes within classes - inheritance and program development

UNIT - IV

12 Periods

Pointers: Addresses and pointer - address-of operator &, pointer and arrays - pointer and fraction - pointer and C-type string. **Memory Management:** New and delete - pointers to objects - debugging pointers. **Virtual Functions:** Virtual functions - friend functions - static functions - assignment and copy initialization - this pointer - dynamic type information. **Streams and Files:** Stream classes - stream errors - disk file I/O with streams - file pointers - error handling in file I/O - file I/O with member functions - overloading the extraction and insertion operators

TEXTBOOKS

1. Robert Lafore. *Object oriented programming in C++*, Pearson Education, Fourth Edition, 2012.

REFERENCES

1. Behrouz A Forouzan, and Richard FGilberg. *Computer Science: A Structured Approach using C++*, Cengage Learning, Second Edition, 2014.
2. Joyce Farrell. *Object Oriented Programming using C++*, Cengage Learning, Fourth Edition, 2014.

ONLINE MATERIAL

1. MIT Courseware - <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-introduction-to-c-and-c-january-iap-2013/index.htm>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Describe the object-oriented concepts • Use basic constructs and functions in C++ for solving a given problem
Unit II	• Employ the concepts of classes and objects for a given application • Develop applications using string class objects for text data
Unit III	• Create user defined data types with overloaded operators • Develop applications by making use of inheritance
Unit IV	• Employ pointers for better memory management • Demonstrate the use of Virtual and friend functions • Develop applications using file streams

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Describe the object-oriented concepts
- Create user defined data types with classes
- Develop applications using string class objects for text data
- Use the Object Oriented Programming concepts like polymorphism, inheritance, virtual and friend functions for developing applications
- Employ pointers for better memory management
- Develop applications using file streams

LIST OF LABORATORY EXERCISES

1. Basic C++ Programs using Input / Output, branching and Loops
2. Programs using functions with default arguments, inline functions and function overloading
3. Programs using classes with constructors and destructors
4. Programs using array as data members of a class and array of objects
5. Programs using String class
6. Programs using operator overloading
7. Programs for data conversion using overloading
8. Programs using inheritance
9. Programs using virtual functions and friend functions
10. Programs using file Input / Output

L	T	P	C
3	1	0	4

Course Code: MAT301R01

Semester: IV

ENGINEERING MATHEMATICS - IV
(Common to all branches except Mechatronics & ICT)

Course Objectives

This course helps the learner to understand the techniques to solve various Partial Differential Equations (PDEs) used in Engineering disciplines, understand properties of Fourier transforms and their applications, understand numerical methods and apply them to solve engineering problems and impart numerical techniques to solve boundary value problems in Engineering computations.

UNIT - I

15 Periods

Partial Differential Equations: Formation of PDE - Eliminating arbitrary constants- Eliminating arbitrary functions – First order nonlinear PDE: $f(p, q) = 0$, $f(z, p, q) = 0$, $f(x, p) = g(y, q)$, Clairaut's equation - Lagrange's linear equation – Homogeneous linear PDE of Higher order with constant coefficients.

Applications: Draining of liquid down the side of a vessel (first-order equations involves the draining of liquid from a vessel, a procedure common to many industrial processes), The transmission-line equations and Deflections of a plate.

UNIT - II

15 Periods

Fourier Transforms: Fourier integral theorem (statement only) – Infinite and finite Fourier and its inverse transform - Fourier sine and cosine transform- Transform of derivatives- Convolution and Parseval's theorem (statement only) - Initial and boundary value problems .

Applications: Electrical Oscillation in simple circuit, Transverse vibration of a continuous string and Transverse vibration of a thin membrane.

UNIT - III

15 Periods

Numerical solution of various equations, Interpolation, Numerical Differentiation and Integration: Newton Raphson method - Gauss Jordan Method - Gauss Seidel Method- Interpolation – Newton's Forward & Backward difference – Lagrange's interpolation – Numerical differentiation – Newton's forward and backward difference formula - Numerical Integration - Trapezoidal's Rule – Simpson's one-third rule and Simpson's three-eighth rule .

Applications: Computation of Velocity and Acceleration by Numerical differentiation. Determine the total quantity of heat and Root Mean square current by Numerical integration.

UNIT - IV

15 Periods

Numerical solution of initial value and boundary value problems: Euler – Improved Euler- Modified Euler - Fourth order Runge-Kutta Method for first and second order ODE –Liebmann's Process for solution of Laplace's equation – Poisson equation – Parabolic equation Bender-Schmidt Method – Crank Nicholson Method – Solution of Hyperbolic equation.

Applications: Swinging pendulum problems and Physical problems governed by elliptic, parabolic and hyperbolic PDE's

TEXTBOOKS

1. T. Veerarajan. *Engineering Mathematics (For IV semester)*, McGraw Hill Publications, 2001.
2. Steven C. Chapra, Raymond. P. Canale. *Numerical Methods for Engineers*, McGraw Hill publications, Sixth edition, 2012.

REFERENCES

1. C.Ray Wylie and Louis C.Barrett. *Advanced Engineering Mathematics*, McGraw Hill, Sixth Edition, 2012.
2. B.S.Grewal. *Higher Engineering Mathematics*, Khanna Publishers, Forty Fourth Edition, 2017.
3. P.Anuradha, and V.Sudhakar, *Transforms and Partial Differential Equations*, Scitech Publications, Second Edition, 2017.

ONLINE RESOURCES

1. <https://nptel.ac.in/courses/111105038/14/>
2. <https://www.youtube.com/watch?v=ujXi29Mf83Q>
3. <https://nptel.ac.in/courses/111105038/31/>
4. https://www.youtube.com/watch?v=Lw_Obk39UzE

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• The learner will be able to form, classify and solve partial differential equations of first and higher orders
Unit II	• The learner will be able to formulate and solve one dimensional heat conduction problems using standard techniques developed using Fourier Transforms
Unit III	• The learner will be able to apply standard techniques of numerical methods to solve boundary value problems
Unit IV	• The learner will be able to determine the values of derivatives of single variable functions using standard numerical techniques and evaluate definite integrals using Trapezoidal and Simpsons' rules

COURSE LEARNING OUTCOMES

Upon the completion of this Course, Learner will be able to

- Solve partial differential equations of first and higher orders
- Solve one dimensional heat conduction problems using Fourier transform methods
- Solve differential equations using numerical methods
- Solve a system of simultaneous equations using numerical techniques

L	T	P	C
3	1	0	4

Course Code: MAT304R01

Semester: IV

MATHEMATICS FOR CYBER SECURITY

Course Objectives

The course aims to train the learner to gain knowledge on the fundamentals of counting principles and probability, comprehend the basic concepts of Number Theory and solve congruence equations, understand the nuances of Algebraic structures and their applications, acquire a working knowledge of various kinds of ciphers

UNIT - I

15 Periods

Fundamentals of Set Theory & Combinatorics: Basics of Counting Principles, The Pigeonhole principle– Generalized Permutations and Combinations – Fundamentals of Probability – Introduction to Random variables (Discrete & Continuous) (Distributions- not included) - Conversion of numbers in decimal to binary and Hexadecimal number systems – Principle of Mathematical Induction & Strong Mathematical Induction.

UNIT - II

15 Periods

Elementary Number Theory: Divisibility, Basic and Extended Euclidean Algorithms, Primality testing algorithms, Properties of prime numbers, functions ($\mu(n)$, $\sigma(n)$, $\tau(n)$, $\varphi(n)$, $s(n)$ and $\pi(x)$) Modular Arithmetic – Linear Congruence equations - Chinese Remainder Theorem. Euler's theorem, Fermat's theorem, and Wilson's theorem (Statements Only) - Determining the GCD of two positive integers using Euclidean Algorithm and expressing it in the form of $ax+by=1$ using Extended Euclidean Algorithm - Problems on existence of unique and multiple solutions of a linear congruence equations - Solving for primitive root of a number and Primality testing using Euler's theorem, Fermat's theorem, and Wilson's theorem- Solving a system of Linear congruence equations using Chinese remainder theorem.

UNIT - III

15 Periods

Fundamentals of Abstract Algebra: Groups: Definitions and fundamental properties of Subgroups, Cosets, Lagrange's theorem- Cyclic groups, Abelian Groups, Permutation groups- Normal Subgroups, Simple groups - Sylow's theorems (Without proof) -Homomorphisms and Isomorphisms - Introduction to Rings, Integral Domains, Fields - Properties.

UNIT - IV

15 Periods

Fundamentals of Cryptography: Applied Number theory in Cryptography and Information security: - Definitions, Block ciphers - Shift Ciphers – Affine Ciphers – Polyalphabetic ciphers - Public Key Cryptography –RSA Cryptosystems - Elliptic curves and Arithmetic operations on the points on an Elliptic curve– Diffie Hellman Key Exchange, Problems on various ciphers and encryption techniques using Public Key Cryptography, RSA and Diffie Hellman Key Exchange.

TEXT BOOKS

1. Rosen. K.H, Discrete Mathematics and its applications, 7th edition, McGraw Hill publications, 2007. (Unit 1)
2. Song Y Yan, Number Theory for Computing, Springer Verlag, New York, 2010. (Unit 2)
3. J. A. Gallian, Contemporary Abstract Algebra, 9th edition, Cengage Learning, USA, 2017.(Unit 3)
4. Mollin, R. A, An Introduction to Cryptography, Chapman and Hall/CRC, 2006. (Unit 4)
5. Lawrence C. Washington, Elliptic curves: Number Theory and Cryptography, Second edition, Chapman & Hall/CRC, 2008. (Unit 4)

REFERENCES

1. Thomas W. Judson, Abstract Algebra: Theory and Applications - Orthogonal Publishing L3c, 2013.
2. Lloyd Jaysingh, Frank Ayres, Schaum's Outline of Abstract Algebra (Schaum's Outlines) McGraw-Hill Education; Second Edition, 2003

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/111/105/111105112/>
2. <https://nptel.ac.in/courses/106/105/106105162/>
3. <https://nptel.ac.in/courses/111/103/111103020/>
4. <https://nptel.ac.in/courses/106/108/106108051/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

Unit I	Analyse and solve problems based on fundamentals of counting principles
Unit II	Identify properties of prime numbers, congruence equations, and solve for unknown values in congruence relations.
Unit III	Grasp the basic concepts such as groups, rings, fields and solve problems in them
Unit IV	Encrypt and decrypt a message based on various ciphers

COURSE LEARNING OUTCOMES

After the successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
CO1	Distinguish between various counting principles and solve problems based on the requirements.	K3
CO2	Learn the applications of fundamentals of Number theory in fields like Cryptography.	K3
CO3	Develop a thorough knowledge of Algebraic structures and Vector spaces.	K3
CO4	Appreciate how fundamentals of Groups and Fields are used in encrypting and decrypting messages based on different methods.	K3
CO5	Encrypt and decrypt a message based on various ciphers and identify.	K3

L	T	P	C
3	1	0	4

Course Code: CSE202
Semester: IV

COMPUTER GRAPHICS USING OPENGL

Course Objectives

This course will help the learner to develop and implement algorithms to create 2-D & 3-D objects. This course will also help the learner to manipulate, view, and render the objects using geometric transformations, viewing procedures, and shading algorithms.

UNIT - I

15 Periods

Overview of graphics systems: Video display devices - Raster Scan Systems - Introduction to OpenGL. **Graphics Output Primitives and Attributes:** Specifying a two-dimensional world coordinate reference frame in OpenGL - OpenGL point functions, OpenGL Line & Polygon functions, Line drawing Algorithms (DDA, BLA) - Circle generating algorithms - Ellipse generating algorithms - Line attributes - Curve attributes - Fill area attributes (Scan-line fill, Boundary Fill, Flood Fill) - Equivalent OpenGL functions including visualization

UNIT - II

15 Periods

2-D Geometric Transformation: Basic 2D transformations - Homogeneous coordinates - Composite transformations - Other 2D transformations. **3-D Geometric Transformation:** Basic 3D transformations - Other 3D transformations, Composite transformations - OpenGL Transformation functions. **2-D Viewing:** 2D viewing pipeline - Clipping window - Normalization and Viewport transformations - OpenGL 2D viewing Function - Point clipping - Line clipping (Cohen Sutherland, Liang Barsky) - Polygon clipping (Sutherland Hodge man, Weiler Atherton) - Curve Clipping - Text clipping

UNIT - III

15 Periods

3-D Viewing: Overview of 3D viewing concepts - 3D viewing pipeline - Transformation from world to viewing coordinate - Projection transformations - Orthogonal projections - Oblique projections - Perspective projection transformation coordinates - Perspective projection equations: Special cases - Vanishing points for perspective projections - Perspective projection Transformation matrix - Viewport transformation and 3D screen coordinates - OpenGL viewing functions. **3-D Representation:** Polyhedra - Spline representation - Hermite interpolation splines - Bezier spline curves and surfaces - B-spline curve equations - Uniform B-spline - B-spline surfaces - OpenGL curve functions

UNIT - IV

15 Periods

Visible surface detection: Classification of visible surface detection Algorithms - Backface detection, Depth buffer method - A-buffer method - Scan-Line method - OpenGL visibility functions. **Illumination models and Surface-Rendering Methods:** Light source - Surface lighting effects - Basic Illumination models - Transparent surfaces - Polygon rendering methods - Bump mapping - Equivalent OpenGL Illumination and Surface-Rendering functions. **Computer animation:** Design of animation sequence - General computer animation functions, raster animation, Computer animation languages, Key frame systems, Motion specifications - OpenGL animation procedures

TEXT BOOK

1. Donald Hearn, Pauline Baker, and Warren Carithers. *Computer Graphics with OpenGL*, Pearson Education, Fourth Edition, 2014.

REFERENCES

1. Shalini Govil - Pai. *Principles of Computer Graphics - Theory and Practice Using OpenGL and Maya*. Springer-Verlag, 2008.
2. F.S. Hill, Jr. *Computer Graphics using OpenGL*. Pearson Education, Second Edition, 2003.

ONLINE MATERIALS

1. NPTEL: <http://nptel.ac.in/courses/Webcourse-contents/IIT-Delhi/Computer%20Graphics/csmain.htm>
2. NPTEL: <http://nptel.ac.in/courses/106106090/>
3. NPTEL: <http://nptel.ac.in/courses/106102065/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Explain the working of display devices and I/O devices • Construct algorithms for standard output primitives and make use of OpenGL functions for creating predefined output primitives
Unit II	• Construct transformation matrices for transforming 2-D and 3-D objects • Apply clipping algorithms on 2-D objects
Unit III	• Construct viewing transformation matrices for viewing 3-D objects (parallel, oblique & perspective projections) • Explain the methods used for 3-D object representation
Unit IV	• Apply Hidden surface removal algorithm to identify and remove back faces. • Explain the algorithms used for illumination models, surface rendering methods, and techniques used for animation

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to:

- Explain the working of display devices and I/O devices
- Apply algorithms to generate 2-D and 3-D output primitives
- Demonstrate 2-D and 3-D transformations
- Apply viewing and clipping algorithms for displaying 2-D and 3-D objects
- Demonstrate transforming of 3-D objects into 2-D using projections
- Create realistic view of objects using surface and colour rendering techniques
- Explore the techniques to animate the 2-D and 3-D objects

L	T	P	C
3	1	0	4

Course Code: CSE307
Semester: IV

OBJECT ORIENTED ANALYSIS & DESIGN

Course Objectives

This course will help the learner to Infer, Analyse and Apply various techniques in Object Oriented Modeling, Analysis, Design and Implementation.

UNIT - I

15 Periods

Introduction: Definitions - OO Themes - Modeling - Abstraction - The three Models. **Class Modeling:** Object and Class concepts - Links and Associations - Generalization and Inheritance - a simple class model - Navigation of Class Models. **Advanced Class Modeling:** Concepts - Association ends - N-ary Association - Aggregation - Abstract class - Multiple Inheritance - Metadata - Reification - Constraints - Derived Data - Packages

UNIT - II

15 Periods

State Modeling: Events - States - Transitions and conditions - State diagrams - Behavior **Advanced State Modeling:** Nested State Diagrams - Signal Generalization - concurrency - Sample State Model - Relation of class and State Models. **Interaction Modeling:** Use Case Models - Sequence Models - Activity Models - Use case relationships - Special constructs for Activity Model

UNIT - III

15 Periods

Process Overview: Development Stages - Development Life Cycle - Devising a system Concept - Elaborating the concept - Preparing the Problem Statement. **Domain Analysis:** Overview - Domain Class Model - Domain State Model - Domain Interaction Model - Iterating the Analysis. **Application Analysis:** Application Interaction Model - Application Class Model - Application State Model - Adding Operations

UNIT - IV

15 Periods

System Design: Overview - Estimating Performance - Making a reuse Plan - Breaking the system into subsystems - Identifying concurrency - allocation of subsystems - Management of data storage - Handling global resources - common Architecture styles - Architecture of the ATM system. **Class Design:** Overview - Bridging the Gap - realizing Use Cases - Designing algorithms - Refactoring - Design Optimization - Organizing a Class Design - ATM Example. **Implementation Modeling:** Overview - fine-tuning classes - Fine-tuning Generalizations - Realizing Associations - Testing

TEXT BOOK

1. Michael R Blaha, James R Rumbaugh. *Object-Oriented Modeling and Design with UML*, Pearson, Second Edition, 2013.

REFERENCES

1. Ali Bahrami. *Object Oriented System Development*, McGraw-Hill Higher Education, 2015.
2. Grady Booch, Robert A. Maksimchuk Michael, W. Engle, Bobbi J. Young, Jim Conallen, and Kelli A. Houston. *Object-Oriented Analysis and Design With Applications*, Addison - Wesley, 2007.

ONLINE MATERIAL

1. <http://ocw.mit.edu/courses/architecture/4-297-special-problems-in-architecture-studies-fall-2000/labs/UML1.pdf>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Infer the basic concepts of Object-Orientation • Infer and Create Class modelling
Unit II	• Infer the concepts of events, state, and transitions • Create State diagrams, various kinds of Interaction diagrams
Unit III	• Infer the overview of Object-Oriented Development • Perform Domain Analysis using various models • Perform Application Analysis using various models
Unit IV	• Perform System Design and Class Design • Infer various Implementation Techniques

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Infer the basic concepts and can create Class modelling
- Infer the concepts of events, state, transitions and interactions
- Create State diagrams and Interaction diagrams
- Perform Domain and Application Analysis
- Perform System Design and Class Design
- Infer various Implementation Techniques

L	T	P	C
3	1	0	4

Course Code: CSE301

Semester: V

THEORY OF COMPUTATION

Course Objectives

This course will help the learner to discuss different classes of formal languages in Chomsky hierarchy, their properties and to design the acceptor machines for those languages.

UNIT - I

15 Periods

Introduction: Preliminaries and notations - Basic concepts - applications - **Finite Automata:** Deterministic FA - Non-deterministic FA - Equivalence - Minimization - **Regular languages and Regular grammars:** Regular expressions - Relation between regular languages and regular expressions – Regular grammars **Properties of Regular Languages:** Closure properties - identifying non-regular languages using pumping lemma

UNIT - II

15 Periods

Context Free Languages: Context free grammars - parsing and ambiguity - Context-free grammars and programming languages **Simplification and Normal Forms:** Methods for Transforming grammars - Chomsky and Greibach normal forms - membership algorithm for CFG. **Push Down Automata:** Non-Deterministic PDA - PDA and CFL - Deterministic PDA and deterministic CFL - Grammars for deterministic CFL

UNIT - III

15 Periods

Properties of CFL: Pumping Lemma for CFL, Closure properties and decision algorithm for CFL. **Turing Machines:** The Standard Turing Machine - combining TM for complicated tasks - Turing's thesis. **Other models of TM:** Minor variations on TM - TM with complex storage - Non-deterministic TM - Universal TM - Linear bounded automata

UNIT - IV

15 Periods

A hierarchy of formal languages and automata: Recursive and recursively enumerable Languages - unrestricted grammars - context sensitive grammars and languages - Chomsky Hierarchy - **Limits of algorithmic computation:** problems that can't be solved by TM - Undecidable problems for recursively enumerable languages - post correspondence problem - Undecidable problems for CFL - **An overview of computational complexity:** Turing Machine models and complexity - Language families and complexity classes - complexity classes P and NP - Some NP problems - Polynomial time reduction - NP-completeness

TEXTBOOK

1. Peter Linz. *An Introduction to Formal Languages and Automata*. Jones and Bartle Learning International United Kingdom, Sixth Edition, 2016.

REFERENCES

1. John E. Hopcroft, Rajeev Motwani and Jeffery D Ullman. *Introduction to Automata Theory, Languages and Computation*, Third Edition, Pearson Education, 2007.
2. Alfred V. Aho, Monica S. Lam, Ravi Sethi and Jeffrey D. Ullman. *Compilers Principles, Techniques & Tools*, Pearson Education, 2007.

3. Susan H. Rodger and Thomas W. Finley. *JFLAP: An Interactive Formal Languages and Automata Package*, Jones & Bartlett Publishers, Sudbury, MA, 2006.
4. Michael Sipser. *Introduction to the theory of computation*, Second Edition, Thomson Course Technology, 2006.

ONLINE MATERIALS

1. <http://nptel.ac.in/courses/106104028/>
2. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-045j-automata-computability-and-complexity-spring-2011/lecture-notes/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- List the properties of regular languages, design regular expressions and construct equivalent automata - Identify and prove a language is regular or not
Unit II	- Design the context-free grammars for context-free languages, transform them into normal forms - Identify a string belongs to the given context-free language or not - Construct PDA for the equivalent context-free grammars
Unit III	- Identify CFL and prove using Pumping Lemma - Demonstrate the properties of CFL - Design Turing Machine for simple and complex tasks - Describe different Turing Machine models
Unit IV	- Summarize Chomsky Hierarchy, and differentiate recursive & recursively enumerable languages - Describe concepts of computational complexity, unsolvable and un-decidable problems

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Design an appropriate automaton for a given language
- Construct equivalent automaton from grammar of languages
- Design an appropriate push down automaton for a given language
- Construct PDA for the equivalent context-free grammars
- Select appropriate Turing Machines for a given problem based on complexity analysis
- Describe concepts of computational complexity, unsolvable and un-decidable problems

L	T	P	C
3	0	0	3

Course Code: CSE308
Semester: V

OPERATING SYSTEMS

Course Objectives

This course will help the learner to understand the modules of an operating system, their functions, process management techniques, scheduling algorithms, problems of concurrent processes and possible solutions, memory management strategies, file and IO system concepts.

UNIT - I

11 Periods

Introduction: What Operating systems do - Structure - Operations - Process management - **Operating system Structure:** Services - User Interfaces - Systems calls and types - Systems programs - Design and implementation - Operating System Structure - System boot

UNIT - II

11 Periods

Process management: Process concept - Scheduling - Operations on processes - Interprocess communication - **Threads:** Overview - Multi-core programming - Multithreading models - **Process scheduling:** Basic concepts - Scheduling criteria - Algorithms: FCFS - SJFS - Priority - RR - Multilevel queue - Multilevel feedback - Thread scheduling: Contention scope - Pthread scheduling - Multiple-processor scheduling - **Process synchronization:** Background - Critical section problem - Peterson's solution - Synchronization hardware - Mutex locks - Semaphore - Classic problems of synchronization

UNIT – III

11 Periods

Deadlocks : System model - Characterization - Methods for handling deadlock - Prevention - Avoidance - Detection - Recovery from deadlocks - **Memory management:** Background - Swapping - Contiguous memory allocation - Paging - Structure of page tables - Segmentation - **Virtual memory:** Background - Demand paging - Copy-on-write - Page replacement - Allocation of frames - Thrashing

UNIT - IV

12 Periods

File system: File concept - Access methods - Directory and disk structure - File system mounting - Protection - File allocation methods - **Mass Storage:** Magnetic disks - Disk structure - Disk scheduling: FCFS - SSTF - SCAN - C-SCAN - LOOK - Selection of an algorithm - **I/O systems:** Overview - I/O hardware - Application I/O interface - Kernel I/O subsystem

TEXTBOOK

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne. *Operating System Concepts*, Ninth Edition, Wiley, 2013.

REFERENCES

1. William Stallings. *Operating Systems: Internals and Design Principles*, Ninth Edition, Pearson, 2017
2. Russ Cox, Frans Kaashoek, and Robert Morris. *X v6: A simple, Unix-like Teaching Operating System*, MIT, 2014

3. Achyut Godbole and Atul Kahate. *Operating System*, Third Edition, Tata McGraw, 2011

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/106106144/2>
2. <https://www.youtube.com/user/MrKeerthikiran/search?query=operating+systems>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Discuss the concept of process and process management • Illustrate the IPC techniques and multithreading • Summarize the operations of operating systems
Unit II	• Distinguish threads from processes and point out the advantages of threads over processes • Demonstrate CPU scheduling algorithms • Compare thread scheduling approaches • Explain the synchronization problems and implement solutions for the classic problems of synchronization
Unit III	• Describe the causes of deadlocks. • Differentiate the three solutions for deadlock handling. • Implement banker's algorithm for deadlock avoidance. • Demonstrate the memory management schemes and compare them.
Unit IV	• Analyze the performance of disk scheduling algorithms • Appraise the concepts of file systems • Summarize the concepts related to I/O systems

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Understand the modules of an operating system and to recognize the services of operating systems
- Experiment with different CPU scheduling algorithms in order to compare and analyzes their performances
- Comprehend the causes and consequences of deadlocks and the possible options to tackle them
- Learn the concepts of memory management and different memory partitioning techniques with their merits and demerits
- Implement various and compare disk scheduling algorithms
- Appraise the file organization and I/O techniques

L	T	P	C
4	0	0	4

Course Code: CSE302

Semester: V

COMPUTER NETWORKS

Course Objectives

This course will help the learner to analyse various application layer protocols and its implication and criticize the TCP flow control, TCP congestion control mechanism and evaluate various inter-domain, intra-domain routing protocols and link layer protocols.

UNIT - I

15 Periods

Computer Networks and the Internet: Introduction about Internet - The Network Edge - The Network Core - Delay, Loss, and Throughput in Packet-Switched Networks - Protocol Layers and Their Service - **Application Layer:** The Web and HTTP - FTP - Electronic Mail in the Internet - DNS - The Internet's Directory Service - Peer-to-Peer Applications

UNIT - II

15 Periods

Transport Layer: Introduction and Transport-Layer Services - Multiplexing and Demultiplexing - **Connectionless Transport:** UDP - Principles of Reliable Data Transfer - **Connection-Oriented Transport:** TCP - Principles of Congestion Control - TCP Congestion Control

UNIT - III

15 Periods

The Network Layer: Introduction - Virtual circuits and datagram networks - Inside a Router - The Internet protocol (IP) forwarding and addressing in the Internet - Routing algorithms - Routing in the Internet - Broadcast and multicast routing

UNIT - IV

15 Periods

The Link Layer: Introduction to the link layer - Error detection and correction techniques - Multiple access links and protocols - Switched Local area networks - Link virtualization - Data centre networking - Wireless link & network characteristics - 802.11 wireless LAN - **Network Management:** The Network management - The infrastructure for Network management - The Internet standard management framework

TEXTBOOK

1. James F. Kurose, and Keith W. Ross. *Computer Networking: A Top-down Approach*. Pearson Education, Seventh Edition, 2017.

REFERENCES

1. William Stallings. *Data and Computer Communications*. Prentice Hall of India, Tenth Edition, 2014.
2. Behrouz A Forouzan and Firouz Mosharraf. *Computer Networks - A Top-Down Approach*. Tata McGraw Hill. Indian Edition, 2012.
3. Larry L Peterson and Bruce S. Davie. *Computer Networks: A Systematic Approach*, Elsevier. Seventh Edition, 2011.
4. Andrew S Tanenbaum and David J. Wetherall. *Computer Networks*. Pearson Education. Fifth Edition, 2010.

ONLINE MATERIALS

1. NPTEL- <https://nptel.ac.in/courses/106105081/>
https://onlinecourses.nptel.ac.in/noc18_cs38/preview
2. MIT Courseware: <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-829-computer-networks-fall-2002/>
3. Coursera course: <https://www.coursera.org/courses?query=computer%20network>

UNITWISE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

Unit I	• Discuss about the networking principles and its Quality of Service parameters • Demonstrate the application layer functionalities
Unit II	• Categorise various routing protocols • Demonstrate IPv4 and IPv6 packet and addressing structures
Unit III	• Assemble Internet- Transport layer services and its significance in networking • Analyse the TCP congestion control strategies
Unit IV	• Describe link layer functionalities and classify various multiple link access protocols • Evaluate the network management framework

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Analyse the networking principles and various network QoS metrics
- Discuss about various application layer protocols
- Compare various routing protocols in the network layer and evaluate Internet packet formatting and forwarding
- Criticise Transport layer services in Internet and evaluate TCP congestion and flow control mechanisms
- Demonstrate link layer functionalities and multiple link access methods
- Analyse Internet network management framework

L	T	P	C
0	0	2	1

Course Code: CSE309

Semester: V

OPERATING SYSTEMS LABORATORY

Course Objectives

This course will help the learner to gain pragmatic knowledge on the different modules of operating system by simulating them and analysing their performance differences.

1. Program to create of a child process from a parent process using fork system call and make them communicate between them using pipe.
2. a. Program to implement IPC using Shared Memory System Call.
b. Program to implement IPC with the help of Message Queues.
3. Programs for the simulation of uni-processor scheduling algorithms and analyze their performances.
4. Program to experiment multi-processor scheduling.
5. Program for the simulation of thread scheduling approaches.
6. Program to implement peterson's algorithm for enforcing mutual exclusion.
7. a. Program to demonstrate for producer-consumer problem using semaphore.
b. Program to apply semaphore for tackling reader-writer problem.
8. Program to apply banker's algorithm for deadlock avoidance.
9. Program to implement deadlock detection algorithm.
10. Program to implement dining philosopher's problem without causing deadlocks.
11. a. Program to simulate page replacement algorithms and to compute number of page faults
b. Program to simulate address translation under paging.
12. Program to implement disk scheduling algorithms.

Additional Experiments:

13. Program to simulate dynamic partitioning and buddy system.
14. Program to demonstrate file allocation techniques.

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Develop program that create parent/child or concurrent process and carry out communication between them using pipe or IPC methods
- Analyze uni-processor, multi-processor and thread scheduling algorithms by simulating them and providing them with sample data
- Implement mutual exclusion based on semaphores to prevent concurrent issues under classic problems of concurrency
- Demonstrate deadlock avoidance and detection strategies
- Simulate memory management schemes and disk scheduling schemes
- Demonstrate file allocation techniques

L	T	P	C
0	0	2	1

Course Code: CSE303
Semester: V

COMPUTER NETWORKS LABORATORY

Course Objectives

This course will help the learner to understand socket programming using TCP and UDP, simulate and analyse various network models. It makes the learner to understand various protocols and standards in networking.

1. Development of a chat application based on socket programming
2. Development of a secure file transfer application
3. Implementation of Remote Method Invocation
4. Simulation and analysis of wired network
5. Simulation and analysis of wireless network
6. Simulation and analysis of TCP flow control and TCP congestion control
7. Simulation and comparative analysis of distance vector routing and link state routing
8. Simulation of Broadcast routing and multicast routing
9. Simulation and analysis of DHCP protocol
10. Simulation and analysis of Ethernet LAN IEEE 802.3
11. Simulation and analysis of Wireless LAN IEEE 802.11
12. Study of SASTRA network infrastructure

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Demonstrate socket programming and RMI
- Evaluate remote method invocation protocol
- Analyze various TCP flow control and congestion control algorithms
- Compare and criticize various routing protocols
- Demonstrate wired LAN using simulator
- Evaluate wireless LAN using various metrics

L	T	P	C
0	0	2	1

Course Code: TNP101
Semester: V

SOFT SKILLS - I

Course Objectives

This course will help the learner to emphasize, impart and improve their soft skills.

S.NO	TOPICS	NO. CLASSES
1	Self-Introduction	2
2	Impromptu	4
3	SWOT Analysis	4
4	Body Language	2
5	Product Launch	2
6	Person I admire / Book I Like / If I were	4
7	Mock Press	2
8	Out of Box Thinking (Creativity & Innovative Thinking)	1
9	Debate on Current Affairs	4
10	General Quiz	1
11	Semester Practical	3
12	Semester Theory Exam	1
TOTAL		30

REFERENCES

1. Hariharan, S. Sundararajan, N. and Shanmuga Priya, S.P. *Soft Skills*, MJP Publishers, 2010.
2. Prof. G. Ravindran, Prof. S.P.B. Elango, and Dr. L. Arockiam, *Success through Softskills*. Institute for Communication & Technology, 2007.
3. Dr. K.Alex. *Soft Skills*. S. Chand & Company, 2009.

LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- The learner will be able to successfully introduce himself before others
- The learner realizes the importance of presence of mind and react sharply and swiftly
- Students are encouraged to do a self introspection of their strengths, weaknesses, opportunities and threats
- The learner realizes the importance of body language in the day to day communication

- The learner understands the importance of marketing skills and the need for sharpening the skill
- This inculcates the habit of reading and know about the greatness of great people
- The learner will enhance the answering skills and the ability to articulate his thoughts
- The learner will develop lateral thinking and encouraged to think differently
- The learner will be exposed to the happenings around the globe through a debate
- The learner will be exposed to the happenings around the globe through a Quiz

L	T	P	C
3	1	0	4

Course Code: MAT305

Semester: V

STATISTICAL FOUNDATIONS FOR COMPUTER SCIENCE

Course Objectives

To understand a random variable that describes randomness or uncertainty in certain realistic situation with probability distributions joint distributions, correlation and regression. To learn the characteristics of sampling and testing of hypothesis of few unknown statistical parameters. To understand the mechanism of queuing system, the characteristics of queue, the mean arrival and service rates, the expected queue length and waiting line.

UNIT - I

15 Periods

Random variables and probability distributions: Random variables - Discrete and continuous. Probability distributions - mass function/ density function of probability distribution. Mathematical Expectation - Moment about origin - Central moments Moment generating function of probability distribution. Binomial - Poisson - normal distributions Erlang and Cauchy Distributions – properties - Moment generating functions of the above three distributions, and hence finding the mean and variance

UNIT - II

15 Periods

Multiple Random variables, Correlation & Regression: Joint probability distributions- Joint probability mass/ density function - Marginal probability mass / density functions - Covariance of two random variables - Correlation Coefficient of correlation - The rank correlation - Regression- Regression Coefficient - The lines of regression and multiple correlation & regression

UNIT - III

15 Periods

Sampling Distributions and Testing of Hypothesis: Introduction to inferential statistics - Procedure for Testing of Hypothesis - Tests of significance for large samples - Difference between sample proportion and population proportion - Difference between two sample proportions - Test of significance for small samples - Student's t-distribution - Difference between sample mean and population - Difference between two sample means - F distribution - Chi Square distribution - Goodness of fit - Independence of Attributes

UNIT - IV

15 Periods

Queuing Theory: Structure of a queuing system - Operating characteristics of queuing system - Transient and steady states - Terminology of Queuing systems - Arrival and service processes - Pure Birth - Death process Deterministic queuing models - M/M/1 Model of infinite queue - M/M/1 model of finite queue

TEXTBOOKS

1. T.Veerarajan. Probability, Statistics and Random Processes, Tata McGraw Hill Publishing Company Ltd, Second Edition, New Delhi, 2003.
2. Jorge I. Aunon and V. Chandrasekar. *Introduction to Probability and Random Processes*, McGraw Hill International Editions, Third Edition, New York, 1998.
3. Sheldon M.Ross. *Probability and Statistics for Engineers and Scientists*, Academic Press, Fourth Edition, 2009.

4. S.D. Sarma, KedarNath Ram Nath. *Operations Research Theory Methods & Applications*, 2017.

REFERENCES

1. S.C.Gupta ,V.K.Kapoor. *Fundamentals of Mathematical Statistics*, Sultan Chand and Sons Publications, Tenth Revised Edition, 2007.
2. George R. Cooper, Clare D. McGillem, *Probabilistic Methods of Signal & System Analysis*, Oxford Press, Third Edition, 2007.

ONLINE RESOURCES

1. <https://ocw.mit.edu/courses/mathematics/18-05-introduction-to-probability-and-statistics-spring-2014/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- The learner will have an understanding of the fundamental concepts of basic Probability Theory, Statistical Distributions to apply in problem situations - Students would be able to identify distribution in certain realistic situation. Also able to differentiate among many random variable involved in the probability models
Unit II	The learner will have an understanding of multiple random variables, Correlation and Regression Analysis before entering into Probabilistic Network Architectures
Unit III	The student would be able to calculate mean and proportions (small and large sample) and to make important decisions from few samples which are taken out of unmanageably huge populations
Unit IV	The students would be able to find the expected queue length, the ideal time, the traffic intensity and the waiting time. These are very useful tools in many engineering and data management problems in industry. It is useful for computer science and engineering

COURSE LEARNING OUTCOMES

Upon the completion of this Course, Learner will be able to

- Apply the probability distribution in a realistic situation
- Use the properties of linear statistical models in different network architectures.
- Make important decisions from few samples which are taken out of unmanageably huge populations.
- Analyze the applications to different queuing models

L	T	P	C
3	0	2	4

Course Code: INT301
Semester: V

NETWORK TOOLS & TECHNIQUES

Course Objectives

This course will help the learners to acquire sufficient knowledge in designing, building well secured systems. The learners are to understand the practical skills for managing un-secured networks and maintain IT infrastructures.

UNIT - I

11 Periods

Basics of Network: Layer - packets - Topology - Internet Protocol (IP) - IPV4 Header - classification of IPV4 - CIDR - subnet masking - problem for subnet masking

Vulnerability scanners: Identify Security holes in your system - Vulnerability scanners to the rescue - Nessus - Metasploit - Consideration for Vulnerability scanning

UNIT - II

11 Periods

Port Scanners: Overview of Port scanners - Consideration of port Scanning - Uses of port scanning - Nmap - THC Amap - NetCat - SoCat

Network Sniffers: Consideration for network sniffers - Tcpdump - Windump - Wireshark - Hping

Network Defences: Intrusion Detection System - NIDS Signature Example - IDS tunnelling - IDS Analysis Tool - Snort - Unique feature - Host Based Intrusion Detection system - Advantage and Disadvantage - Tripwire

UNIT - III

12 Periods

Firewall: Firewall and Packet Filters - The Basics - Packet Filters Vs Firewall - How a Firewall protects a Network - Packet characteristics to Filter - Stateless Vs Statefull Firewalls - Windows firewall - Linux Iptables - Installing Iptable - Creating Iptable - Using Iptable - Creating Iptable Firewall - IP Masquerading with Iptable - Network Address Translation (NAT) - Basics of Virtual private Networking - Tunneling-protocols - DMZ - Port forwarding and redirection - Understanding ports and services - Secure Shell (SSH) - Datapipe - Fpipe - WinRelay - Password Attack - Password cracking tool - Hashcat - Kismet - Airtight - WEP, WPA & WPA2

UNIT - IV

11 Periods

Brief History of Malware - Computer Viruses - Malware - Riskware Cloak of the Rootkit Types of Rootkit - Rootkit Techniques - Tackling Rootkit - Rise of the BotNet: Main characteristics - Components - BotNet Usage - BotNet Protective Mechanism - Fight against BotNet - The threat Ecosystem - Technical Element - Human Element - Evolution - Advanced Persistent Threat

TEXT BOOKS

1. Peter L Dordal. *Introduction to Computer Network*, Loyola University Chicago, 2016.
2. Christopher C. Elisan. *Malware ,Rootkit and BotNetA Beginner's guide*, McGraw-Hill, 2013.
3. Shema,Mike and Bradley C Johnson. *Anti-hacker Tool Kit*, Mc-Graw-Hill/Obsorne, 2006.
4. Tony Howlett. *Open Source Security Tools-Practical Application for Security*, Prentice Hall, 2004.

REFERENCES

1. Thomas G. Robertazzi. *Introduction to Computer Networks*, Springer, 2017.
2. Christopher C. Elisan Michael A. Davis Sean M. Bodmer Aaron LeMasters. *Hacking Exposed Malware & Rootkits: Security Secrets and Solutions*, McGraw Hill Professional, Second Edition, 2016.

ONLINE MATERIALS

1. https://onlinecourses.nptel.ac.in/noc18_cs38/preview
2. https://onlinecourses.nptel.ac.in/noc19_cs29/announcements
3. <https://www.oreilly.com/library/view/malware-rootkits/9780071792066/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Infer the basic ideology about Networks • Perceive the knowledge of vulnerability scanning
Unit II	• Develop the idea behind port scanner and network sniffers • Explain the concepts of intrusion detection systems
Unit III	• Demonstrate the knowledge on firewall concepts • Elaborate VPN usage and Password attacks
Unit IV	• Develop the malware computer virus • Outline the threat ecosystem

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Infer the basic ideology about the Networks and vulnerability scanning
- Explain the concepts of IDS, port scanner and network sniffers
- Demonstrate the knowledge on firewall concepts
- Importance of NAT and VPN
- Interpret the various types of malware
- Explain the features of ecosystems

LIST OF LABORATORY EXERCISES

1. Installing Nessus for port scanning
2. Launching attacks using metasploit framework
3. Remote accessing service through Netcat commands
4. Identifying ports and protocols using Nmap commands
5. Redirecting traffic through datapipe and win relay
6. Perform packet sniffing using TCPdump
7. Capturing and analysing packets through wireshark
8. Cracking password using hashcat
9. Intrusion detection using Kismet in wireless LAN
10. WEP cracking using Aircrack-ng

L	T	P	C
3	1	0	4

Course Code: INT310
Semester: V

ARTIFICIAL INTELLIGENCE

Course Objectives

This course will help the learner to gain knowledge in various problem solving methods using artificial intelligence and design an intelligent agent that can solve ontological problems and derive rational decisions.

UNIT - I

15 Periods

Introduction: Intelligent Agents - Agents and environments - Good behaviour - Nature of environments - The structure of Agents

Solving Problems by Searching: Problem solving agents - Example problems - Searching for solutions - Uniformed search strategies - Informed search strategies - Heuristic Functions

UNIT - II

15 Periods

Beyond Classical Search: Local search algorithms and Optimization problems - Searching with nondeterministic actions - Searching with partial observations

Adversarial Search: Games - Optimal decisions in games - Alpha-Beta Pruning

Constraint Satisfaction Problems: Defining constraint satisfaction problems - Constraint propagation - Backtracking search for constraint satisfaction problems - Local search for constraint satisfaction problems

UNIT - III

15 Periods

First Order Logic: Combining the best of formal and natural languages - Syntax and Semantics for first order logic (FOL) - Using FOL - Knowledge Engineering in FOL - Inference in FOL - Propositional Vs. FOL - Unification and Lifting - Forward chaining - First Order definite clauses - A simple forward chaining algorithm - Backward chaining - Backward chaining algorithm - Resolution CNF for FOL - The resolution inference rule - Example Proofs

UNIT - IV

15 Periods

Classical Planning: Definition - Algorithm for planning as state space search - Planning graphs - Other classical planning approaches - Analysis of planning approaches

Probabilistic Reasoning: Quantifying uncertainty: Acting under uncertainty - Basic probability - Bayes' rule and its use - Representing knowledge in an uncertain domain - The semantics of Bayesian networks - Exact and approximate inference in Bayesian networks

TEXT BOOK

1. Stuart Russell and Peter Norvig. *Artificial Intelligence - A Modern Approach*, Pearson Education, Prentice Hall of India, Third Edition, 2014.

REFERENCES

1. Munesh Chandra Trivedi. *A Classical Approach to Artificial Intelligence*, Khanna Publishing, 2018.
2. Elaine Rich, Kevin Knight and Shiva shankar B. Nair. *Artificial Intelligence*, Tata McGraw - Hill, Third Edition, 2009.

3. George F. Luger. *Artificial Intelligence-Structures and Strategies for Complex Problem Solving*, Pearson Education / Prentice Hall of India, 2009.
4. Nils J. Nilsson. *Artificial Intelligence: A new Synthesis*, Morgan Kaufmann, 2009.

ONLINE MATERIALS

1. NPTEL- <https://nptel.ac.in/courses/106105077/>
2. MIT Courseware - <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-034-artificial-intelligence-fall-2010/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Classify categories of agents based on their relationship with the environment • Solve problems involving repeated states and identify ways of removing them • Infer Admissible Heuristics for solving problems
Unit II	• Choose an appropriate searching technique for the given problem • Determine the optimal search technique for the given constraint to solve problem
Unit III	• Apply and use knowledge and logic in various ontological problems • Judge, which representation is suitable under which circumstance
Unit IV	• Explain the Classical Planning approaches and analyse them • Discuss the ways to extract knowledge from uncertain domain and employ efficient probabilistic inferences

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Classify intelligent agents based on their relationship with the environment
- Infer Admissible Heuristics for solving problems
- Choose an appropriate search strategy for problem solving
- Choose appropriate ontology and logic for knowledge representation and inference
- Employ classical planning approaches and analyse them
- Apply probabilistic inference for uncertain domain of real world problems

L	T	P	C
3	0	2	4

Course Code: CSE304
Semester: V

PYTHON PROGRAMMING WITH WEB FRAMEWORKS

Course Objectives

This course will help the learner to develop python applications using predefined and user defined objects. The learner will be able to develop client server programs and web applications using Django.

UNIT - I

11 Periods

Introduction: Introduction to Python - use IDLE to develop programs - basic coding skills - work with data types and variables, numeric data, string data - Python functions - code control statements

Functions and modules: Defining and using functions - create and use modules - using standard modules - lists and tuples

UNIT - II

12 Periods

File I/O, Exceptions: Introduction to file i/o - text files and csv files - handling single and multiple exceptions

Other concepts and skills: Work with numbers, strings, dates and time - dictionaries - recursion and algorithms

UNIT - III

12 Periods

Object Oriented Programming: Introduction to classes and objects - define class - object composition - encapsulation - inheritance

Internet Client Programming: Internet Clients - Transferring files - Network news - E-mail - Related modules

UNIT - IV

11 Periods

Web Clients and Servers: Introduction - Python Web client tools - Web Clients - Web (HTTP) Servers

Web Frameworks - Django: Web Frameworks - Django - projects and apps - Blog - Add database service - Python application shell - Django administration app - Blog's user interface - improving output - working with user input - Forms and Model Forms

TEXT BOOKS

1. Michael Urban, Joel Murach. *Murach's Python Programming*, Mike Murach & Associates, First Indian Reprint, 2017.
2. Wesley J. Chun, *Core PYTHON Applications Programming*, Prentice Hall, Third Edition, 2013.

REFERENCE

1. Mark Lutz, *Learning Python*, O'Reilly Media, Fifth Edition, 2013

ONLINE MATERIALS

1. https://www.amazon.com/Murachs-Python-Programming-Michael-Urban/dp/1890774979#reader_1890774979
2. <https://www.oreilly.com/library/view/core-python-applications/9780132779371/>
3. <https://github.com/halterman>
4. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016/>
5. <https://nptel.ac.in/courses/106106182/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Write programs using predefined python objects and functions • Develop functions and modules using standard modules
Unit II	• Demonstrate file manipulation and exception handling • Operate on string, dictionary objects and develop recursive applications
Unit III	• Solve problems using object oriented concepts involving inheritance • Develop applications for internet client programming
Unit IV	• Develop programs for web client and server interaction • Design web applications using Django framework

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Develop applications using predefined objects of Python
- Create user defined functions and modules for application development
- Schedule read and write operations using files
- Demonstrate exception handling techniques and recursive algorithms
- Use object oriented programming concepts to solve inheritance problems
- Develop applications for file transfer across web clients and web applications using Django framework

LIST OF LABORATORY EXERCISES

1. Programs using numeric data types and string data
2. Programs using selection and iteration structures
3. Programs using functions and modules (built-in and user defined)
4. Programs using list, tuple, dictionary
5. Programs for file i/o with text and csv files
6. Programs for exception handling
7. Programs to demonstrate recursive algorithms
8. Programs to demonstrate inheritance
9. Programs to demonstrate file transfer between web clients and servers
10. Programs to host web applications using Django

L	T	P	C
3	0	2	4

Course code: CSE203

Semester: V

LINUX PROGRAMMING

Course Objective

This course will help the learner to describe the features of Linux operating system and develop applications in Linux environment.

UNIT - I

15 Periods

Introduction and shell Programming : Starting with Linux shells - Looking into the Linux kernel - GNU utilities- Basic bash Shell Commands - Listing Files and Directories - Handling Files - Managing Directories - Viewing File Contents - Working with data files - Using Linux Environment Variables - Basic Script building - Creating script file - Redirecting input and output - Pipes - Performing Math - exit command

UNIT - II

15 Periods

Using structured Commands - Conditionals and looping statements - Creating functions. Advanced Shell Programming: Regular Expressions - Types of regular expressions- Defining BRE patterns - Extended regular expressions SED editor basics - Understandings Linux file Permissions - Linux security - Using Linux groups - Decoding file permissions - Changing security settings - Sharing files

UNIT - III

15 Periods

File I/O :Opening Files - Reading via read() - writing with write() - Synchronized I/O - Direct I/O - Closing files - Seeking with lseek() - Positional read and write -Truncating files - Multiplexed I/O - **Memory Management**: Process address space -Managing the Data segment - Anonymous memory mappings - Advanced memory allocation - Debugging Memory allocation - Stack based memory allocation

UNIT - IV

15 Periods

Process Management: The Process ID-Running a New Process - Terminating a Process - Waiting for Terminated Child Processes - Users and Groups - Sessions and Process groups - **Threading** - Binaries, Processes and Threads - Multithreading - Concurrency, Parallelism and Races - Synchronization - pthreads

TEXT BOOKS

1. Richard Blum and Christine Bresnahan. *Linux Command and Shell Scripting Bible*, John Willey & Sons Inc., Third Edition, Indianapolis, 2015.
2. Robert Love. *Linux System Programming*, O'Reilly Media Inc., Second Edition, 2013.

REFERENCES

1. Richard Petersen. *Linux: The Complete References*, Tata McGraw Hill, Sixth Edition, 2008.
2. Michael K.Jhonson and Erik W.Troan, *Linux Application Development*, Pearson Education, 2005.
3. Robert Love. *Linux Kernal Development*, Pearson Education, Third Edition, 2010.

ONLINE MATERIAL

1. https://nptel.ac.in/courses/106108101/pdf/Lecture_Notes/Mod%2020_LN.pdf

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Write Shell scripts using structure commands • Apply functions in Linux environment
Unit II	• Construct regular expressions • Create group of users and choose secure access sharing of files between the users
Unit III	• Apply file management programs. • Demonstrate memory management
Unit IV	• Develop applications for process management • Apply threads and pthreads.

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Develop Shell script to control hardware, software, process and the whole system
- Apply basic command helps to manage the system activities
- Create group and maintain the group of users
- Identify the role of “Linux system Administrator”
- Apply Process Management program to handle and control the process
- Apply threading program helps to understand parallel, concurrency and synchronization

LIST OF LABORATORY EXERCISES

1. Working with basic Linux Commands, File commands
2. Working with File permission, Searching commands
3. Working with system information commands, file compression and exit commands
4. Write a shell script that accept a file name starting and ending line numbers as arguments and display all the lines between given line no.
5. Write a shell script that delete all lines containing a specified word
6. Write a shell script that accept a list of file names as arguments count and report the occurrence of each word.
7. Write a C program that makes a copy of a file using standard I/O and system calls
8. Write a C program to list every file in a directory, its inode number and file name
9. Write a C program that illustrates two processes communicating using shared memory
10. Write a program to handle threads and multi threads

L	T	P	C
3	1	0	4

Course Code: CSE401

Semester: V

DIGITAL IMAGE PROCESSING

Course Objectives

This course will help the learner to understand the principles and concepts of various image processing and image understanding techniques also expose the students to solve real problems in the field of digital image processing.

UNIT - I

15 Periods

Fundamentals of Image Processing: Introduction - Origin - Application - Fundamental Steps in Digital Image Processing

Digital Image Fundamentals: Elements of visual perception - Image Formation in the Eye, Brightness Adaptation and Discrimination - Image Sensing and Acquisition - Image Sampling and Quantization - Basic Relationship between Pixels - Linear and Nonlinear operations, Spatial Operations, Matrix Operations

UNIT - II

15 Periods

Image Enhancement: Basic Intensity Transformations - Histogram Processing - Fundamentals of Spatial Filtering - Smoothing Spatial Filters - Sharpening Spatial Filters - Highpass, Bandreject, and Bandpass Filters from Lowpass

Filtering in the Frequency Domain - Image Transforms - Fourier and Discrete Fourier transform and its properties - Discrete Cosine Transform - Wavelet transforms in Two Dimension - Frequency Domain Filters - The Basics of Filtering in the Frequency Domain - Image Smoothing Using Lowpass Frequency Domain Filters - Image Sharpening Using Highpass Filters - Selective Filtering

UNIT - III

15 Periods

Image Restoration and Reconstruction: A Model of the Image Degradation/Restoration Process - Noise Models - Restoration in the Presence of Noise Only - Spatial Filtering - Mean Filters, Order Statistics Filters, Adaptive Filters - Periodic Noise Reduction by Frequency Domain Filtering - Linear, Position-Invariant Degradations

Morphological Image Processing: Preliminaries - Erosion and Dilation - Opening and Closing - The Hit-or-Miss Transform - Some Basic Morphological Algorithms - Morphological Reconstruction - Grayscale Morphology.

Color Image Processing: Color Models, Pseudo color Image Processing - Basics of Full-Color Image Processing

UNIT - IV

15 Periods

Image segmentation: Point, line, edge detection - More Advanced Techniques for Edge Detection - Thresholding - Basic global Thresholding - Otsu's Thresholding - Multiple Thresholds - Variable Thresholding - Segmentation by Region Growing and by Region Splitting and Merging - Region Segmentation Using Clustering and Superpixels - The Use of Motion in Segmentation

Feature Extraction: Boundary Preprocessing - Boundary Approximations Using Minimum-Perimeter Polygons - Boundary Feature Descriptors - Region Feature Descriptors - Principal Components as Feature Descriptors

TEXT BOOK

1. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, Pearson Education, Second Edition, 2018.

REFERENCES

1. William K. Pratt, *Digital Image Processing*, John Wiley & Sons, Fourth Edition, 2007.
2. Anil K. Jain, *Fundamentals of Digital Image Processing*, Prentice Hall of India, Fourth Edition, 2010.
3. Gregory A. Baxes, *Digital Image Processing Principles and Application*, John Wiley & Sons, 1994.

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/106105032/>
2. <https://swayam.gov.in/courses/4774-july-2018-digital-image-processing>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Infer image basics, applications and components of digital system • Demonstrate image sampling and quantization, representation • Apply basic pixel based operations on images
Unit II	• Apply image enhancement techniques • Demonstrate image smoothing and sharpening techniques in spatial domain • Demonstrate image smoothing and sharpening techniques in frequency domain
Unit III	• Interpret the image restoration and reconstruction using filters • Apply Morphological algorithms on gray scale images • Demonstrate image processing techniques on color image processing
Unit IV	• Apply Segmentation algorithms on images • Demonstrate the efficiency of thresholding • Describe and representation of various features extraction techniques on images

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Infer image acquisition, sampling and basic operations on image pixel and image
- Relate the mathematical operation with the image processing application on the image
- Apply transformations and represent image in spatial and frequency domain
- Estimate the difference between the spatial and frequency based enhancements
- Employ restoration and reconstruction techniques on binary, Gray and color images
- Use image segmentation methods to identify and represent the regions of interest

L	T	P	C
3	1	0	4

Course Code: CSE310
Semester: V

SYSTEM SOFTWARE

Course Objectives

This course will introduce the different categories of system software and their features to the learner; also emphasize the role and functioning of assembler, linker, loader and macro-processor.

UNIT - I

15 Periods

Introduction to System software and Software tools: Goals - System programs and System programming - Views of System software - Kinds of language processors - Language processing activities - Benefits of Interpretation - Overview of Interpretation - Software tools for program development - Editors - Debug monitors - Programming environment - User interfaces - Introduction to SIC

UNIT - II

15 Periods

Assemblers: Basic assembler functions - A simple SIC Assembler - Assembler algorithm and data structures - Machine dependent assembler features - Instruction formats and addressing modes - Program relocation - Machine independent assembler features - Literals - Symbol-defining statements - Expressions - One pass assemblers and multi pass assemblers - Implementation example - MASM Assembler

UNIT - III

15 Periods

Loaders and Linkers: Basic Loader functions - Design of an Absolute loader - A simple Bootstrap loader - Machine dependent loader features - Relocation - Program linking - Algorithm and data structures for linking loader - Machine-independent loader features - Automatic library search - Loader options - Loader design options - Linkage editors - Dynamic linking - Bootstrap loaders - Implementation example - MS DOS linker

UNIT - IV

15 Periods

Macro Processors: Basic Macro processor functions - Macro definition and expansion - Macro processor algorithm and data structures - Machine-independent macro processor features - Concatenation of macro parameters - Generation of unique labels - Conditional macro expansion - Keyword macro parameters - Macro within macro-implementation example - MASM Macro processor - ANSI C macro language

TEXTBOOKS

1. D. M. Dhamdhere. *Systems Programming*, Tata McGraw Hill Education, First Edition, 2011.
2. Leland L. Beck and Manjula D. *System Software - An Introduction to Systems Programming*, Pearson Education Asia, Third Edition, 2011.

REFERENCE

1. John J. Donovan. *Systems Programming*. Tata McGraw Hill, First Edition, 2017.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• List the kinds of language processors and state their uses & features. • Explain the overview and benefits of interpretation. • Describe about various software tools for program development.
Unit II	• Translate an assembly language program into its equivalent object program • Describe the working of single pass and multi pass assemblers • Explain the features of assembler • Classify whether a feature of the assembler is machine dependent or independent
Unit III	• Elucidate how the object program is represented in memory and the process of linking the related object programs • Explain the features of loader and classify whether a feature is machine dependent or independent
Unit IV	• Apply the macro expansion procedure for a macro call • Differentiate macro features in the source program

COURSE LEARNING OUTCOMES

Upon successful completion of the course, the learner will be able to

- List the language processors and explain their activities
- Describe about various software tools for program development
- Translate an assembly language program into its equivalent object program
- Describe the working of single pass and multi pass assemblers
- Explain how the object program is loaded and represented in memory and the process of linking the related object programs
- Explain the working of a macro processor

L	T	P	C
3	1	0	4

Course Code: CSE311
Semester: V

SYSTEM MODELLING & SIMULATION

Course Objectives

This course will help the learner to know the concept of system modelling and specific about statistical models in simulation with performances

UNIT - I

15 Periods

Introduction: Simulation is the appropriate tool - not appropriate - Advantages and disadvantages of Simulation - Areas of application - Some recent applications of simulation - Systems and system environment - Components of a system - Discrete and continuous systems - Model of a system - Types of Models - Discrete-Event System Simulation - Steps in a Simulation Study. **General Principles:** Concepts in Discrete Event Simulation - List Processing

UNIT - II

15 Periods

Statistical Models in Simulation: Review of terminology and concepts: Discrete random variables-continuous random variables-cumulative distribution function-Expectation -The mode - **Useful statistical models - Discrete distributions:** Bernoulli trials and the Bernoulli distribution-Binomial distribution-Geometric and Negative Binomial distribution-Poisson distribution - **Continuous distributions:** Uniform distribution-Exponential distribution-Gamma distribution-Erlang distribution-Normal distribution- Weibull distribution-Triangular distribution-Lognormal distribution-Beta distribution - **Poisson process :** Properties of a Poisson Process-Non stationary Poisson Process - Empirical distributions

UNIT - III

15 Periods

Random-Number Generation, Random-Variate Generation: Properties of random numbers - Generation of pseudo-random numbers - Techniques for generating random numbers - Tests for Random Numbers, **Random-Variate Generation:** Inverse transform technique - Acceptance Rejection technique - Special properties

UNIT - IV

15 Periods

Input Modeling: Data Collection - Identifying the distribution with data - Parameter estimation - Goodness of Fit Tests - Fitting a Non stationary Poisson Process - Selecting Input Models without Data - Multivariate and Time-Series Input Models. **Verification, Calibration, and Validation of Simulation Models:** Verification, Calibration, and Validation - Verification of Simulation Models - Calibration and Validation of Models

TEXT BOOKS

1. Jerry Banks, John S. Carson II, Barry L. Nelson and David M. Nicol: *Discrete - Event System Simulation*, Pearson Education, Second Edition, 2010.
2. Averill M. Law: *Simulation Modeling and Analysis*, Fourth Edition, Tata McGraw-Hill, 2007.
3. Lawrence M. Leemis, Stephen K. Park: *Discrete - Event Simulation: A First Course*, Pearson Education, 2006.

REFERENCE

1. Altiook T., & Melamed B., *Simulation, Modeling and Analysis with ARENA*, Academic Press, 2007.

ONLINE MATERIAL

1. https://onlinecourses.nptel.ac.in/noc17_me35/preview

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Identify the basic concepts in system environment modeling and simulation • Discuss about the behaviour of a system as it evolves over time by developing a simulation model • Study the general principles governing system simulation
Unit II	• Analyze statistical model and distribution types • Relate the statistical terminology and model with real time applications • Discuss variety simulation models and select the suitable model for any given problem
Unit III	• Study the properties of Random Number Generation • Identify the various random number generation techniques • Develop and test random number variate and apply them to develop simulation models
Unit IV	• Design a model to fit any given input data • Analyze the model performance and design optimum model • Analyze output data produced by a model and test validity of the model

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Understand the principles of modeling and simulation
- Identify system types and simulation models
- Apply different statistical models and distribution types
- Understand the principles of random number generation
- Construct the system model with estimating and tuning parameters
- Analyze the model and validate with optimization parameters

L	T	P	C
3	0	2	4

Course Code: CSE312
Semester: V

SCRIPT PROGRAMMING

Course Objectives

This course will help the learner to develop client side and server side scripts and design rich internet applications using AJAX.

UNIT - I

11 Periods

Introduction to HTML5: First Example - W3C HTML5 validation service-Headings - Linking-Images - Lists - Tables - Forms - Internal Linking - HTML form input Types. **Javascript:** Introduction - Your First Script - Obtaining user Input with prompt Dialogs - Arithmetic - Decision Making - Javascript control statements: Algorithms - Control statements - if, if - else selection statement - while repetition statement - Formulating Algorithms based on counter controlled repetition - Sentinel - Controlled Repetition - Nested Control Statements - Assignment Operators - Increment and Decrement operators - for Repetition statement

UNIT - II

11 Periods

Control Structures: Switch Multiple-Selection Statement - do - while Repetition statement - break and continue Statements - Logical Operators - Functions: Function Definitions - Random Number Generation - Scope Rules - Javascript Global Functions - Recursion-Recursion Vs Iteration. **Arrays:** Declaring and Allocating Arrays - Examples using Arrays-Objects: Math - String - Date - Boolean and Number objects - document Object Model - Modeling a Document - Traversing and Modifying a DOM Tree - DOM Collections - Dynamic Styles and Timer to create Animated Effects

UNIT - III

11 Periods

Ajax: Introduction - Evolution of the Web - The Real Ajax - Ajax Principles - Technologies behind Ajax - Ajax and Web 2.0. **Ajax Basics:** HTTP Primer - Ajax Communication Techniques - Cache Control. **Ajax patterns:** Communication Control Patterns - Fallback Patterns. Ajax Libraries: JQuery. **Request Management:** Priority Queues - The Request Management Object - Using Request Manager. **JSON** - Introduction

UNIT - IV

12 Periods

Introduction to JSP: The Problem with Servlets - The Anatomy of a JSP Page - JSP Processing - JSP Application Design with MVC. **Setting Up and JSP Environment:** Installing the Java Software Development Kit - Tomcat Server & Testing Tomcat JSP Application Development - Generating Dynamic Content - Using Scripting Elements - Implicit JSP Scripting Objects - Using Scriptlets - Using Expressions - Using Declarations - Mixing Action Elements and Scripting Elements. **Processing Input and Output:** Reading Request Parameter Values - Validating User Input - Formatting HTML Output. **Error Handling and Debugging:** Dealing with Syntax Errors - Debugging a JSP Application - Dealing with Runtime Errors. **Accessing a Database:** Accessing a Database from a JSP Page - Validating Complex Input without a Bean - Using Transactions - Application - Specific Database Actions

TEXTBOOKS

1. Paul Deitel, Harvey Deitel and Abbey Deitel. *Internet and World Wide Web How to Program*, Pearson Higher Education, Fifth Edition, 2011.
2. Nicholas C Zakas and Jeremy McPeak. *Professional Ajax*, Wiley India, Second Edition, 2007.
3. Hans Bergsten. *Java Server Pages*, O'Reilly, Third Edition, 2003.

REFERENCES

1. Jeremy McPeak and Paul Wilton. *Beginning JavaScript*. Wrox (Wiley Brand), Fifth Edition, 2015.
2. Tom Negrino and Dori Smith. *JavaScript*, Peachpit Press, Eighth Edition, 2009.
3. Pekowsky. *Java Server Pages*, Addison-Wesley Professional, Pearson, Second Edition 2004.
4. Duane K. Fields, Mark A. Kolb and Shawn Bayern. *Web Development with Java Server Pages*, Manning Publications, Second Edition, 2001.

ONLINE MATERIALS

1. <http://nptel.ac.in/courses/106105084/13>
2. <http://nptel.ac.in/courses/106105084/25>
3. <http://nptel.ac.in/courses/106105084/26>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit the learner will be able to

Unit I	• Create Webpages using HTML5 • Design simple applications using JavaScript • Apply mark up languages for processing, identifying, and presenting of information in web pages
Unit II	• Design and implement dynamic web page with validation using JavaScript objects • Apply different event handling mechanisms
Unit III	• Design and implement rich internet enabled applications using AJAX. • Understand how AJAX requests and use the response to dynamically update the web pages
Unit IV	• Create server side scripts using Java Server Pages • Develop JSP applications by proper error handling, database connectivity

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Design HTML forms and build rich web applications
- Experiment with client side and server side scripting
- Design Interactive WebPages using JavaScript
- Build dynamic website using AJAX
- Design applications for Communication control patterns using AJAX
- Develop a server-side JSP application with Interactive database connectivity

LIST OF LABORATORY EXERCISES

1. Creation of a webpage using tables, frames, forms and lists
2. Working with controls in HTML 5
3. Developing simple applications using Javascript (Using alert box, confirm, prompt, control structures, and arrays)
4. Create an application using DOM
5. Using a timer and dynamic styles to create animated effects
6. Creating a simple AJAX application using XMLHttpRequest object
7. Implementing communication techniques and cache control using AJAX
8. Creating a jQuery application for customer management system using GET and POST request
9. Create a JSP application for Form validation
10. Develop JSP applications for employee management system and establish database Connectivity

L	T	P	C
3	1	0	4

Course Code: CSE313
Semester: VI

CRYPTOGRAPHY & NETWORK SECURITY

Course Objectives

This course will help the learner to question the need for cryptographic and network security algorithms, apply security principles to system design. This course will also help the learner to identify and investigate network security threats.

UNIT - I

15 Periods

Computer and Network Security Concepts: Computer security concepts - OSI security architecture - Security attacks - Security service - Security mechanisms - Fundamental security design principles - Attack surfaces and attack trees - A model for network security - Standards - **Classical Encryption Techniques:** Symmetric cipher model - Substitution techniques - Transposition techniques - Rotor machines - Steganography - **Block Ciphers and Data Encryption Standard (DES):** Traditional block cipher structure - DES - A DES example - Strength of DES - Block cipher design principles

UNIT - II

15 Periods

Introduction to Number Theory: Divisibility and the division algorithm - The Euclidean algorithm - Modular arithmetic - Prime numbers - **Finite Fields:** Groups - Rings - Fields - Finite fields of the form $GF(p)$ - Polynomial arithmetic - Finite fields of the form $GF(2^n)$ - **Advanced Encryption Standard (AES):** Finite field arithmetic - AES structure - AES transformation functions - AES key expansions - An AES example - AES implementation - **Block Cipher Operation:** Multiple encryption and triple DES - Electronic codebook - Cipher block chaining mode - Cipher feedback mode - Output feedback mode - Counter mode - XTS-AES mode for block-oriented storage devices

UNIT - III

15 Periods

Public-key cryptography and RSA: Principles of public-key cryptosystems - The RSA algorithm - **Other public-key cryptosystems:** Diffie-Hellman key exchange - Elgamal cryptographic system - Elliptic curve arithmetic - Elliptic curve cryptography - Pseudorandom number generation based on asymmetric cipher - **System Security:** Malicious software - Intruders - Firewalls - **Cyber law and security:** Cyber crime and Criminal Justice - Concept of cyber crime and the IT Act, 2000 - Hacking - Teenage Web Vandals - Cyber Fraud and cyber cheating - Defamation, Harassment and E-mail abuse - Cyber Pornography - Other IT Act offences - Monetary Penalties, Adjudications and Appeals under IT Act, 2000 - Jurisdiction and cyber crimes - Nature of cyber criminality

UNIT - IV

15 Periods

Cryptographic hash functions: Applications of cryptographic hash functions - Two simple hash functions - Requirements and security - Hash functions based on cipher block chaining - Secure Hash Algorithm (SHA) - SHA-3 - **Message Authentication Codes (MAC):** Message authentication requirements - Message authentication functions - Requirements for MAC - Security of MAC - HMAC - DAA and GCM - Key wrapping - Pseudorandom number generation using hash functions and MAC - **Digital Signatures:** Digital signatures - Elgamal digital signature scheme - Schnorr digital signature scheme - NIST, Elliptic curve and RSA-PSS digital

signature algorithm - **Transport level security:** Web security considerations - Transport layer security - HTTPS - Secure shell - **Electronic mail security:** Internet mail architecture - Email formats - Email Threads and Comprehensive Email security - S/MIME - Pretty Good Service - DNSSEC - DNS based authentication of named entities - Sender policy framework - **IP security:** Overview - Policy - Encapsulating security payload - Combining security associations - Internet key exchange - Cryptographic suites

TEXTBOOKS

1. William Stallings. *Cryptography and Network Security Principles and Practice*, Pearson, Seventh Edition, 2017.
2. Vivek Sood. *Cyber Laws Simplified*, Tata McGraw Hill, 2001.

REFERENCES

1. Behrouz A Forouzan and Debdeep Mukhopadhyay. *Cryptography and Network Security*, McGraw Hill Education, Third Edition, 2018.
2. Bruce Schneier. *Applied Cryptography: Protocols, Algorithms and Source Code in C*, John Wiley & Sons, Second Edition, 2017.
3. Alfred J. Menezes, Paul C. Van Oorschot and Scott A. Vanstone. *Handbook of Applied Cryptography*, CRC Press, 2000.

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/106105031/>
2. <https://nptel.ac.in/courses/106105162/>
3. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-875-cryptography-and-cryptanalysis-spring-2005/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- Recall security concepts, security attacks - Classify and compare cryptographic algorithms - Choose an appropriate one for a given application
Unit II	- Solve modular arithmetic and finite fields - Demonstrate block cipher operations
Unit III	- Explain and classify public key algorithms - Investigate Cyber Laws and Cyber Ethics - Illustrate the need of ethical hacking for a given scenario
Unit IV	- Choose cryptographic hash functions - Select suitable authentication codes - Formulate digital signatures

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Employ number theory concepts and finite fields in cryptography
- Choose an appropriate cryptosystem for ensuring confidentiality, integrity, authenticity and non-repudiation
- Identify appropriate key exchange algorithm for secure key exchange
- Explain cyber security laws and ethical hacking
- Justify enforcing of security in internet communication
- Defend the importance of enforcing security in internet communication

L	T	P	C
3	0	0	3

Course Code: CSE314R01
Semester: VI

SOFTWARE ENGINEERING PRACTICES

Course Objective:

This course will help the learner to design a real world project in a systematic manner and also assess the possible problems involved during System Design

UNIT - I

11 Periods

Software Engineering: The Nature of Software - Software Process - Engineering Practice - A Generic Process Model - Defining a Framework Activity - Identifying a Task Set - Process Patterns - Process Assessment and Improvement. **Process Models:** Prescriptive Process Models - Specialized Process Models - The Unified Process - Personal and Team Process Models. **Agile Model:** Agility and Cost of Change - Agile Process - Extreme Programming - Other Agile Process Models - CMMI.

Case Study*: Agile Manifesto, Pass on the Secret Session, Waterfall model Session.

UNIT - II

11 Periods

Requirement Engineering: Requirement Engineering - Establishing the Groundwork - Eliciting Requirements - Developing Usecases - Building the Analysis Model - Negotiating and Validating Requirements. **Software Project Estimation:** Project Planning Process - Resources - Decomposition Techniques - Empirical Estimation Models - Estimation for OO Projects - **Requirement Modeling:** Scenario-based methods - class-based methods - Creating a Behavioral Model - Identifying Events with the Use Case - State Representations.

Case Study*: Cause Effect Session, Project Inception, Customer Collaboration, Speed Boat Session

UNIT - III

11 Periods

Design Engineering: Design within the Context of Software Engineering - The Design Process - Design Concepts - The Design Model. **Architectural Design:** Software Architecture - Architectural Genres - Architectural Styles - Architectural Considerations - Architectural Decisions - Representing System in Context - Defining Archetypes - Refining the Architecture into Components - Describing the Instantiations of the System.

Component Design: Traditional, Object-oriented and Process related View of Components - Designing Class-based Components - Conducting Component-Level Design - Component-based Development. User **Interface Design:** The Golden Rules - User Interface Analysis and Design - Interface Analysis & Design Steps - Design Evaluation.

UNIT - IV

12 Periods

Testing: Testing Strategies - Strategic Approach - Strategic Issues - Test Strategies for Conventional and Object-oriented Software – Validation Testing – System Testing – Art

of Debugging. **Conventional Testing:** White-Box Testing - Basis Path Testing - Control Structure Testing -Black-Box Testing –Model-based Testing – Testing for Real-Time Systems. **Object-oriented Testing:** Testing OOA and OOD Models - OO Testing Strategies – OOTesting Methods -Testing Methods Applicable at Class Level - Interclass Test Case Design.

***Case studies are meant only for tutorial sessions**

TEXT BOOK

1. Roger S, Pressman and Bruce R, Maxim. *Software Engineering A Practitioner's Approach*, McGraw Hill, Eighth Edition, 2015.

REFERENCES

1. Pankaj Jalote. *An Integrated Approach to Software Engineering*, Narosa Publishing House, Third Edition, 2014.
2. Rajib Mall. *Fundamentals of Software Engineering*, Prentice Hall of India, Third Edition, 2009.

ONLINE MATERIALS

1. <http://nptel.ac.in/courses/106101061/>
2. http://nptel.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Soft%20Engg/New_index1.html

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Build software projects using different process models • Implement agile software process model
Unit II	• Dramatize the activities involved in analysing the requirements • Estimate LOC and FP based efforts in person/months • Design usecase, data, class and behavioural models for the software
Unit III	• Illustrate architectural and component models • Create effective user interfaces
Unit IV	• Test software code based on functional and non-functional aspects of the software • Prepare test plans and strategies for conventional and OO software

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Select an appropriate process model for software product requests
- Implement agile software process models for a given scenario
- Identify the stakeholder's requirements and employ the models for analysing there requirement
- Estimate software cost and efforts in person/months
- Construct Use cases, Activity, Collaboration and state transition diagrams that maps software requirements
- Design architecture for developing a project based on user requirements
- Create test plans and employ testing strategies for verifying functional and non-functional requirements of the project

L	T	P	C
4	0	0	4

Course Code: CSE402
Semester: VI

COMPILER ENGINEERING

Course Objectives

This course will help the learner to explain various phases in translating source language to target language construct scanner and parser, design Adhoc syntax directed translator, and identify the opportunities for optimization.

UNIT - I

15 Periods

Overview: Introduction - Compiler structure - Overview of translation **Scanners:** Introduction - Implementing Scanners - **Using Lex:** Regular expressions - A word count program **Parsers:** Introduction - Review of top down parsing - Bottom up parsing

UNIT - II

15 Periods

Using YACC: A YACC parser-The lexer - Arithmetic expression and ambiguity - Variables and typed tokens **Context Sensitive Analysis:** Introduction - Adhoc syntax directed translation. **Intermediate Representations:** Introduction - Graphical IRs - Linear IRs - Mapping values to names - Symbol tables

UNIT - III

15 Periods

Introduction to optimization: Introduction - Background - Scope of optimization - Local optimization - Regional optimization - Global optimization - Inter procedural optimization. **Instruction Selection:** Introduction - Code generation - Extending the Simple Tree Walk scheme - Instruction selection via Tree Pattern matching - Instruction selection via Peephole Optimization

UNIT - IV

15 Periods

Instruction Scheduling: Instruction Scheduling problems - Local List scheduling - Regional Scheduling: **Register Allocation:** Background issues - Local Register Allocation and Assignment - Global register Allocation and Assignment

TEXTBOOKS

1. Keith D.Cooper and Linda Torczon. *Engineering a Compiler*, Morgan Kauffman Publishers, Second Edition, 2013.
2. Levine, John R., Tony Mason, and Doug Brown. *Lex & yacc*, O'Reilly Media, Inc., Second Edition, 2013.

REFERENCES

1. Dick Grune, Kees Van Reewijk, Henry E.Bal, C. J.H. Jacobs, Koen G. Langendoen, *Modern Compiler Design*, Springer, Second Edition, 2012.
2. Das, Vinu V. *Compiler Design using FLEX and YAC*, Prentice Hall of India Learning Pvt.Ltd, 2007.
3. Alfred V.Aho, Ravi Sethi, Jeffrey D. Ullman, Monica S. Lam. *Compilers: Principles, Techniques and Tools*, Pearson Education, Second Edition, 2006.

ONLINE MATERIALS

1. <http://nptel.ac.in/courses/Webcourse-contents/IIT-KANPUR/compiler-desing/ui/TOC.html>
2. <http://nptel.ac.in/courses/106108052/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Describe the phases of compiler • Design and develop scanners using Lex • Construct LL and LR parsers
Unit II	• Develop parser using Lex & YACC • Describe the significance of ad-hoc syntax directed translation • Construct intermediate representations
Unit III	• Identify the different techniques for code optimization for compiler construction • Demonstrate the techniques used for instruction selection in the backend phase of compiler
Unit IV	• Implement the local list scheduler and explain about regional scheduler • Elucidate the register allocation process in the backend phase of a compiler

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Demonstrate the scanner construction from using Lex
- Develop parser using Lex & YACC
- Apply context sensitive analysis for type Inferencing
- Construct intermediate code representation for a given source code
- Identify appropriate techniques for code optimization
- Explain about the instruction selection, instruction scheduling, and register allocation components in the backend phase of a compiler

L	T	P	C
0	0	2	1

Course Code: CSE315R01

Semester: VI

SOFTWARE ENGINEERING PRACTICES LABORATORY

Course Objectives

The course will help the learner to accustom the steps and utilities of various tools available for software engineering processes and practices.

1. CASE Tools Study: Study of Rational Unified Process
2. Prepare Software Requirement Specification (SRS) Report
3. Design Use case and Activity Diagrams
4. Draw Sequence, Collaboration and Class Diagram
5. Demonstrate Forward and Reverse Engineering for JAVA programs
6. Execute Memory Leak Testing
7. Perform Code Coverage Testing
8. Find Speed and Efficiency of Application Programs
9. Prepare Automated Test Plans
10. Create Manual Test Scripts
11. Build Test Scripts for Object Insertions and Alphanumeric Verification Points
12. Generate Test Scripts to receive inputs from External Data Sources
13. Automate Test Scripts for Data Pools
14. Implement Test Scripts for Website Testing

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to:

- Identify use cases and map them to the corresponding software features.
- Prepare software requirement specification document in Rational Requisite Pro.
- Create use case, activity, sequence, collaboration and class diagrams.
- Employ black box and memory leak testing of the code.
- Execute the portions of application that dominates the execution time.
- Construct test plans to write manual and automatic test cases

L	T	P	C
0	0	2	1

Course Code: CSE403

Semester: VI

COMPILER ENGINEERING LABORATORY

Course Objectives

The learners will be able to design and implement the following phases of compiler like scanning and parsing, ad-hoc syntax directed translation, code generation, and code optimization for any formal language using Lex and YACC tools.

1. Develop a scanner using LEX for recognizing the token types in a given C program
2. Develop a program to find the FIRST and FOLLOW sets for a given Context Free Grammar
3. Extend the outcome of Experiment 2 to implement a LL(1) parser in C or Java to decide whether the input string is valid or not
4. Implement a LR(1) bottom up parser in C or Java to decide whether the input string is valid or not (Context- Free Grammar, Action and GOTO tables are supplied as inputs)
5. Develop a parser for all branching statements of 'C' programming language using LEX & YACC
6. Develop a parser for all looping statements of 'C' programming language using LEX & YACC
7. Develop a parser for complex statements in 'C' programming language with procedure calls and array references using LEX & YACC
8. Identify the structure of if-else statements in 'C' programming language that can be transformed into their equivalent switch-case statements. Design and implement the Adhoc syntax directed translation system for the identified case
9. Use LEX and YACC to create two translators that would translate a given input (compound expression used in experiment 9) into three-address and postfix intermediate codes. The input and output of the translators should be file
10. Write an optimizer pass in C or Java that does common-sub expression elimination on the three address intermediate code generated in the previous exercise
11. Implement Local List Scheduling Algorithm
12. Use Lex & YACC to write a back end that traverses the three address intermediate code and generates x86 code

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Develop Scanners using Lex
- Develop Parsers with and without using Lex and Yacc
- Generate Intermediate codes for High level Language compound expressions.
- Make use of optimization techniques and generate machine level code.
- Implement a source to source translator
- Implement local list scheduling

L	T	P	C
0	0	2	1

Course Code: TNP102
Semester: VI

SOFT SKILLS - II

Course Objectives

This course will help the learner to emphasize, impart and improve their soft skills.

S.NO	TOPICS	NO. CLASSES
1	Employers Expectation and Selection Process	2
2	Experiential sharing and writing of Statement of Purpose	2
3	Resume Writing & Presentation	2
4	Aptitude Test Verbal Logical Numerical Analytical	1
		1
		1
		1
5	Getting ready for Interview	1
6	Technical Quiz (Core Subject)	2
7	Technical Interview (Core & Programming)	2
8	Group discussion on Current Affairs	4
9	Mock GD	3
10	Mock Interview	4
11	Semester Practical	3
12	Semester Theory Exam	1
TOTAL		30

REFERENCES

1. Hariharan, S. Sundararajan, N. and Shanmuga Priya, S.P. *Soft Skills*, MJP Publishers, 2010.
2. Prof. G. Ravindran, Prof. S.P.B. Elango, and Dr. L. Arockiam, *Success through Softskills*. Institute for Communication & Technology, 2007.
3. Dr. K.Alex. *Soft Skills*. S. Chand & Company, 2009.
4. Raghu Palat . *Interview Tips*, Sanman & Co, 2009.
5. Mark Peterson. *Group Discussions*, Lotus Press, 2008.
6. Anand Ganguly and Pustak Mahal. *Group Discussion for Admission & Jobs*, First Edition

7. Dr. R.S.Aggarwal. *A modern approach to verbal and Non verbal reasoning*, S. Chand 2012.

LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- The learner gets to know the expectations of employer from the prospective future employees
- The learner becomes familiar about selection process through any senior student who will share his/her experience about their interview. Learner becomes familiar about how to prepare SOP for higher studies / research
- The learner learns the art of writing a successful resume
- The learner undergoes Verbal, Logical, Numerical and Analytical test
- The learner comes to know the ways to equip himself to face a job interview
- The learner will understand the importance of domain knowledge to face the technical round in job interview
- The learner will understand the importance of domain knowledge to face the technical round in job interview
- The learner realizes the importance of knowledge on current affairs through Group Discussion
- The learner understands the Do's and Don't's of a group discussion
- The learner evaluates his personal preparedness for the future job interview

L	T	P	C
4	0	0	4

Course Code: INT302
Semester: VI

GREEN COMPUTING

Course Objectives

This course will help the learner to apply green computing practices to minimize negative impacts on the environment, design applications by incorporating energy saving practices in the use of hardware, examine technology and tools that can reduce paper waste and carbon footprint by user, and understand how to minimize equipment disposal requirements.

UNIT - I

15 Periods

Introduction: Green IT Fundamentals - Business - IT and the Environment. **Green computing:** carbon foot print, scoop on power. **Green IT Strategies:** Drivers, Dimensions, and Goals - Environmentally Responsible Business: Policies, Practices, and Metrics

UNIT - II

15 Periods

Green Assets and Modeling: Buildings, Data Centers, Networks and Devices - **Green Business Process Management:** Modeling, Optimization, and Collaboration - Green Enterprise Architecture - Environmental Intelligence - Green Supply Chains - **Green Information Systems:** Design and Development Models

UNIT - III

15 Periods

Green Framework: Buying a Green Computer - Choosing Earth-Friendly Peripherals - Recycling Your Computer. **Greener Under the Hood:** Optimization of computer power management - Greening mobile devices. **Telecommuting, teleconferencing and teleporting:** Making the Case for Telecommuting - The Green Home Office - Collaborating and Cloud Computing - Making the Connection - Virtual Presence - Best Ways to Make Your Computer Greener

UNIT - IV

15 Periods

Green Compliance: Socio-cultural aspects of Green IT - Green Enterprise Transformation Roadmap - **Green Compliance:** Protocols, Standards, and Audits - **Emergent Carbon Issues:** Technologies and Future. **Case Studies:** Applying Green IT Strategies and Applications to a Home - Hospital - Packaging Industry - Telecom Sector

TEXTBOOKS

1. Bhuvan Unhelkar. *Green IT Strategies and Applications-Using Environmental Intelligence*, CRC Press, 2011.
2. Woody Leonhard and Katherine Murray. *Green Home computing for dummies*, Wiley Publisher, 2009.

REFERENCES

1. Wu Chun Feng. *Green computing: Large Scale energy efficiency*, CRC Press, 2012
2. Alin Gales, Michael Schaefer and Mike Ebberts. *Green Data Center: steps for the Journey*, IBM Redbook, 2011.
3. Carl speshocky. *Empowering Green Initiatives with IT*, John Wiley & Sons, 2010.

4. John Lamb. *The Greening of IT*, Pearson Education, 2009.
5. Harris, Jason. *Green computing and green it best practices on regulations and industry initiatives, virtualization, power management, materials recycling and telecommuting*, Emereo Pty Ltd, 2008.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Identify the impact of Green Computing in the environment • Describe business policy decisions, practices and metrics
Unit II	• Describe Green assets and business process management • Classify Enterprise Architecture and development models
Unit III	• Describe green IT in relation to technology and relate green IT to sustainable development • Evaluate technology tools that can reduce paper waste and carbon footprint by the stakeholders
Unit IV	• Recognize the socio-cultural aspects of Green IT and design strategies in overcoming problems in environmental applications • Apply green IT strategies in designing and maintaining green framework

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Acquire knowledge to adopt green computing practices to minimize negative impacts on the environment
- Identify the impact of Green Computing in the environment and business policy decisions
- Classify Green assets, Modelling and their Enterprise Architecture
- Interpret green IT in relation to technology and relate green IT to sustainable development
- Apply green IT strategies in designing and maintaining green framework
- Recognize the socio-cultural aspects of Green IT and design strategies in overcoming problems in environmental applications

L	T	P	C
4	0	0	4

Course Code: INT303
Semester: VI

DATA WAREHOUSING & DATA MINING

Course Objectives

This course will help the learner to construct a data warehouse for analytical processing and apply data mining techniques to uncover interesting patterns.

UNIT - I

15 Periods

Data warehousing And OLAP: Basic concepts - Data Warehouse Modeling - Data cube and OLAP - **Data cube technology:** Data cube computation - Preliminary concepts. **Data Mining:** Introduction - Data Objects and Attribute Types - Basic Statistical Descriptions of Data, Measuring data similarity and dissimilarity. **Data Preprocessing:** Overview - Data cleaning - Data Integration - Data Reduction - Data Transformation and Data Discretization

UNIT - II

15 Periods

Mining Frequent Patterns, Associations, and Correlations: Basic Concepts - Frequent Itemset Mining Methods - Pattern evaluation methods, Pattern Mining in Multilevel, Multidimensional Space. **Classification:** Basic concepts - Bayes classification methods - Decision tree induction

UNIT - III

15 Periods

Cluster Analysis: Cluster Analysis - Partitioning Methods - Hierarchical Methods - Density Based Methods - Grid based methods - Evaluation of clustering. **Outlier Analysis:** Introduction - Extreme Value Analysis - Probabilistic Models - Clustering for Outlier Detection - Distance-Based Outlier Detection - Density-Based Methods

UNIT - IV

15 periods

Applications: Mining Text Data: Introduction - Document Preparation and Similarity Computation - specialized Clustering Methods for Text - Topic Modeling. **Mining Time Series Data:** Introduction -Time Series Preparation and Similarity - Time Series Forecasting - Time Series Motifs. **Mining Graph Data:** Introduction - Matching and Distance Computation in Graphs Transformation-Based Distance Computation - Frequent Substructure Mining in Graphs

TEXT BOOKS

1. Charu C. Aggarwal. *Data Mining: The Text Book*, Springer International Publishing, Switzerland, 2015.
2. Jiawei Han, Micheline Kamber, and Jian Pei. *Data Mining: Concepts and Techniques*, Morgan Kaufman, Third Edition, 2012.

REFERENCES

1. Mohammed J. Zaki, and Wagner Meira Jr. *Data Mining and Analysis - Fundamental Concepts and Algorithms*, Cambridge University Press, 2014.
2. Alex Berson, and Stephen J. Smith. *Data Warehousing, Data Mining and OLAP*, TataMcGraw Hill, 2012.

3. Ian H. Witten, Eibe Frank and Mark A. Hall. *Data Mining - Practical Machine Learning Tools and Techniques*, Morgan Kaufman, Third Edition, 2011.
4. Vikram Pudi and P.Radhakrishna. *Data Mining*. Oxford University Press: 2009.
5. Margaret H. Dunham. *Data Mining: Introductory and Advanced Topics*. Pearson Education, Sixth Impression, 2009.

ONLINE MATERIAL

1. <https://nptel.ac.in/courses/106105174/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- Describe the basics of data warehousing, data cube computation - Evaluate preprocessing techniques for sample data
Unit II	- Illustrate frequent patterns from data - Analyze data mining functionality such as classification
Unit III	- Apply any kind of basic clustering techniques for knowledge discovery - Illustrate outlier detection techniques to identify abnormal data
Unit IV	- Choose mining techniques to other kinds of data such as text data, time series data and Graph data - Analyze merits and demerits of mining techniques to other kinds of data

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Build a data warehouse, data cube and generate results and insights for analytical queries
- Evaluate preprocessing techniques on data of a given task.
- Illustrate frequent item sets and patterns from any structured data
- Compare different classification and clustering techniques
- Demonstrate outlier detection techniques to identify anomalies
- Recommend mining techniques to other kinds of data such as text, time series and Graph data

L	T	P	C
3	0	2	4

Course Code: INT309
Semester: VI

DOMAIN-CENTRIC SECURITY

Course Objective

This course will help the learner to evaluate security in web applications, database, Operating system, Cloud & IoT applications.

UNIT - I

12 Periods

Web Application Security Principles: Authentication - Authorization. **Browser Security Principles:** Cross-Site Scripting & Cross-Site Request Forgery - Remote file Upload - Web Vulnerability scanning and Penetration testing.

UNIT - II

12 Periods

Database Security: Introduction - Why Care about Database Security **MySQL:** MySQL Architecture - Discovery - Attack and Defense - Moving Further into the Network - Securing MySQL - Authentication and Authorization - SQL Injection attacks

UNIT - III

12 Periods

Security in IOTs & Cloud - Threats and Attacks: Internet of Things (IoT) as Interconnection of Threats (IoT) - Attack - Defense and Network Robustness of Internet of Things - Sybil Attack Detection in Vehicular Networks - Malware Propagation and Control in Internet of Things - A Solution-based Analysis of Attack Vectors on Smart Home Systems - Cloud Security Baselines - Securing the cloud Infrastructure - Cryptographic Key Management for Data Protection. **Cloud Security Access Control:** Distributed Access Control - Cloud Security Key Management - Cloud User Controls. **Advanced Cloud Computing Security:** Advanced Security Architectures for Cloud Computing - Side-Channel Attacks and Defenses on Cloud Traffic

UNIT - IV

12 Periods

Operating System Security - Multics: Multics History - The Multics System - Multics Security - Multics Vulnerability Analysis. **Security in Ordinary Operating Systems:** System Histories - UNIX Security - Windows Security - Verifiable Security Goals - Information Flow - Information Flow Secrecy Models - Information Flow Integrity Models. **Covert Channels:** Channel Types - Non-interference

TEXT BOOKS

1. Sullivan, Bryan, and Vincent Liu. *Web Application Security, A Beginner's Guide*. McGraw-Hill Education Group, 2011.
2. Litchfield, David, Chris Anley, John Heasman, and Bill Grindlay. *The database hacker's handbook: defending database servers*, John Wiley Sons, 2005.
3. Vacca, John R. *Cloud Computing Security: Foundations and Challenges*, CRC Press, 2016.
4. Giuliano, Romeo, Franco Mazzenga, Alessandro Neri, and Anna Maria Vegni. *Security and Privacy in Internet of Things (IoT) Models, Algorithms, and Implementations*, 2015.
5. Trent Jaeger. *Operating System Security*. Morgan & Claypool Publishers, 2008.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Explain vulnerabilities and attacks against web applications
Unit II	• Identify security threats in databases • Build secure mechanisms to address the threats
Unit III	• Appraise the need of security in IoT & Clouds
Unit IV	• Recognize the importance of ethics and security principles in operating system • Differentiate security in windows and Linux operating systems

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Explain vulnerabilities and attacks against web applications
- Identify security threats in databases
- Build secure mechanisms to address the threats
- Appraise the need of security in IoT & Clouds
- Recognize the importance of ethics and security principles in operating system
- Differentiate security in windows and Linux operating systems

LIST OF LABORATORY EXERCISES

1. Identify CSRF attacks in web applications.
2. Induce a Cross-Site Scripting (XSS) attack on websites.
3. Inject stored and reflected XSS attacks on vulnerable web sites.
4. Scan the web vulnerabilities of web sites using penetration testing tools.
5. Generate Structured Query Language (SQL) injection attack and implement security countermeasures.
6. Demonstrate the use of SSL in IOT ecosystem for secure communication.
7. Encrypt and transfer data to an intended recipient in Cloud.
8. Design an RBAC for a given set of users in a Linux Environment
9. Experiment and detect side-channel attacks in a Cloud.
10. Analyze the security vulnerabilities of Multics Operating System (MOS).

L	T	P	C
3	1	0	4

Course Code: ICT307

Semester: VI

NETWORK ADMINISTRATION & MANAGEMENT TOOLS

Course Objectives

This course will help the learner to familiarise with the basics of networks and Microsoft specialized network tools (DNS, DHCP, Netmon and ADDS).

UNIT - I

15 Periods

Introduction to Networks: Overview of TCP/IP - Core Protocols of TCP/IP - IP Security - TCP/IP filtering - IP Addressing and Routing - TCP/IP Utilities - IP Subnetting. DNS - Introduction to DNS - Architecture - DNS Domain Names - DNS Domain Namespaces - Domain Name Hierarchy - Types of DNS - DNS and Internet Domains - Resource Records - DNS Database - Zone Files and Delegation - Types of Zone File Replication - DNS Query Types - Time to Live - Root Hints - Common DNS Resource Record Types

Hands on Session: Identifying various TCP/IP settings in a system - IP subnetting and IP routing - Install and configure DNS in Windows Server

UNIT - II

15 Periods

Introduction to DHCP: DHCP terms and definitions - Architecture - Client functionality - Automatic IP configuration - Server responsibilities - Scopes - Lease duration - Interaction between client and server - DHCP lease process - DHCP options - RPC- Introduction to RPC - Terms and definitions - Architecture - components - process, supported network protocols - security context multiplexing, Ports and RPC - NetMon Tool - Introduction to Netmon tool - settings and options - capturing network traces - parsing network traces - using filters

Hands on Session: Install and Configure a DHCP Server, Practice various available options in Microsoft DHCP Server - RPC and End point mapper - Install and configure NetMon tool - Using NetMon capture TCP/IP handshake - DNS request and response for a web site - DNS request for an MX record

UNIT - III

15 Periods

Introduction to Microsoft Active Directory Services (ADS): Microsoft Active Directory - LDAP - Roles of ADS - Common terminologies and ADS concepts - Administrative hierarchy - Group policy etc - Active Directory Structure and Storage Technologies - Active Directory Domains and Forests - Logical and physical Structure of Active Directory - Network Ports Used by Domains and Forests

Hands on Session: Install Active Directory Domain Services in Windows - Understanding Active Directory Domain Services Functional Levels - Domain Functional Levels - Forest Functional Levels - Guidelines for raising domain and forest functional levels.

UNIT - IV

15 Periods

Active Directory Administration: Introduction to various AD Snap-ins and their functions - Active Directory Users and Computers - Active Directory Domains and Trusts - Active Directory Sites and Services - ADSI Edit - Schema Manager - Group Policy Management Console

Backup/Restore of Active Directory: Backing Up Active Directory Domain Services - Recovering Active Directory Domain Services

Hands on Session: Create Users - Groups - Computer Accounts and OUs and explore concepts learnt so far explore the physical components of AD - Backup and Restore

TEXT BOOK

Study materials provided by Microsoft

REFERENCES

For Unit 1 and Unit 2

1. [http://technet.microsoft.com/en-us/library/cc737682\(v=ws.10\).aspx](http://technet.microsoft.com/en-us/library/cc737682(v=ws.10).aspx) - The TCP/IP Overview
2. [http://technet.microsoft.com/en-us/library/cc772774\(v=ws.10\).aspx](http://technet.microsoft.com/en-us/library/cc772774(v=ws.10).aspx) - How DNS Works
3. [http://technet.microsoft.com/en-us/library/cc738291\(v=ws.10\).aspx](http://technet.microsoft.com/en-us/library/cc738291(v=ws.10).aspx) - How RPC Works

For Unit 3 and Unit 4

4. [http://technet.microsoft.com/en-us/library/cc759186\(v=WS.10\).aspx](http://technet.microsoft.com/en-us/library/cc759186(v=WS.10).aspx) - Active Directory Structure and Storage Technologies
5. [http://technet.microsoft.com/en-us/library/cc794908\(v=ws.10\).aspx](http://technet.microsoft.com/en-us/library/cc794908(v=ws.10).aspx) - Administering Active Directory Domain Services
6. [http://technet.microsoft.com/en-us/library/cc816751\(v=WS.10\).aspx](http://technet.microsoft.com/en-us/library/cc816751(v=WS.10).aspx) - Recovering Active Directory Domain Services

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Recall IP addressing and routing • Discuss the DNS configuration
Unit II	• Elaborate the concepts of DHCP and RPC • Demonstrate the NetMon tool to trace network functionalities
Unit III	• Examine the importance of ADDS • Distinguish the logical and physical structure of ADDS
Unit IV	• Demonstrate the ADDS installation procedure • Identify the significance of backup and recovery of ADDS

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Recall the concepts of networks, install and apply Microsoft Specialized network topologies
- Describe concepts of IP addressing, subnetting, routing protocols, transport protocols, and DNS
- Discuss the DHCP protocol and its architecture
- Elaborate the RPC Concepts in Microsoft platform
- Examine the network performance using NetMon tool
- Describe the ADDS structure, its importance and applications
- Demonstrate backup and restoring of Active Directory

L	T	P	C
4	0	0	4

Course Code: ICT403
Semester: VI

WIRELESS SENSOR NETWORKS

Course Objectives

This course will help the learner to define the principles of Wireless Sensor Networks (WSN), routing technologies, interference of WSNs with other communication systems, sensor node fusion and event detection, RFID architecture and Internet of Things(IoT) with WSN.

UNIT - I

15 Periods

Principles of Wireless Sensor Networks: Challenges in the design and Implementation of WSNs -IEEE802.15.4 Standard and Wireless Sensor Network - Constructing WSN with IEEE802.15.4 - ZigBee and Wireless Sensor Network - 6LoWPAN and Wireless Sensor Network. **Routing Technologies in WSN:** Routing basics - Classification of Routing Protocols in WSNs - AODV Routing Protocols - Cluster - Tree Routing Protocol - Energy Aware Routing Protocols

UNIT - II

15 Periods

Hardware design for WSNs: General Wireless Sensor Node Architecture - System on Chip and Component based Design - Design Guidelines - Design Case - Power Management - Energy Scavenging. **Embedded Software Design for WSNs:** Embedded Software Design for WSNs - Sensor Driver Development - Implementing a WSN with IEEE 802.15.4 - Bridging WSNs with an External Public Network

UNIT - III

15 Periods

Interference of WSNs with IEEE802.11b Systems: Wireless Coexistence and Interference in WSNs - Performance Metrics - Coexistence Mechanism of IEEE 802.15.4 - Mitigating Interference Between IEEE 802.11b and IEEE 802.15.4 - Advanced Mitigation Strategies - Empirical Study. **Sensor node Fusion and Event Detection:** Sensor Data Fusion Techniques - Event Detection - Generic Sensor State Model - Sensor State Model Based Event Detection - Sensor Network as a Database. **Hybrid RFID/WSNs for Logistics Management:** RFID - Hybrid RFID/Sensor Network - Generic Hybrid RFID/Sensor Network Architecture - Possible Use in Humanitarian Logistics Management

UNIT - IV

15 Periods

Internet of Things with WSNs: Challenges and Features of the IoT - Connecting WSNs with the Internet - IoT Service Oriented Architecture - Possible Implementations in Emergency Response. **Mobile Target Localization and Tracking:** Distance Determination - Localization Methods - Improving Tracking Accuracy - Multiple Mobile Targets Tracking. **Optimization of Sink node positioning:** Challenges of Sink node Positioning - Categories of Sink Node Positioning Approaches-Optimizing location of Static Multiple Sink Nodes - Solving optimal location Problems

TEXT BOOK

1. Shuang Hua-Yang. *Wireless Sensor Networks Principles, Design and Applications*, Springer Verlag, London, 2014.

REFERENCES

1. Anna Forster, *Introduction to Wireless Sensor Networks*, IEEE Press, Published by John Wiley & Sons, 2016.
2. Robert Faludi. *Building Wireless Sensor Networks*, O'Reilly Media, 2011.
3. Ian F.Akyildiz. *Wireless Sensor Networks*, John Wiley & Sons, 2010.
4. Jun Zheng, Abbas Jamalipour. *Wireless sensor Networks: A Networking perspective*, John Wiley & Sons, 2009.
5. Kazem Sohraby, Daniel Minoli and Taieb Znati. *Wireless Sensor Network Technology - Protocols and Applications*, John Wiley & Sons, 2007.

ONLINE MATERIAL

1. https://nptel.ac.in/courses/nptel_download.php?subjectid=106105160
2. http://ocw.mit.edu/courses/engineering-systems-division/esd-290-special-topics-in-supplychain-management-spring-2005/conference-videos/daeyoungkim_rfid.pdf

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Define the principles of WSN with IEEE 802.15.4 standard and ZigBee technology • Identify various types of routing technologies and protocols for WSN
Unit II	• Illustrate the importance of hardware and software design and components available with sensor • Design Embedded software for WSN
Unit III	• Identify the wireless coexistence and interference in WSNs • Interpret the concept of sensor data fusion and event detection and also hybrid RFID/WSNs for logistics management
Unit IV	• Combine the IoT with WSNs • Interpret the concept of mobile target localization and tracking and optimization of sink node positioning

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Identify the best technologies suitable of WSN routing and develop the new protocols
- Design the WSN with IoT for better processing of the collected data
- Develop the event detection process and also perform data fusion in the sink node
- Design the WSN to interoperate with other related wireless technologies like ZigBee and low power personal area networks
- Identify the challenges of sink node positioning for optimization locations problems
- Illustrate mobile target localization and tracking for distance determination

L	T	P	C
3	1	0	4

Course Code: CSE404
Semester: VI

EVOLUTIONARY COMPUTING

Course Objectives

This course will help the learner to understand the various aspects of Evolutionary algorithms, in particular GA, GP, MA, SI, and will concentrate on the basic concepts of representation, operators and overall control, followed by examples of the use of these concepts in important applications.

UNIT - I

15 Periods

Introduction: Definition - Components of Evolutionary Algorithms - An Evolutionary Cycle by Hand - Example Applications - The Operation of an Evolutionary Algorithm - Evolutionary Computing, Global Optimization, and other search algorithms - **Representation, Mutation and Recombination:** Binary, Integer, Real-valued, Permutation and Tree Representations

UNIT - II

15 Periods

Fitness, Selection and Population Management: Parent Selection - Survivor Selection - Selection Pressure - Multimodal Problems, Selection and need for diversity. **Popular Evolutionary algorithms variants:** Genetic algorithms - Evolution Strategies - Evolutionary Programming - Genetic Programming - Learning Classifier system - Differential Evolution. **Parameters and Parameter Tuning:** EA Parameters - EA Instances - Designing EAs - The Tuning Problem - Tuning Methods. **Parameter Control:** Examples of changing parameters - Classification of Control Techniques - Examples of varying Parameters

UNIT - III

15 Periods

Hybridization with other techniques- Memetic Algorithms: Motivation for hybridizing EA - Local search - Structure of a Memetic Algorithm - Adaptive Memetic Algorithms - design issues for Memetic algorithms - Example application. **Non stationary and Noisy function Optimization:** Characterization -The Effect of different sources of Uncertainty - Algorithmic approaches. **Multi Objective Evolutionary Algorithms:** Dominance and Pareto Optimality - EA Approaches - Example Application. **Constraint Handling** - Two Main types - Approaches - Example Application. **Interactive Evolutionary Algorithms:** Characteristics - Algorithmic Approaches - Example Application

UNIT - IV

15 Periods

Swarm Intelligence - Ant Colony Systems - Development of Ant Colony System - Applications of Ant Colony Intelligence - the working of Ant Colony Systems - Particle Swarm Intelligence Systems - The basic PSO Method - Characteristic Features - Procedure of the Global Version - Parameter Setting - Comparison with other EC Techniques

TEXT BOOKS

1. Eiben, A.E. and Smith, J.E. *Introduction to Evolutionary Computing*, Springer, Second edition, 2015.
2. N.P.Padhy. *Artificial Intelligence and Intelligent Systems*, Oxford University Press, 2014.

REFERENCE BOOKS

1. Xinjie Yu and Mitsuo Gen. *Introduction to Evolutionary Algorithms*, Springer, 2010.
2. Kenneth A. De Jong. *Evolutionary Computation - A Unified Approach*, MIT Press, 2006.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Acquire the fundamental concepts of Evolutionary Algorithms • Solve simple optimization problems
Unit II	• Demonstrate the ability to design Evolutionary and Genetic Programming • Apply Parameter Tuning and Control Techniques
Unit III	• Acquire concept on Hybridization of EA • Design and implement Multi Objective and Interactive EAs
Unit IV	• Apply Ants Colony System to Optimization Problems • Illustrate the concept of Particle Swarm Intelligence for optimization problems

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Apply fundamental concepts of EA
- Demonstrate the ability to design Evolutionary and Genetic Programming
- Apply Parameter Tuning and Control Techniques
- Acquire concept on Hybridization of EA
- Design and implement Multi Objective and Interactive EAs
- Apply Ants Colony System and Particle Swarm Intelligence to Optimization Problems

L	T	P	C
3	1	0	4

Course Code: CSE405R01
Semester: V

NATURAL LANGUAGE PROCESSING

Course objective: This course will help the learner to describe the phases of Natural Language Processing (NLP) and to apply the concepts in specific applications.

UNIT - I

15 Periods

Introduction: Elementary Probability Theory - Essential Information Theory - **Linguistic Essentials:** Parts of Speech and Morphology - Phrase structure - Semantics and Pragmatics - **Collocations:** Frequency - Mean and Variance - Hypothesis Testing - Mutual Information. **Statistical Inference:** Statistical Estimators - **Word Sense Disambiguation:** Supervised Disambiguation - Dictionary-Based Disambiguation.

UNIT - II

15 Periods

Lexical Acquisition: Evaluation Measures - Verb Subcategorization - Attachment Ambiguity - Selectional Preferences - Semantic Similarity - **Markov Models:** Hidden Markov Models - The Three Fundamental Questions for HMMs - **Probabilistic Context Free Grammars:** Features of PCFGs - Questions for PCFGs - The Probability of a String

UNIT - III

15 Periods

Neural Networks and Neural Language Models: Units - The XOR problem - Feedforward Neural Networks - **Feedforward networks for NLP:** Classification - Feedforward Neural Language Modeling - Training Neural Nets - **Deep Learning Architectures for Sequence Processing:** Recurrent Neural Networks - RNNs as Language Models - RNNs for other NLP tasks - The LSTM - Contextual Generation and Summarization.

UNIT - IV

15 Periods

Machine Translation and Encoder-Decoder Models: Language Divergences and Typology - The Encoder-Decoder Model - Encoder-Decoder with RNNs - Attention - Beam Search - Encoder-Decoder with Transformers - Some practical details on building MT systems - MT Evaluation. **Information Extraction:** Relation Extraction - Relation Extraction Algorithms - Extracting Times - Extracting Events and their Times - Template Filling.

TEXT BOOKS

1. Manning, C. D., and Schutze, H, *Foundations of Statistical Natural Language Processing*, MIT press, 2000.
2. Dan Jurafsky, and James H. Martin, *Speech and Language Processing*, Stanford University, 2022.

REFERENCES

1. Hobson Lane, Cole Howard, and Hannes Max Hapke, *Natural Language Processing in Action: Understanding, Analyzing and Generating Text with Python*, Manning Publications, 2019.
2. Bird, Steven, Ewan Klein, and Edward Loper, *Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit*, O'Reilly, 2009.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- Recall the fundamental mathematical models in the field of Natural Language Processing - Solve the collocation and word sense disambiguation problem using Natural Language Processing Techniques
Unit II	Construct Neural Language Models for solving classification problem
Unit III	- Demonstrate different Natural Language Processing task using Recurrent Neural Networks - Explain machine learning techniques used in NLP, such as hidden Markov models and probabilistic context-free grammars.
Unit IV	Illustrate the Machine Translation and Information Extraction using machine learning algorithms

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

CO No	COURSE OUTCOME	Knowledge Level
1	Recall the fundamental mathematical models in the field of Natural Language Processing	K1
2	Solve the collocation and word sense disambiguation problem using Natural Language Processing Techniques	K3
3	Construct Neural Language Models for solving classification problem	K3
4	Demonstrate different Natural Language Processing task using Recurrent Neural Networks	K2
5	Explain machine learning techniques used in NLP, such as hidden Markov models and probabilistic context-free grammars.	K2
6	Illustrate the Machine Translation and Information Extraction using machine learning algorithms	K2

LIST OF EXERCISES

1. Perform the following task using NLTK: Tokenize and tag some text, identify named entities, display a parse tree and find the ambiguity of the sentence using parse tree.
2. Perform t-Test and Chi-Square test to check whether a given sequence of words is a collocation or not.
3. Implement the decision rule-based Naïve Bayes disambiguation method to find the sense of an ambiguous word with the given training set.
4. Implement the Hindle and Rooth algorithm for solving the attachment ambiguity problem.
5. Implement the forward and backward procedures using Hidden Markov Model to find the probability of a word sequence.
6. Implement the Viterbi algorithm to find the probability of a word sequence, and infer the best tag sequence using Hidden Markov Model.
7. Implement the Probabilistic Context Free Grammar (PCFG) and find the inside probability of a word sequence using the CYK algorithm.
8. Construct the parse tree for the given sentence using Viterbi PCFG algorithm.
9. Apply the Multi-Layer Perceptron to the task of classifying surnames to their country of origin.

10. Design text generator using LSTMs and generate 30 words based on the input provided by the users with available corpus.
11. Design RNN based Python machine translation system.
12. Perform the relationship extraction/information extraction using Bootstrapping method.

L	T	P	C
3	1	0	4

Course Code: CSE406

Semester: VI

ETHICAL HACKING

Course Objectives

This Course will help the learner to understand ethics behind hacking and exploit the vulnerabilities related to computer system and networks using open source tools and technologies.

UNIT - I

15 Periods

Basics of Security: Understanding the importance of security - Introduction to Ports & Protocols - Trojans & Backdoors - Viruses & Worms - **Introduction to Hacking:** Important Terminologies - Hacking - Types of Hackers - Phases of Hacking - Virtualization and Introduction to Kali Linux - **Footprinting:** Google Hacking and Google Dorking - Scanning - **Information Gathering:** Active, Passive and Sources of information gathering, Copying Websites Locally, NeoTrace, Cheops-ng, Intercepting a Response, Netcraft, Basic Parameters, Xcode Exploit Scanner

UNIT - II

15 Periods

Vulnerability Assessment: Vulnerability - OWASP Top 10 Vulnerabilities - Prevention - Damn Vulnerable Web Application (DVWA): Installation and Testing - OWASP ZAP - **Basics of Exploitation:** Remote Exploitation: Understanding Network Protocols - Server Protocols - Attacking Network Remote Services - Common Target Protocols - THC Hydra - OpenSSH Username Discovery Bug - Cracking SSH with Medusa - Basic Syntax - Cracking an RDP with Ncrack - Combining Nmap and Ncrack for Optimal Results - Attacking SQL Servers - MS SQL Servers - Metasploit - **Client Side Exploitation:** Client Side Exploitation Methods - Introduction - PDF Launch Action - Creating a PDF Document with a Launch Action - Origami Framework - **Post Exploitation:** Acquiring situation awareness - Privilege Escalation - Cracking the Hashes to Gain Access to Other Services - Backdoors - MSFPayload/MSFEncode - MSFVenom - Dumping the Hashes - Windows Exploit Development Basics

UNIT - III

15 Periods

Performing the Hack: Hacking into System - Sniffing - Packet analysis - types of sniffing - active and passive sniffing techniques - Session Hijacking - Types and methods of session hijacking - Wireless Hacking - Web Hacking - **Social Engineering:** Process of Social Engineering - Identity Theft - Human based social engineering techniques - Computer based social engineering - Phishing process - Phishing attacks - Social Engineering Toolkit - Site Cloning and Mass Mailer using social engineering toolkit - prevention of social engineering - **Deep Web and Dark Net:** Surface Web - Deep Web - Dark Net - Bitcoin - The Onion Router - TOR - Using TOR

UNIT- IV

15 Periods

Advance Hacking: Denial of Service - **Web application & Web server penetration testing:** Web application attack phase - **Web application security vulnerabilities:** SQL Injection vulnerabilities - Cross-Site scripting vulnerabilities - Web application testing using tools - WPScan - **SCADA Attacks:** SCADA - SCADA Protocols - SCADA Fuzzing - **Hacking Malware:** Trends in Malware - De-obfuscating Malware - Reverse-Engineering Malware - Stuxnet malware

TEXTBOOKS

1. Harsh Bothra. *Hacking: Be a Hacker with Ethics*, Khanna Publishing, 2017.
2. Rafay Baloch. *Ethical Hacking and Penetration Testing Guide*, CRC Press, 2015.
3. Regalado D., Harris S., Harper A., Eagle C., Ness J., Spasojevic B., Linn R., and Sims S. *Gray Hat Hacking The Ethical Hacker's Handbook*, McGraw-Hill Education Group, 2015.

REFERENCES

1. Weidman, Georgia. *Penetration testing: a hands-on introduction to hacking*, No Starch Press, 2014.
2. Stuttard D., Pinto M. *The web application hacker's handbook: Finding and exploiting security flaws*, John Wiley & Sons, 2011.

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/109106136/39>
2. <https://www.cybrary.it/course/ethical-hacking/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Identify the basic terminologies of security & hacking • Illustrate the ethical hacking process and analyze the information gathering techniques regarding the target's online presence
Unit II	• Identify different vulnerabilities using open source vulnerability assessment tools • Defend offensive situations in case of compromised operating system(s)
Unit III	• Classify various methods to gain unauthorized access and manipulate organizational data • Apply malicious code in social network applications via ethical hacking in both known and anonymous environments
Unit IV	• Identify diverse applications and it's vulnerabilities through penetration testing • Categorize the attacks based on protocols in SCADA systems

COURSE LEARNING OUTCOMES

After successful completion of the course, student will be able to

- Identify the legal and ethical requirements related to hacking
- Infer hands-on experience on vulnerability detection tools and packet sniffers
- Develop the process and hierarchy of ethical hacking
- Identify attacks of the compromised host/network and logging them for incident-handling
- Discover the key issues plaguing the information security world, incident management process, and penetration testing
- Defend against future attacks

L	T	P	C
4	0	0	4

Course Code: CSE408
Semester: VI

GRAPH THEORY & ITS APPLICATIONS

Course Objectives

This course will help the learner to define how graphs serve as models for many standard problems and to discuss the concept of graph, tree, Euler graph, cut set and to represent graphs using matrices.

UNIT - I

15 Periods

Introduction - Graph Terminologies - Types of Graphs - Sub Graph- Multi Graph - Regular Graph - Isomorphism - Isomorphic Graphs - Sub-graph - Euler graph - Hamiltonian Graph - Related Theorems-Travelling Salesman problem

UNIT - II

15 Periods

Trees -Properties- Distance and Centres - Types - Rooted Tree - Labeled Tree - Unlabeled Tree - Spanning Tree - Fundamental Circuits - Finding all spanning trees of a graph-spanning trees in a weighted graph- Cut Sets - Properties - Fundamental Circuit and Cut-set- Connectivity-Separability and its applications

UNIT - III

15 Periods

Network Flows - Planar Graph - Representation - Detection - Dual Graph - Geometric and Combinatorial Dual - Related Theorems - Digraph - Properties - Euler Digraph-paired comparisons and Tournaments

UNIT - IV

15 Periods

Matrix Representation - Adjacency matrix- Incidence matrix- Circuit matrix - Cut-set matrix - Path Matrix - Properties - Related Theorems - Correlations. Graph Coloring - Chromatic Polynomial - Chromatic Partitioning - Matching - Covering-Graph Algorithms- Connectedness and Components - Spanning Tree- Fundamental Circuits- Cut Vertices- Directed Circuits- Shortest Path

TEXTBOOKS

1. NarsinghDeo. *Graph Theory with Application to Engineering and Computer Science*, Prentice-Hall of India Pvt.Ltd, 2004.

REFERENCES

1. Bondy, John Adrian, and Uppaluri Siva Ramachandra Murty. *Graph Theory with Applications*, North Holland Publication, 2008.
2. West, Douglas Brent. *Introduction to Graph Theory*, Pearson Education, 2011.
3. John Clark, and Derek Allan Holton. *A First Look at Graph Theory*, World Scientific Publishing Company, 1991.
4. Diestel, R. *Graph Theory*, Springer, Third Edition, 2006.

ONLINE RESOURCES

1. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-042j-mathematics-for-computer-science-fall-2010/video-lectures/lecture-6-graph-theory-and-coloring/>
2. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-042j-mathematics-for-computer-science-fall-2010/readings/MIT6_042JF10_chap05.pdf
3. <https://nptel.ac.in/courses/111106050/2>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Understand the basic concepts of graph theory • Understand Eulerian graphs and Hamiltonian graphs and apply these concepts to solve travelling salesman problem
Unit II	• List the properties of trees and use these properties to find minimum spanning trees of a graph • Find the difference between connectivity of a graph and cut sets of graph
Unit III	• Characterize planar graph and construct dual of planar graph • Apply directed graph concepts in decision making
Unit IV	• Represent graphs using matrices and design algorithms using graph theoretical concepts for various applications • Apply graph colouring and matching principles in network theory and assignment problems

COURSE LEARNING OUTCOMES

Upon the completion of this Course, Learner will be able to

- Create graph models for real life problems
- Model and solve travelling salesman problem using Hamiltonian graphs
- Identify planarity in graphs and construct its dual
- Represent graphs in machines as different types of matrices
- Apply the concepts of domination and colouring in network models
- Solve assignment problems using matching techniques

L	T	P	C
3	1	0	4

Course Code: INT430

Semester: VI

INFORMATION SECURITY

Course Objectives

This course will help the learner to understand the key features of information systems security, security issues of wired/wireless network, biometrics access control, cryptographic systems, digital signatures, information security model/metrics, and cyber laws

UNIT - I

15 Periods

Information systems in global context: globalization of business and the need for distributed information systems, global information systems: role of Internet and web services, information systems security and threats, threats to information systems: new technologies open door to the threats, information-level and network-level threats, **information systems security:** threats and attacks, classifications of threats and assessing damages, protecting information systems security, information security management in organizations, **building blocks of information security,** information security risk analysis, Security challenges posed by mobile devices

UNIT - II

15 Periods

Security considerations in mobile and wireless computing: organizational measures for handling mobile devices related security issues, laptops: physical security countermeasures, use of RFID in mobile commerce and information asset protection, **security in cloud computing:** protecting information security and data privacy in cloud computing, smartphone security, **security of wireless network:** wired world versus wireless world - putting wireless networks in information security context, security and privacy challenges - understanding security and privacy issues in **IoT**, intelligent buildings: security threats, **smart cities:** privacy and security

UNIT - III

15 Periods

Biometrics for security: Biometrics identification and authentication techniques, biometrics system: architectural design issues, biometric measurement issues, benefits of biometrics over traditional authentication methods, **network security in perspective:** need for security in the networked world, establishing security perimeter for network protection, **cryptography and encryption:** role of cryptography in information security, **digital signature** - a method for information security, cryptographic algorithms, **intrusion detection (ID)** for securing the networks: ID for information systems security, **firewalls for network protection:** why firewalls are needed? examining firewalls in the context of IDS, use firewalls effectively

UNIT - IV

15 Periods

Security models, frameworks, standards and methodologies: ISO 27001, ISMMM, SSE-CMM, IA-CMM, **methodologies:** IAM, IEM, OCTAVE, OSSTMM, SIPES, **security metrics:** security metrics basics, classifications, importance, implementing security metrics program, key success factors in implementing infosec metrics, pitfalls and challenges in organizational security metrics program, **laws, and legal framework for information security:** understanding the laws for information security, the Indian IT Act, laws for IPR, patent law, copyright law, Indian Copyright Act, privacy issues and laws in Hong Kong, Japan and Australia, European outlook on laws for IS, data protection act in Europe, HIPPA, GLBA, building security into software/system development life cycle

TEXTBOOKS

1. Nina Godbole, Information systems security - security management, metrics, Frameworks and Best Practices, Wiley India Pvt. Ltd., 2nd ed., 2017
2. Sood, Cyber Laws Simplified, Mc Graw Hill, 2017

REFERENCES

1. Umesha Nayak and UH Rao, The InfoSec Handbook: An Introduction to Information Security, Apress; 1st ed. edition (10 September 2014)
2. Gaurav Gupta and Sarika Gupta, Information Security and Cyber Laws, Khanna Publishing; First edition (2019)
3. Mark Rhodes-Ousley, Information Security: The Complete Reference, McGraw-Hill Education; 2 edition (16 September 2012)

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

UNIT I	• Comprehend various threats to the information systems. • Become aware of the information security protection mechanism.
UNIT II	• Know mobile and wireless computer security considerations. • Grasp the protection and privacy concerns in cloud comp., and IoT.
UNIT III	• Understand biometrics identification and authentication techniques. • Understand different cryptographic systems, IDS, and firewalls.
UNIT IV	• Become aware of different info. security models and metrics. • Gain knowledge of the global laws for information security.

COURSE LEARNING OUTCOMES

Upon successful completion of the course, the learner will be able to

- Explain the globalization of business and the need for distributed information systems.
- Understand the various information security techniques for protecting distributed information systems.
- Comprehend various security issues of wired/wireless networks, cloud computing, smart-phone, smart-cities, and IoT devices.
- Understand the biometric identification and authentication techniques for preventing impersonation attacks.
- Understand some common cryptographic systems and digital signatures for information security.
- Grasp the information security models/metrics, and different global laws for information systems security.

L	T	P	C
0	0	4	2

MINI PROJECT

Course Code: CSE300

Semester: VI

Course Objectives

Mini Project empowers the learner to

- Develop technical skills such as ability to design and construct a system meeting stated requirements
- Enhance interpersonal skills, including ability to lead and to work cooperatively in a team
- Improve oral and written technical communication skills
- Build solutions by applying software engineering principles

Course Description:

- The learner undertakes mini project course during the 6th semester. They work as a team of consisting of two to three members. This course enhances their group working skills. The learner can choose any problem and find a solution using the knowledge gained up to the 6th semester applying the state-of-the-art techniques.
- The learner carries out formulation of the problem statement, system design specification, Implementation, interface development and testing. The learner keeps in mind, societal and environmental impact of the system under development throughout all these phases.
- Individual performance of the learner is continuously assessed by periodic reviews of the progress of the project work.
- Final assessment is done at the end of the semester. The learner has to prepare documentation of the work in the form of a project report and demonstrates the outcome of the work to a panel of examiners

COURSE LEARNING OUTCOMES

Upon successful completion of Mini project, the learner will be able to

- Identify the functional and non-functional requirements of the system
- Identify and distribute tasks among the team members
- Use technical literature for designing cost effective solution based on acquired knowledge
- Implement the design by using appropriate software tools
- Evaluate the system by using appropriate testing strategies

Write a report explaining the technical concepts of the developed system

L	T	P	C
3	0	0	3

Course Code: INT404R01
Semester: VII

BIG DATA ANALYTICS

Course Objective:

This course will help the learner to discuss the fundamental techniques of data science and data analysis suitable for big data and solve it using Hadoop.

UNIT - I

12 Periods

Getting Started with Big Data: Grasping the Fundamentals of Big Data. **An Operating System for Big Data:** Basic Concepts - Hadoop Architecture - Working with a Distributed File System - Working with Distributed Computation - Submitting a MapReduce Job to YARN. **A Framework for Python and Hadoop Streaming:** Hadoop Streaming - A Framework for MapReduce with Python - Advanced MapReduce

UNIT - II

11 Periods

MapReduce and the New Software Stack: Map Reduce Algorithms - Communication Cost Model. **Mining Data Streams:** Stream Data Model - Sampling Data in a Stream - Filtering Streams - Counting Distinct elements in a Stream - Estimating moments - Counting ones in a Window - Decaying Windows. **Link Analysis:** Pagerank Algorithm - Efficient Computation of Pagerank - Topic-Sensitive PageRank - Link Spam.

UNIT - III

11 Periods

Recommendation Systems: A Model for Recommendation System - Content Based Recommendations - Collaborative Filtering - Dimensionality Reduction. **Mining Social Network Graphs:** Social Networks as Graphs - Clustering of Social Network Graphs - Direct Discovery of Communities - Partitioning of Graphs - Finding overlapping communities - Simrank - Counting Triangles.

UNIT - IV

11 Periods

Structured Data Queries with Hive: The Hive Command-Line Interface (CLI), Hive Query Language, Data Analysis with Hive, HBase. **Data Ingestion:** Relational Data with Sqoop. **Analytics with Higher-Level APIs:** Pig, Spark's Higher-Level APIs. **Machine Learning:** Scalable Machine Learning with Spark

TEXTBOOKS

1. Jure Leskovec, Anand Rajaraman, and Jeffrey D. Ullman, *Mining of Massive Data Sets*, 2019.
2. Benjamin Bengfort, and Jenny Kim, *Data Analytics with Hadoop*, O'Reilly, 2016.
3. Judith Hurwitz, Alan Nugent, Dr. Fern Halper, and Marcia Kaufman, *Big Data for Dummies*, John - Wiley and Sons, 2013.

REFERENCES

1. Jimmy Lin, Chris Dyer, *Data-Intensive Text Processing with MapReduce*, Morgan & Claypool Publishers, 2010.

2. Paul C. Zikopoulos, Chris Eaton, and Dirk deRoos, Thomas Deutsch, George Lapis, *Understanding Big Data: Analytics for Enterprise Class Hadoop and Streaming Data*, The McGraw-Hill Companies, 2012.
3. Srinath Perera, and Thilina Gunarathne, *Hadoop MapReduce Cookbook*, Packt Publishing, 2013.
4. Tim McGovern, *Big Data Now: 2014 Edition*, O'Reilly Media, Inc., 2015.
5. Jason Venner, *Pro Hadoop*, Apress, 2009.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- Discuss the fundamental concepts of big data and Hadoop - MapReduce framework - Demonstrate the MapReduce algorithms using python and Hadoop Streaming
Unit II	- Explain the methods of processing and filtering of data streams - Apply link analysis for page rank computation
Unit III	- Describe the methods for developing a model for recommendation system. - Explain dimensionality reduction and its implementation - Solve community detection problems in social network graphs by applying algorithms for clustering and partitioning
Unit IV	Illustrate the different programming languages in Hadoop such as Hive, HBase, Pig, Spark

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

CO No.	Course Outcomes	Knowledge Level
1	Discuss the fundamental concepts of big data and Hadoop - MapReduce framework	K3
2	Demonstrate the MapReduce algorithms using python and Hadoop Streaming	K4
3	Explain the methods of processing and filtering of data streams and page rank algorithms	K3
4	Describe the methods for developing a model for recommendation system with dimensionality reduction	K3
5	Solve community detection problems in social network graphs by employing algorithms for clustering and partitioning	K3
6	Illustrate the different programming languages in Hadoop such as Hive, HBase, Pig, Spark	K3

L	T	P	C
3	1	0	4

Course Code: CSE407R01

Semester: VII

CLOUD COMPUTING

Course Objective:

This course will help the learner to understand the various models, cloud technologies, the role of virtual machines and containers. Also, the learner will explore the importance of Cloud for analytics, machine learning, mobile computing and IOT with highlights on security challenges, and will acquire knowledge on applications of cloud computing in various fields with case studies.

UNIT - I

15 Periods

Foundation of Cloud Computing: Introduction - Cloud computing characteristics - Advantages and disadvantages - Comparison of traditional and cloud computing paradigms - Future of the cloud. **Cloud Services and Deployment Models:** Types of Cloud - Infrastructure-as-a-Service - Platform-as-a-Service - Software-as-a-Service. **Managing data in the cloud: Storage as a Service:** Storage Models - The Cloud Storage Landscape - Access Methods: Portals and APIs - Amazon Cloud Storage Services - Microsoft Azure Storage Services - Google Cloud Storage Services.

UNIT - II

15 Periods

Computing in the Cloud: Computing as a Service: Virtual Machines and Containers - Advanced Computing Services - Serverless Computing - Pros and Cons of Public Cloud Computing. **Using and Managing Virtual Machines:** Historical Roots - Amazon's Elastic Compute Cloud - Azure VMs - Google Cloud VM Services - Jetstream VM Services. **Using and Managing Containers:** Container Basics - Docker and the Hub - Containers for Science - Creating Your Own Container. **Scaling Deployments:** Paradigms of Parallel Computing in the Cloud - SPMD and HPC-style Parallelism - Many Task Parallelism - MapReduce and Bulk Synchronous Parallelism - Graph Dataflow Execution and Spark - Agents and Microservices - HTCondor.

UNIT - III

15 Periods

The Cloud as Platform: Data Analytics in the Cloud: Hadoop and YARN - Spark - Amazon Elastic MapReduce - Azure HDInsight and Data Lake - Amazon Athena Analytics - Google Cloud Datalab. **Machine Learning in the Cloud:** Spark Machine Learning Library - Azure Machine Learning Workspace - Amazon Machine Learning Platform - Deep Learning: Introduction - Amazon MXNet Virtual Machine Image - Google TensorFlow in the Cloud - Microsoft Cognitive Toolkit. **Wireless Internet and Mobile Cloud Computing:** - Mobile Devices and Internet Edge Networks - Wi-Fi, Bluetooth, and Wireless Sensor Networks - Cloudlet Mesh for Mobile Cloud Computing - Mobile Clouds and Colocation Clouds.

UNIT - IV

15 Periods

IoT Sensing and Interaction with Clouds: Local and Global Positioning Systems - Cloud-Based RAN for Building Mobile Networks - IoT Interaction Frameworks with Clouds and Devices. **Cloud Computing in Social Media Applications:** Social Media Big-Data Industrial Applications - Social Networks and API for Social Media Applications - Social Graph Properties and Representations - Social Graph Analysis on Smart Clouds. **Multicloud Mashup Architecture and Service:** Cloud Mashup Architecture for Agility and Scalability - Multicloud

Mashup Service Architecture - Skyline Discovery of Mashup Services - Dynamic Composition of Mashup Services. **Security and Privacy:** Thinking about Security in the Cloud - Role-based Access Control - Secure Data in the Cloud - Secure Your VMs and Containers - Secure Access to Cloud Software Services. **Applications and Case studies:** Cloud computing in education, healthcare, politics, business, agriculture - Cloud computing adoption in Sub-Saharan Africa - Cloud computing adoption in India.

TEXT BOOKS

1. Hiran KK, Doshi R, Fagbola T, Mahrishi M. *Cloud computing: Master the Concepts, Architecture and Applications with Real-world Examples and Case Studies*, BPB Publications, 2019.
2. Foster I, Gannon DB. *Cloud Computing for Science and Engineering*. MIT Press, 2017.
3. Hwang K. *Cloud Computing for Machine Learning and Cognitive Applications*, MIT Press, 2017.

REFERENCES

1. Anthony T. Velte, Toby J. Velte and Robert Elsenpeter. *Cloud Computing: A Practical Approach*, McGraw Hill, 2010.
2. Dinkar Sitaram and Geetha Manjunat. *Moving To The Cloud: Developing Apps in the New World of Cloud Computing*, Elsevier, 2012.

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/106104182>
2. <https://archive.nptel.ac.in/courses/106/105/106105167/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	Gain an insight into service, deployment models and storage as a service
Unit II	Explore an idea on computing as a service, management of virtual machines, containers and scaling deployment
Unit III	Know how cloud computing can be a platform with regard to Data Analytics, Machine Learning and Mobile networks
Unit IV	Learn the integration of cloud computing with IOT, Social Media and apply the concepts in various case studies

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

CO No.	Course Outcomes	Knowledge Level
1	Distinguish various cloud services and deployment models	K4
2	Explain the various ways of storing and accessing the data from the service provider's perspective	K3
3	Identify the differences between virtual machine and the container and explore various VM services	K4
4	Examine the various approaches of parallelism for scalable deployment	K4
5	Evaluate how cloud computing can be used as a platform for data analytics, machine learning and mobile cloud.	K4
6	Explore the ways of using cloud computing as a secured platform for IOT, social media and multicloud mashup service architecture	K4

L	T	P	C
4	0	0	4

Course Code: CSE409
Semester: VII

PARALLEL & DISTRIBUTED SYSTEMS

Course Objectives

This course will help the learner to design parallel and distributed algorithms and demonstrate them using CUDA.

UNIT - I

15 Periods

Heterogeneous Parallel Computing with CUDA: Parallel Computing - Heterogeneous Computing - **CUDA Programming Model:** Timing Your Kernel - Organizing Parallel Threads - **Global Memory** - CUDA memory model - Memory Management - **Shared Memory and Constant Memory:** Shared Memory Allocation - Banks and Access Mode - Configuring the Amount of Shared Memory - Synchronization - Constant Memory - **Streams and Concurrency:** Introducing Streams and Events - **Tuning Instruction-level primitives:** CUDA Instructions

UNIT - II

15 Periods

Parallel Processing: Introduction - Parallel Processing Terminology - The Sieve of Eratosthenes - **PRAM Algorithms:** Parallel Reduction - Prefix sums - List Ranking - Pre-order Tree Traversal - Merging of two sorted Lists - Graph coloring - **Matrix Multiplication:** Algorithms for processor Arrays - **Sorting:** Enumeration sort - Odd Even transposition sort - Parallel Quick sort - Hyper quick sort

UNIT - III

15 Periods

Introduction: Design goals - **Types of distributed systems:** High performance distributed computing - Distributed information systems - Pervasive systems - **Architecture:** System architecture - **Communication:** Message-oriented communication - Simple transient messaging with sockets - Advanced transient messaging - Message-oriented persistent communication - **Multicast communication:** Application-level tree-based multicasting - Flooding-based multicasting - Gossip-based data dissemination

UNIT - IV

15 Periods

Coordination: Clock Synchronization - Logical clocks - **Mutual Exclusion:** Centralized algorithm - Distributed Algorithm - Token-ring algorithm - Decentralized Algorithm - **Election Algorithms:** Bully algorithm - Ring algorithm - Elections in wireless environment and large scale systems - **Fault Tolerance:** Introduction to fault tolerance - Concepts - Failure models - Failure masking by redundancy - **Reliable client server communication:** Point to point communication - RPC semantics in the presence of failures - **Reliable Group Communication:** Atomic multicast - Distributed commit - Recovery

TEXTBOOKS

1. Maarten van Steen and Andrew S. Tanenbaum. *Distributed Systems*, Prentice Hall of India, Third Edition, 2017.
2. John Cheng, Max Grossman and Ty McKercher. *Professional CUDA C programming*, John Wiley & Sons Inc., 2014.
3. Michael J. Quinn. *Parallel Computing Theory and Practice*, McGraw Hill, Second Edition, 2011.

REFERENCES

1. Jason Sanders and Edward Kandrot. *CUDA by Example: An Introduction to General Purpose GPU Programming*, Addison-Wesley, 2011.
2. Andrew S. Tanenbaum. *Distributed Operating System*, Prentice Hall of India, 2006.
3. Ananth Grama, Anshul Gupta, George Karypis and Vipin Kumar. *Introduction to Parallel Computing*, Pearson Education, Second Edition, 2003.

ONLINE MATERIALS

1. <http://nptel.ac.in/courses/106102114/>
2. <http://nptel.ac.in/courses/106106107/>

LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

Unit I	• Recognize the properties of CUDA enabled device • Design kernel using different types of memory with multiple blocks and multiple threads
Unit II	• Classify the concepts of parallel computing • Distinguish various PRAM algorithms and parallel algorithms for matrix multiplication and sorting
Unit III	• Understand the need and nature of distributed concepts • Evaluate transient and persistence message oriented communication • Understand the multi-cast communication
Unit IV	• Demonstrate clock synchronization with fault tolerance • Formulate group communication • Construct distributed commit and recovery procedures

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Write programs to explore parallel programming, thread synchronization and atomics using CUDA environment
- Apply PRAM algorithms and parallel algorithms for matrix multiplication and sorting
- Appraise the software and hardware characteristics of distributed systems
- Discuss the concept of system architecture and style of distributed systems
- Demonstrate message oriented communication in distributed environment
- Formulate group communication with clock synchronization, fault tolerant mechanisms, distributed commit and recovery procedures

L	T	P	C
3	1	0	4

Course Code: CSE415R01

Semester: VII

INTERNET OF THINGS

Course Objective

This course will help the learner to describe the fundamental of IoT, to create small low-cost embedded systems to illustrate to IoT concepts in real world scenarios.

UNIT - I

15 Periods

Introduction to IoT: Definition and Characteristics of IoT - Physical Design of IoT: Things in IoT - IoT Protocols - Logical Design of IoT: Function blocks - Communication Models - Communication APIs - IoT Enabled Technologies - IoT Levels and Deployment Templates - **Domain Specific IoTs:** Home Automation, Cities, Environment, Energy, Retail, Logistics, Agriculture, Industry, Health and Lifestyle. **Components in IoT:** Control Units - Sensors - Communication modules - Power Sources - Communication Technologies - RFID - Bluetooth - Zigbee - Wifi - Rfinks - Mobile Internet – Wired Communication.

UNIT - II

15 Periods

IoT and M2M: M2M - Difference between IoT and M2M - Software Defined Networks - Network Function virtualization - IoT System Management with NETCONF - YANG - Need for IoT systems management - SNMP - Network Operator requirements - IoT systems management. **Developing IoT:** IoT Design Methodology: Purpose & Requirements Specification - Process Specification - Domain Model Specification - Information Model Specification - Service Specifications - IoT Level Specifications - Functional View Specification - Operational View Specification - Device & Component Integration - Application Development - Case study on IoT for Weather Monitoring

UNIT - III

15 Periods

IoT Physical Devices and Endpoints: Motivation for using Python, Python Packages of Interest for IoT, Building blocks of an IoT Device, Exemplary device: Raspberry Pi, Linux on Raspberry Pi, Raspberry Pi Interfaces - Serial, SPI, I2C, Programming Raspberry Pi with Python - Other IoT Devices

Realization of IoT Ecosystem for Wireless technologies: Architecture of IoT using mobile devices - Mobile Technologies for supporting IoT Ecosystem - Energy Harvesting for Power Conservation in the IoT System - Mobile Application Development Platforms - Low Power Wide Area Technologies. **Infrastructure and service-oriented Discovery protocols for the IoT Eco system:** Layered architecture for IoT - Protocol Architecture of IoT - Infrastructure protocols - Protocols for IoT service Discovery

UNIT - IV

15 Periods

The Enablement Platform for IoT applications and Analytics: The IoT building Blocks - IoT Application Enablement Platforms - Characterizing IoT or M2M Application Platforms - IoT AEPs - Azure IoT Hub - Amazon Web service IoT Platform - the Axeda IoT Platform - IoT data Analytics Platforms - IoT Data Virtualization Platforms - IoT Data Visualization Platform – IoT Edge Data Analytics **The Next generation Clouds for IoT Applications and Analytics:** Motivation for Cloud Enabled Environments - IoT and cloud Inspired Smarter environments - Hybrid Clouds - Emergence of Edge/Fog Clouds - Building Blocks of Software Defined Networking - Software Defined Storage. **Expounding the Edge/Fog Computing paradigm:**

Introduction to Edge/Fog Computing - Fog Computing Concepts - Use Cases of Fog/Edge Computing - Edge Analytics

TEXT BOOKS

1. ArshdeepBahga, Vijay Madiseti, *Internet of Things: A Hands of Approach*, ArshdeepBagha& Vijay Madiseti, 1st Edition, 2014.
2. Pethuru Raj, Anupama C.Raman, *The Internet of Things: Enabling Technologies, Platforms, and use cases*, CRC Press, 1st Edition, 2017.
3. CharalamposDoukas, *Building Internet of Things with the Arduino*, Volume 1, Create Space Independent Publishing Platform, 2012.

REFERENCES

1. HonbaZhou, *The Internet of things in the Cloud: A Middleware Perspective*, CRC Press, 1st Edition, 2012.
2. Adrian McEwen, Hakim Cassimally, *Designing the Internet things*, John Wiley and Sons, 1st Edition, 2014.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will able to

Unit I	- Describe the physical design, logical design, IoT enabled technologies and their components - Illustrate the applications in specific domains
Unit II	- Differentiate IoT and M2M - Describes a generic design methodology for IoT Platforms
Unit III	- Design a cost effective, energy efficient and environment friendly IoT application using embedded boards - Explain the wireless technologies and infrastructure protocols specific to IoT systems
Unit IV	- Identify the various intelligent platforms - Illustrate the role of Edge and Fog computing in IoT

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will able to

- Describe the physical and logical designs of IoT-enabled technologies
- Differentiate IoT and M2M
- Describes a generic design methodology for IoT Platforms
- Design a cost effective, energy efficient and environment friendly IoT application using embedded boards
- Explain the wireless technologies and infrastructure protocols specific to IoT systems
- Illustrate the role of Edge and Fog computing in IoT

L	T	P	C
0	0	2	1

Course Code: CSE411
Semester: VII

PARALLEL & DISTRIBUTED SYSTEMS LABORATORY

Course Objective

The course will help the learner to implement parallel algorithms using CUDA programming and develop applications in a Distributed Computing Environment.

1. Compute Dot product of two vectors using parallel Reduction.
2. Program on Matrix transpose using shared memory.
3. Find 1-D Stencil using Constant Memory.
4. Program to implement Pre order tree traversal using PRAM algorithm.
5. Program to implement Odd even Transposition sort.
6. Program to implement Quick sort algorithm.
7. Design Chat server application with multiple clients.
8. Program to implement Mutual Exclusion in distributed environment.
9. Program to implement Group Communication.
10. Program to implement Clock synchronization.
11. Program to design a Leader Election algorithm.
12. Program to implement fault tolerance mechanism using Byzantine Agreement.

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Design basic parallel programs in CUDA using parallel reduction
- Use various CUDA memories and implement PRAM algorithms such as matrix multiplication and sorting
- Construct chat server application for multiple clients in distributed environment
- Employ MPI programming to implement group communication and clock synchronization concepts with multiple processors
- Develop various leader election algorithms using different processors
- Experiment fault tolerance mechanism using byzantine agreement

L	T	P	C
0	0	2	1

Course Code: CSE431

Semester: VII

INTERNET OF THINGS LABORATORY

Course Objectives

This course will help the learner to understand the networking mechanism of an IoT and to design IoT applications with Arduino (ATmega), ESP 32; Raspberry Pi & ARM based microcontroller boards.

List of Experiments

(Using IoT Boards)

1. GPIO and Associated Peripheral Interfacing using Rpi
2. Camera Module Interface using Rpi
3. Deploying local server using Raspberry pi
4. Digital I/O Interface – Multicolour Led, IR Sensor, PIR, Slot Sensor
5. Analog Read and Write – Potentiometer, Temperature Sensor, Led Brightness Control
6. Implementation of I2C / CAN
7. Configure and connection establishment of Wi-Fi with IoT development board (AP/RE/GW/Packages)
8. MQTT – Publish and Subscribe with IoT development board (Local Host as Broker)
9. MQTT – Publish and Subscribe with IoT development board (Web Broker)
10. Import Sensor data into cloud database (AWS/Azure/Firebase)

Using RTOS

11. Task Context Maintenance
12. Mutual Exclusion

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

CO No.	Course Outcomes	Knowledge Level
1	Experiment by interfacing Peripherals with Raspberry pi	K3
2	Connect two boards/devices using I2C / CAN	K3
3	Establish communication using MQTT protocols	K3
4	Incorporate Database for IoT applications	K3
5	Deploy Local/ web server to deliver the request from the user	K3

L	T	P	C
3	1	0	4

OPTIMIZATION TECHNIQUES FOR COMPUTING

Course Code: MAT328

Semester: VII

Course Objectives

The objective of the course is to make the students to understand and articulate the various technical and mathematical issues involved in the traditional optimization techniques and it also aims at improving the optimization problem formulation and solving skills to the students.

UNIT - I

15 Periods

Linear programming - formulation - Graphical and simplex methods - Big-M method - Two phase method - Dual simplex method - Primal Dual problems

UNIT - II

15 Periods

Unconstrained one dimensional optimization techniques -Necessary and sufficient conditions - Unrestricted search methods - Fibonacci and golden section method -Quadratic Interpolation methods, cubic interpolation and direct root methods

Unconstrained n dimensional optimization techniques - direct search methods -Random search -pattern search and Rosen brock's hill claiming method- Descent methods-Steepest descent, conjugate gradient, Quasi - Newton methods

UNIT - III

15 Periods

Constrained optimization Techniques- Necessary and sufficient conditions -Equality and inequality constraints-Kuhn-Tucker conditions-Gradient projection method-cutting plane method-penalty function method

UNIT - IV

15 Periods

Dynamic programming- principle of optimality- recursive equation approach - application to shortest route, cargo-loading, allocation and production schedule problems

TEXT BOOKS

1. Rao, S.S., *Optimization: Theory and Application*. Wiley Eastern Press, 1978
2. Taha,H.A, *Operations Research- An Introduction*. Prentice Hall of India, 1970

REFERENCES

1. Fox, R.L., *Optimization methods for Engineering Design*, Addition Welsey, 1971
2. KantiSwarup, Gupta, P.K and Man Mohan, *Operation Research*, Sultan Chand, Eleventh edition, 2003
3. Bazaraa, S.M., Sherali, H.D., Shetty, C.M., *Nonlinear Programming: Theory and Algorithms*, Wiley Eastern Press, 2006

ONLINE MATERIALS

1. NPTEL: http://nptel.ac.in/courses/Webcourse-contents/IIScANG/OPTIMIZATION%20METHODS/New_index1.html

UNITWISE LEARNING OUTCOMES

Unit I	• The learner will be able to appreciate and formulate an optimization problem • The learner will be able to solve linear programming problems using various mathematical techniques
Unit II	• The learner will be able to compare and contrast the problem solving procedure involved in the single dimensional and multi dimensional optimization problem. • The learner will be able to articulate and interpret the various problem solving stages involved
Unit III	• The learner will be able to solve problems related constrained optimization techniques • The learner can explain how the problem formulation and solution will become different when a constraint is introduced in the problem
Unit IV	• The learner can understand and explain the mathematical rigor behind the dynamic programming techniques • The learner can apply the idea of dynamic programming for various standard optimization problems

COURSE LEARNING OUTCOMES

Upon the completion of this course learner will be able to

- Formulate Linear Programming problem.
- Solve a Linear Programming problem graphically and algebraically using simplex method.
- Solve unconstrained n - dimensional optimization problems using Unrestricted and interpolation methods
- Solve constrained optimization problems using Lagrangian and Kuhn-Tucker conditions
- Solve Non linear programming problems using cutting plane method and penalty function method.
- Solve the Linear programming problems and Non linear Programming problem using dynamic programming techniques

L	T	P	C
3	1	0	4

Course Code: INT405R01

Semester: VII

MACHINE LEARNING TECHNIQUES

Course Objectives

This course aims at introducing the basics of Machine Learning, its scope and applications. It further helps the learners to understand and analyse simplest algorithms such as linear regression to recent deep learning algorithms

UNIT - I

15 Periods

Machine Learning Basics: Random Variables, Probability Distributions, Marginal Probability, Conditional Probability, The Chain Rule of Conditional Probabilities, Independence and Conditional Independence, Expectation, Variance and Covariance, Common Probability Distributions, Useful Properties of Common Functions. **Learning Algorithms** - Capacity - Overfitting and Underfitting - Hyperparameters and Validation Sets - Estimators, Bias and Variance - Maximum Likelihood Estimation - Bayesian Statistics - Supervised Learning Algorithms - Unsupervised Learning Algorithms - Gradient-Based Optimization - Constrained Optimization - Example: Linear Least Squares, Stochastic Gradient Descent

UNIT - II

15 Periods

Linear Models for Classification: Discriminant Functions - Two classes - Multiple classes - Least squares for classification - Fisher's linear discriminant - Relation to least squares - Fisher's discriminant for multiple classes - The perceptron algorithm - **Probabilistic Generative Models:** Continuous inputs - Maximum likelihood solution - Discrete features - Exponential family - **Probabilistic Discriminative Models:** fixed basis functions - Logistic regression - Iterative reweighted least squares - Multiclass logistic regression - **Sparse Kernel Machines:** Maximum Margin Classifiers - Overlapping class distributions - Relation to logistic regression - Multiclass SVMs - SVM for regression.

UNIT - III

15 Periods

Neural Networks: Feed -forward Network Functions - Network Training - parameter optimization - Use of gradient information - Gradient descent optimization - Error Backpropagation - A simple example. **Regularization for Deep Learning:** Dataset Augmentation - Noise Robustness - Semi - Supervised Learning - Multi -Task Learning - Early Stopping - Parameter Tying and Parameter Sharing - Sparse Representations - Bagging and Other Ensemble Methods - Dropout. **Convolutional Networks:** The Convolution Operation - Motivation - Pooling - Convolution and Pooling as an Infinitely Strong Prior - Variants of the Basic Convolution Function.

UNIT - IV

15 Periods

Sequence Modeling: Markov Models - Hidden Markov Models - Maximum likelihood for the HMM - The forward-backward algorithm - The Viterbi algorithm. **Recurrent and Recursive Nets:** Unfolding Computational Graphs - Recurrent Neural Networks - The Challenge of Long-Term Dependencies - The Long Short-Term Memory and Other Gated RNNs

TEXTBOOKS

1. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, Second Edition, 2010

2. Ian Good Fellow, YoshuaBengio, and Aaron Courville, *Deep Learning*, The MIT Press, First Edition, 2016

REFERENCES

1. Nikhil Buduma, *Fundamentals of Deep Learning*, O'REILLY Media, First Edition, 2017
2. M. Mohri, A. Rostamizadeh, and A. Talwalkar, *Foundations of Machine Learning*, MIT Press, Second Edition, 2018
3. Kevin P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, First Edition, 2022
4. D. Barber, *Bayesian Reasoning and Machine Learning*, Cambridge University Press, First Edition, 2012

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Understand the importance of probability concepts and distributions • Describe the nature of different categories of machine learning techniques
Unit II	• Apply and analyse any generative and discriminative learning algorithms • Demonstrate the effectiveness of SVM for classification and regression
Unit III	• Implement simple neural network for classification problems • Apply convolutional neural network for any ML task with images as input
Unit IV	• Demonstrate the use of a traditional sequential data modelling technique • Compare HMM with Recurrent Neural network

COURSE LEARNING OUTCOMES

Upon successful completion of the course, the learner will be able to

CO No.	Course Outcomes	Knowledge Level
1	Identify the fundamental issues and challenges of machine learning with respect to data, model selection, model complexity, etc.	K2
2	Demonstrate the strengths and weaknesses of many popular machine learning approaches.	K3
3	Influence the importance of mathematical relationships within and across Machine Learning algorithms and the paradigms of machine learning and deep learning.	K3
4	Develop machine learning solutions to classification and regression problems and apply various regularization and optimization techniques	K4
5	Design and implement various machine learning and deep learning algorithms for real-world applications	K4
6	Evaluate and interpret the results of the machine learning and basic deep learning algorithms	K4

L	T	P	C
4	0	0	4

Course Code: INT406
Semester: VII

HIGH PERFORMANCE COMPUTING

Course Objectives

This course will help the learner to realize the need of supercomputing in science and engineering domains. This course also helps to learn various parallel algorithms, debugging strategies, MPI primitives, check points, and realize the occurrences of deadlocks due to bad parallel programming.

UNIT- I

15 Periods

Introduction: High Performance Computing Disciplines - Impact of Supercomputing on Science - Society and Security - Anatomy of a Supercomputer - Computer Performance

HPC Architecture: Systems and technologies - Key Properties of HPC Architecture - Parallel Architecture Families-Flynn's Taxonomy - Enabling Technology - Vector and Pipelining - Single-Instruction - Multiple Data Array - Multiprocessors - Introduction to graphics processing units

UNIT- II

15 Periods

The Essential OpenMP: Introduction - Overview of OpenMP Programming Model - Parallel Threads and Loops - Synchronization - Reduction

The Essential MPI: Introduction - Message-Passing Interface Standards - Message-Passing Interface Basics - MPI_Init - MPI_Finalize - Message-Passing Interface Example Hello world - Communicators-Size-Rank-Example - Point-to-Point Messages - MPI Send-Message - Passing Interface Data Types - MPI Recv - Example - Synchronization Collectives - Overview of Collective Calls - Barrier Synchronization - Example - Communication Collectives - Collective Data Movement - Broadcast-Scatter - Gather - Allgather - Reduction Operations - All to all-Nonblocking Point-to-Point Communication - User-Defined Data Types

UNIT - III

15 Periods

Parallel Algorithms: Introduction - Fork-Join-Divide and Conquer - Manager-Worker Embarrassingly Parallel - Halo Exchange - The Advection Equation Using Finite Difference - Sparse Matrix Vector Multiplication - Permutation - Cannon's Algorithm

Task Dataflow: Breadth First Search - Finding the maximum - Parallel Quick sort algorithm - Parallel Random Number generation - Parallel Conjugate Gradient method - Floyd's algorithm

UNIT - IV

15 Periods

Libraries: Linear Algebra - Basic Linear Algebra Subprograms - Linear Algebra Package - Scalable Linear Algebra Package

Benchmarking: Introduction - Key Properties of an HPC Benchmark

Debugging -Tools: The GNU Debugger - Break Points - Watch Points and Catch Points-Back Trace - Setting a Variable

Debugging OpenMP example: accessing an unprotected Shared variable

Debugging MPI example: Deadlock - compiler flags for debugging

Check Pointing: Introduction - System-Level Check pointing - Application-Level Check points

TEXTBOOKS

1. Sterling, T., Anderson, M., & Brodowicz, M. *High Performance Computing: Modern Systems and Practices*, Morgan Kaufmann, 2017.
2. Quinn, M. J. *Parallel Programming in C with MPI and OpenMP*, Mc-Graw Hill, 2003.

REFERENCES

1. Cook, R. *An Introduction to Parallel Programming with OpenMP, PThreads and MPI*, Amazon Digital Service, 2011.
2. Bertsekas, D. P., & Tsitsiklis, J. N. *Parallel and Distributed Computation: Numerical methods*, Prentice Hall, 2003.
3. Snir, M., Otto, S., Huss-Lederman, S., Dongarra, J., & Walker, D. *MPI--the Complete Reference: The MPI core*. Vol. 1, MIT press, 1998.
4. Quinn, M. J., & Quinn, M. J. *Parallel computing: theory and practice*. Tata McGraw- Hill, 1994
5. JaJa, J. *An introduction to parallel algorithms* Vol. 17. Reading: Addison-Wesley, 1992.

ONLINE MATERIALS

1. <http://web.stanford.edu/class/cme194/cgi-bin/>
2. <https://nptel.ac.in/courses/106108055/40>
3. <https://www.openmp.org/resources/>
4. <https://www.openmp.org/wp-content/uploads/omp-hands-on-SC08.pdf>
5. <http://web.mit.edu/dimitrib/www/pdc.html>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit-I	● Explain the need of Supercomputing in Sciences and Engineering domains ● Classify various High Performance Computing Architectures ● Choose appropriate architecture for solving real world problems
Unit-II	● Choose OpenMP and MPI primitives for implementing parallel programs ● Find the Deadlock occurrences in parallel programming ● Interpret the challenges in implementation of parallel programming
Unit-III	● Develop parallel Divide and Conquer mechanisms ● Make use of parallel algorithms for Graph explorations ● Develop parallel algorithms for Dense and Sparse Linear Algebra operations
Unit-IV	● Apply various linear algebra packages ● Outline the Benchmarking for High Performance Computing ● Construct debugging strategies for parallel programs

COURSE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

- Select appropriate High Performance Computing Architectures
- Design parallel algorithms for real world problems
- Implement MPI and OpenMP programs
- Choose appropriate debugging strategies in parallel programming
- Create check points at system and application level
- Solve linear algebra problems using BLAS packages

L	T	P	C
3	0	2	4

Course Code: INT409
Semester: VII

EMBEDDED SECURITY

Course Objectives

This course will help the learner to implement of security solutions on reconfigurable hardware and embedded processors, understand the impact of possible attacks on hardware systems and relevant countermeasures, and apply machine learning principles for analyzing information security strategies.

UNIT - I

12 Periods

Realization of Security Algorithms on Embedded Processor: Review of processor architecture - Embedded C Programming fundamentals.

Cryptography Implementation on ARM Platform: Random Number Generators - Symmetric Block Ciphers - DES - AES - Stream Ciphers - RC5 - Public Key Ciphers - Case studies.

UNIT - II

12 Periods

Realization of Security Algorithms on Reconfigurable Hardware: Review of FPGAs - Fine grained and course grained architectures - Verilog HDL Programming.

Cryptography implementation on FPGA: Symmetric Block Ciphers - DES - AES - Stream Ciphers - RC5 - Public Key Ciphers - Ring Oscillators - Case studies.

UNIT- III

11 Periods

Hardware Attacks and Countermeasures: Classification - Invasive Vs Non-Invasive - Active Vs passive - Origin of power leakage - Measurement setup - Simple power Analysis - Differential Power Analysis - Differential attack against DES - Countermeasures - Attacks and countermeasures for embedded microcontrollers and FPGAs - Case studies.

UNIT - IV

10 Periods

Machine Learning for Information Security: Basic concepts of machine learning - Supervised and Unsupervised learning - Representations, models and algorithms - Ensemble methods - Sequence learning - Malicious executables detection through learning - Data collection - Classification methodology - Instance based learning - TFIDF classifier - Naïve Bayes - Support Vector Machines.

REFERENCES

1. D.Mukhopadhyay, R.S.ChakrabortY. *Hardware Security - Design, Threats and safeguards*, CRC Press, 2015.
2. Mohammad Tehranipoor and Cliff Wang. *Introduction to Hardware Security and Trust*, Springer, 2012.
3. Benoit Badrignans et al. *True Random Number Generators in FPGA*, Security Trends for FPGAs, Chapter 5, Springer, 2011.
4. Francis-Xavier Standaert. *Introduction to Side-Channel Attacks*, Chapter 2, Springer, LLC 2010.
5. Marcus A. Maloof. *Machine learning and data mining for computer security*, Springer-Verlog London Limited, 2006.

6. Andrew N. Sloss, Dominic Symes, Chris Wright. *ARM System Developer's Guide: Designing and Optimizing System Software*, Elsevier Inc., 2004.
7. Clive "Max" Maxfield. *Design Warriors Guide to FPGA*, Elsevier, 2004.
8. Michael D.Ciletti. *Advanced Digital Design with the Verilog HDL*, PHI, 2003.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Use Embedded C for optimized realization of data security algorithms on ARM processor
Unit II	• Use Verilog for implementing optimized data security algorithm • Demonstrate data security algorithms on FPGA and analyze the hardware utilization, power and timing requirements
Unit III	• Classify hardware attacks • Describe and analyze the effects of attacks on hardware devices • Formulate design strategies to counterattack the attacks on embedded microcontrollers and FPGA
Unit IV	• Describe supervised and unsupervised learning algorithms • Comprehend the importance of machine learning algorithms and data analysis for designing efficient information security architecture

COURSE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

- Realize security algorithms on embedded microcontrollers and FPGA platforms
- Analyze the effects of various attacks on hardware and to formulate countermeasures

LIST OF EXPERIMENTS

1. Implementation of Simple ciphers like Caesar, Hill and Transposition on ARM Platform
2. Implementation of Simple ciphers like Caesar, Hill and Transposition on FPGA Platform
3. Implementation of PRNG on ARM and FPGA Platforms
4. Implementation of Block Cipher (AES / DES) on ARM Platform
5. Implementation of Block Cipher (AES / DES) on FPGA Platform
6. Implementation of RC5 on ARM Platform
7. Implementation of RC5 on FPGA Platform
8. Hardware and security analyses on ARM platform
9. Hardware and security analyses on FPGA platform
10. Hamming distance and Hamming weight studies
11. Machine learning algorithm on ARM platform
12. Machine learning algorithm on FPGA platform

L	T	P	C
3	1	0	4

Course Code: ICT401
Semester: VII

MOBILE COMPUTING

Course Objectives

This course will help the learner to comprehend mobile computing paradigm, telecommunication systems with strong emphasize on issues and solutions to medium access, network, transport and application layer. It also provides an exposure to mobile application development environments and tools.

UNIT - I

15 Periods

Introduction: Mobile computing Vs. Wireless networking - Applications - Characteristics - Structure of mobile computing application

Communication Systems: Global System for Mobile communications (GSM) - General Packet Radio Service (GPRS) - Universal Mobile Telecommunications System (UMTS) - Software Defined Radio (SDR) - DECT

MAC Protocols: Properties - Issues - Taxonomy - Fixed assignment schemes - Random assignment schemes - Reservation based scheme - 802.11 MAC standards - MAC protocols for ad hoc networks

UNIT - II

18 Periods

Wireless LANs: IEEE 802.11 - System architecture - Protocol architecture - Physical layer - MAC layer - MAC management. Bluetooth - Scenarios - Architecture - Radio layer - Baseband layer - LMP - L2CAP - IEEE 802.15

Mobile Network Layer: Mobile IP - Goals, Entities, IP packet delivery, Agent discovery, Registration, Tunnelling and Encapsulation, Optimizations, Reverse tunnelling, Dynamic Host Configuration Protocol (DHCP)

Mobile ad hoc networks: Routing - Destination sequenced distance vector routing - Dynamic source routing - Alternative metrics - Overview of ad hoc routing protocols

UNIT - III

15 Periods

Mobile Transport Layer: Traditional TCP, Indirect TCP, Snooping TCP, Mobile TCP, Fast retransmit/fast recovery, Transmission /time-out freezing, Selective retransmission, Transaction oriented TCP

Wireless Application Protocol: Architecture - Wireless datagram protocol - Wireless transaction protocol - Wireless session protocol - Wireless application environment

UNIT - IV

12 Periods

Operating systems for mobile computing: Basic concepts - Constraints and requirements - Windows mobile - Palm OS - Symbian OS - iOS - Android - Blackberry OS - Comparison.

Application development protocols: HDML - J2ME - Android application development

TEXT BOOKS

1. Prasant Kumar Pattnaik Rajibmall. *Fundamentals of Mobile Computing*, PHI, Second Edition, 2016.
2. Jochen Schiller. *Mobile Communications*, Pearson Education, Second Edition, 2008.

REFERENCES

1. Asoke. K Talukder, Roopa R. Yavagal, Asoke K. Talukder. *Mobile Computing Technology, Applications and Service Creation*, Tata McGraw Hill, 2008.
2. Raj Kamal. *Mobile Computing*, Oxford university press, Second Edition, 2007.

ONLINE MATERIAL

1. <http://nptel.ac.in/courses/106104124/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Describe the mobile computing architecture and existing communication technologies • Appraise and choose an appropriate MAC protocol for an application
Unit II	• Comprehend the typical wireless networking infrastructure • Examine the issues and develop appropriate solutions in network layer • Distinguish the concepts of ad hoc networks from infrastructure based networks
Unit III	• Apply fundamental technologies to improve the performance of TCP • Utilize protocols for wireless application and session management
Unit IV	• Synthesize mobile applications using various operating systems and tools

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Describe the typical mobile networking infrastructure and communication systems
- Manage shared medium access using wireless MAC schemes
- Interface mobile computing systems to hardware and networks
- Design effective mobile interfaces using wireless technologies and principles
- Assess and utilize the right operating system for an application
- Demonstrate various layouts of OS

L	T	P	C
3	1	0	4

Course Code: CSE413R01

Semester: VII

COMPUTER VISION & PATTERN RECOGNITION
(Prerequisite any image processing course)

Course Objective

This course will help the learners to acquire knowledge on the 2D and 3D feature representation and to represent structure from motion, understand the 3D reconstruction and recognition techniques and adapt the various classifiers to pattern recognition problems

UNIT - I

15 periods

2D and 3D Features: 2D and 3D feature-based alignment - Pose estimation - Geometric intrinsic calibration. Structure from motion: Triangulation - Two-frame structure from motion - Factorization - Bundle adjustment - Constrained structure and motion. **Case study*** on POSIT: Pose Estimation and Structure from motion using OpenCV.

UNIT - II

15 periods

3D Reconstruction and Recognition: Shape from X - Active range finding - Surface representations - Point-based representations - Volumetric representations - Model-based reconstruction - Recovering texture maps and albedos. Recognition: Object detection - Face recognition - Instance recognition - Category recognition - Context and scene understanding - Recognition databases and test sets. **Case study*** on Face detection using OpenCV.

UNIT - III

15 periods

Introduction to Pattern Recognition: Machine Perception - Pattern Recognition Systems - Design Cycle - Learning and Adaption - Maximum Likelihood Estimation - Bayesian Estimation - Bayesian Parameter Estimation: Gaussian Case - Problems of Dimensionality - Hidden Markov Models - Non-Parametric Techniques: Density Estimation - Parzen Windows - K Nearest Neighbour Estimation - Nearest Neighbour Rule - Metrics and Nearest Neighbour Classification - Relaxation Methods - Fisher Linear Discriminant - Multiple Discriminant Analysis - **Case Study*** on Naïve Bayes Classifier using OpenCV.

UNIT - IV

15 periods

Pattern Classifiers: Stochastic Search - Boltzmann Learning - Decision Trees - CART - Grammatical Methods - Unsupervised Learning and Clustering: Mixture Densities and Identifiability - Maximum-Likelihood Estimates - Unsupervised Bayesian Learning - Data Description and Clustering - Criterion Functions for Clustering - Hierarchical Clustering - Graph Theoretic Methods - Component Analysis - Low Dimensional Representations and Multidimensional Scaling. **Case Study*** on Binary Decision Trees using OpenCV.

***Case studies are meant for tutorial session only and not for end-semester examinations**

TEXTBOOKS

1. Richard Szeliski, *Computer Vision: Algorithms and Applications*, Second Edition, Springer, 2022.
2. Richard O. Duda, Peter E. Hart, David G. Stork, *Pattern Classification*, Second Edition, Wiley Interscience, 2001.

- Gary Bradski, Adrian Kaehler, *Learning OpenCV*, First Edition, O'Reilly Publisher, 2008.

REFERENCES

- Richard Hartley, Andrew Zisserman, *Multiple View Geometry in Computer Vision*, Second Edition, Cambridge University Press, March 2004.
- K. Fukunaga, *Introduction to Statistical Pattern Recognition*, 2nd Edition, Academic Press, Morgan Kaufmann, 1990.
- D. A. Forsyth, J. Ponce, *Computer Vision: A Modern Approach*, Pearson Education, 2003
- M. Narasimha Murty, V. Susheela Devi, *Pattern Recognition: An Algorithmic Approach*, Springer, 2011
- Robert J. Schalkoff, *Pattern Recognition: Statistical, Structural and Neural Approaches*, John Wiley & Sons, 2001.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- Identify the 2D and 3D features of an image - Deduct geometric structure from motion pictures
Unit II	- Explain 3D reconstruction techniques for computer vision based applications - Illustrate 3D recognition techniques for computer vision system
Unit III	- Explain the pattern recognition system and its design cycle - Elaborate parametric and non-parametric pattern recognition techniques
Unit IV	Classify objects using stochastic, nonmetric and unsupervised classifiers

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

CO No.	Course Outcome	Knowledge Level
1	Identify 2D and 3D features and extract geometric structure from motion pictures	K2
2	Explain 3D reconstruction techniques for computer vision based applications	K3
3	Illustrate 3D recognition techniques for computer vision system	K3
4	Explain the pattern recognition system and its design cycle	K2
5	Elaborate parametric and non-parametric pattern recognition techniques	K2
6	Classify objects using stochastic, nonmetric and unsupervised classifiers	K3

L	T	P	C
3	1	0	4

Course Code: CSE414

Semester: VII

SOFTWARE TESTING & QUALITY ASSURANCE

Course Objectives

The main objective of the course is to impart students the knowledge and learning about software testing and software quality. This course will help the learner to discuss the types of testing techniques, strategies for generating test cases and examine the characteristics of tools used for test automation

UNIT - I

15 Periods

Testing Basics: Principles of testing - Testing - Verification - Validation. **Test Case Design Strategies:** White box testing - Static testing - Static analysis tools - Structural testing - Unit/code functional testing - Code coverage testing - Code complexity testing - Black box testing - Boundary value analysis - Decision tables - Equivalence partitioning - Graph based testing - Domain testing

UNIT - II

15 Periods

Phases of Testing: Integration testing - Integration testing as a type of testing - Scenario testing - System testing - Functional versus Non-functional testing - Functional system testing - Non-functional testing - Scalability testing - Stress testing - Acceptance testing. **Levels of Testing:** Regression testing - Types of regression testing - Smoke or Sanity test - Understanding the criteria for selecting test cases - Classifying test cases - Methodology for selecting test cases - Best practices in regression testing - Agile and extreme testing

UNIT - III

15 Periods

Test Management: Test planning - Test management - Test process - Test reporting. **Test Automation:** What is test automation - Terms used in automation - Skills needed for automation - Scope of automation - Design and architecture for automation - Process model for automation - Selecting a test tool - Automation for extreme programming model - Challenges in automation

UNIT - IV

15 Periods

Software Quality: The software quality challenge - Software quality assurance - Software quality factors - Product operation - Product revision - Product transition - alternative models of software quality factors - Software compliance with quality factors - The SQA system - an SQA architecture. **Quality Management Standards:** The scope of quality management standards - ISO 9001 and ISO 9000-3 - Capability Maturity Models - CMM and CMMI assessment methodology. (Exposure to tools like JMeter, and Selenium*)

TEXT BOOKS

1. Daniel Galin. *Software Quality Assurance: From Theory to Implementation*, Pearson Education, Second Edition, 2017.
2. Srinivasan Desikan, and Gopalaswamy Ramesh. *Software Testing: Principles and Practices*, Pearson Education, 2014.

REFERENCES

1. Naresh Chauhan. *Software Testing: Principles and Practices*, Oxford University Press, 2017.

2. Aditya P. Mathur. *Foundations of Software Testing*, Pearson Education, 2013.
3. Kshirasagar Naik and Priyadarshi Tripathy. *Software Testing And Quality Assurance: Theory and Practice*, Wiley publications, 2008.
4. Jeff Tian. *Software Quality Engineering: Testing, Quality Assurance, and Quantifiable*, Wiley, 2005.

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/106105150/>
2. <https://ocw.mit.edu/ans7870/6/6.005/s16/classes/03-testing/>
3. <https://ocw.vu.edu.pk/CourseDetails.aspx?cat=Computer+Science%2FInformation+Technology+&course=CS706>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- Identify the objectives and principles of testing for the verification and validation of any software product - Design and apply test case for black box and white box testing for any application
Unit II	- Discuss the types of software testing techniques and choose a suitable technique for the given application software testing - Classify the test cases and select the appropriate one for the given application
Unit III	- Create test strategies and plans, prioritize and execute them - Compare and contrast available test automation tools and their applications
Unit IV	- Examine strategies central to the assurance of quality and various standards within different software development environments. - Evaluate the standards, models and techniques for quality assurance in a software development environment

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Design test cases suitable for a software development for different domains
- Identify suitable testing methodologies to be carried out for any application
- Make Use of automatic testing tools to develop bug free product
- Obtain the knowledge about various quality management standards
- Identify the needs of software test automation, and define and develop a test tool to support test automation
- Select the suitable software testing methods and modern software testing tools for their testing projects

L	T	P	C
4	0	0	4

Course Code: CSE410
Semester: VII

EMBEDDED SYSTEMS

Course Objectives:

This course will help the learner to understand the essential concepts like embedded computing, program design and analysis, Real Time Operating Systems for designing the efficient Embedded System

UNIT - I

15 Periods

Embedded Computing: Introduction - Complex systems and microprocessors - The embedded system design process - CPUs - Programming input and output - Supervisor mode, exceptions, and traps - Coprocessors - Memory system mechanisms - CPU performance - CPU power consumption

UNIT - II

15 Periods

Program Design and Analysis: Components for embedded programs – Models of programs - assembly, linking, and loading - Compilation techniques - Program optimization - Program-level performance analysis - Software performance optimization - Program-level energy and power analysis and optimization - Analysis and optimization of program size - Program validation and testing. **Multiprocessors:** CPU'S and accelerators - multiprocessor performance analysis - consumer electronics architecture - networks for embedded systems - network-based design - internet-enabled systems - elevator controller - System Design Techniques

UNIT - III

15 Periods

Programming concepts and embedded programming in C, C++ and Java : Program elements : Macros and functions - Program elements : data types, data structures, modifiers, statements, loops and pointers - Use of loops, infinite loops and conditions - Use of function calls - Multiple function Calls in cyclic order - Function pointers and function queues. **Device drivers and interrupts service mechanism:** Port for Device Accesses without Interrupts Servicing Mechanism Using Programmed I/O - Interrupt driven Input and Output - Interrupt Sources - Hardware Interrupts -Software Interrupts - Interrupts Servicing Mechanism - Multiple Interrupts - **Embedded software development process and tools:** Embedded software development process and tools - Host and target machines - Linking and locating software - Issues in hardware software design and co-design

UNIT - IV

15 Periods

Program modeling concepts: Program models - Data-Flow Graph-Based Program Models - State Machine programming models for event controlled programs. **Real time operating systems:** Multiple processes in an application - Multiple threads in an application - Tasks - Task and thread states - Task and data - Clear-cut distinction between functions, ISR, IST, and task by their characteristics - Inter-process communication and synchronization - Signals - Concepts of semaphores - Disabling and enabling functions - Shared data problem - Queues and mailboxes - Pipe and socket functions - Remote procedure call functions - Process Management - Timer functions - Event functions - Memory management - Device, file and I/O subsystems management - Interrupt routines in RTOS environment and handling of interrupt-

source calls - Introduction to RTOS - Basic design using a RTOS - RTOS task scheduling models

TEXT BOOKS

1. Wayne Wolf, *Computers as Components: Principles of Embedded Computing System Design*, Elsevier, 2nd Edition, 2008.
2. Raj Kamal, *Embedded systems - Architecture, Programming and design*, Tata McGraw Hill, 3rd Edition, 2014.

REFERENCES

1. Marilyn Wolf, *Computers as Components: Principles of Embedded Computing System Design*, Elsevier - Morgan Kaufmann, 4th Edition, 2017
2. Daniel W. Lewis, *Fundamentals of Embedded Software with the ARM Cortex-M3*, Second Edition, Pearson Education, 2013

ONLINE MATERIALS

1. <http://nptel.ac.in/courses/108102045>.
2. http://nptel.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Embedded%20systems/New_index1.html

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Define and recall the concepts of an embedded system design process related to CPU performance • Analyse the computing platform for designing an embedded system • Explain the hardware components essential for designing embedded systems
Unit II	• Discuss the concepts of program analysis and design • Analysis and optimization of program size • Describe the networked embedded systems for connecting the embedded systems
Unit III	• Interpret the concepts of Exception and interrupt handling • Presents the programming concepts and tools for designing the embedded systems • Describes the Embedded software development process and tools
Unit IV	• Presents the program modeling for embedded systems • Explain the necessary concepts in real time operating systems for managing multiple task in designing embedded systems • Infer the significance of RTOS for an embedded system with its applications

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Explain the hardware components essential for designing embedded systems
- Discuss the concepts of program analysis and design
- Describe the networked embedded systems for connecting the embedded systems
- Presents the programming concepts and tools for designing the embedded systems
- Describe the interrupt service routines and device drivers for handling I/O devices
- Explain the necessary concepts in real time operating systems for managing multiple task in designing embedded systems

L	T	P	C
3	1	0	4

Course Code: CSE416
Semester: VII

VIRTUALIZATION TECHNIQUES

Course Objectives

This course will help the learner to understand the types, methods and benefits of virtualization and enable them to manage the resources by creating virtualized infrastructure in data centres.

UNIT - I

15 Periods

Virtualization Basics: Introduction - A virtual history - Virtualization workload candidates - Consolidating web services with VMs - Practical aspects of virtualization. **Comparing Virtualization Techniques:** Guest OS/Host OS - Hypervisor - Emulation - Kernel-level - Shared kernel. **Server Virtualization:** Configuring dedicated server with virtualization - Deploying server appliances - Adjusting and tuning virtual Servers - Securing Virtual Servers - VM Backup - Migrating VMs to New Servers - Migrating physical to virtual

UNIT - II

15 Periods

Storage Virtualization: Enhanced storage and Data services - Implementing Storage virtualization - Achieving High Availability - Performance - capacity - Storage virtualization and SNIA storage management initiative - Policy based service level agreement

UNIT - III

15 Periods

Network Functions Virtualization - Virtualizing Appliances - Virtualizing Core Networking Functions - Scalability and Performance. Modern Networking Approaches to Virtualization - OpenFlow - VMware Nicira - Open Stack. **Software Defined Networks:** The evolution of the Data Center network - Review of Networking - Understanding SDN - SDN Controllers - Business case for SDN

UNIT - IV

15 Periods

Building the Virtual Infrastructure: Hardware's Role in Virtualization - Form factor Choices and their implications. **Managing the Virtual Infrastructure:** The Planning Stage - Deployment - VMware server - Citrix XenServer - Microsoft Hyper-V

TEXT BOOKS

1. Jim Doherty. *SDN and NFV Simplified: A Visual Guide to Understanding Software Defined Networks and Network Function Virtualization*, Pearson Education, First Edition, 2016.
2. Kenneth Hess and Amy Newman. *Practical Virtualization Solutions*, Pearson Education Inc., First Edition, 2010.
3. Frank Bunn, Nik Simpson, Robert Peglar, and Gene Nagle. *Storage Virtualization*, Storage Networking Industry Association, 2003.

REFERENCES

1. Paul Goransson and Chuck Black. *Software Defined Networks: A Comprehensive Approach*, Elsevier, First Edition, 2014.

2. Dan Kuznetsky. *Virtualization: A Manager's Guide*, O'Reilly Media Inc., First Edition, 2011.
3. William Von Hagen. *Professional Xen Virtualization*, Wiley Publishing Inc., First Edition, 2009.
4. James E. Smith and Ravi Nair. *Virtual Machines Versatile Platforms for Systems and Processes*, Morgan Kaufmann Publishers, First Edition, 2005.

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Explain the need of virtualization and its techniques • Illustrate the creation and configuration of virtualized servers
Unit II	• Examine the technology used to provide storage virtualization • Analyze the use of storage virtualization in Data centers
Unit III	• Interpret the significance and functioning of Network Functions Virtualization • Describe the functionalities of Software Defined Networks
Unit IV	• Develop the virtual infrastructure using existing hardware • Compare the virtualization products available for x86 architecture

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Explain the need of virtualization and different techniques
- Illustrate the creation and configuration of virtualized servers
- Analyse the technology used to provide storage virtualization and its effective usage
- Examine the technology and the impact on networking by adopting network function virtualization and software defined networks
- Interpret the existing virtualization product
- Build the virtual infrastructure to manage the resources using x86 architecture-based virtualization products

L	T	P	C
3	1	0	4

Course Code: CSE417

Semester: VII

ROBOTICS

Course Objectives

This course will help the learner to develop programs for robotic system by understanding the fundamentals of robot kinematics and dynamics and conceptually how to associate the different paradigms of artificial intelligence techniques to the robot system.

UNIT - I

15 Periods

Fundamental of robotics: Introduction - Automation and Robotics - A brief history of robotics - Robotics market and the future prospects - Fundamentals of robot technology, programming and applications - Robot anatomy - Work volume - Robot drive systems - Control Systems and dynamic performance - Precision of movement - End effectors - Robotic sensors - Robot programming and work cell control - Robot applications

The robot motion analysis and control: Introduction to manipulator kinematics - Homogeneous transformations and robot kinematics - Manipulator path control - Robot dynamics - Configuration of a robot controller

UNIT - II

15 Periods

Robot programming: Methods of robot programming - Lead through programming methods - A robot programming as a path in space - Motion interpolation - WAIT, SIGNAL, and DELAY commands - Branching - Capabilities and limitations of lead through methods

Robot languages: The textual robot languages - Generation of robot programming languages - Robot language structure - Constants, variables and other data objects - Motion commands - End effector and sensor commands - Computations and operations - Program control and subroutine - Communications and data processing - Monitor mode commands

UNIT - III

15 Periods

Robotic Paradigms - A Brief History of Robotics - Industrial manipulators - Space robotics and the AI approach - Tele-operation - Tele-presence - Semi-autonomous control

The Seven Areas of AI Hierarchical Paradigm - Overview - Attributes of the Hierarchical Paradigm - Strips - More realistic Strips - Closed World Assumption and the Frame Problem - Representative Architectures - Nested Hierarchical Controller - NIST RCS - Evaluation of hierarchical architectures - Advantages and Disadvantages Programming Considerations

UNIT - IV

15 Periods

The Reactive Paradigm : Overview - Attributes of Reactive Paradigm - Characteristics and connotations of reactive behaviors - Advantages of programming by behavior - Representative architectures - Subsumption Architecture - Potential Fields Methodologies - Visualizing potential fields - Magnitude profiles Introduction to AI Robotics Evaluation of Reactive Architectures.

The Hybrid Deliberative/Reactive Paradigm : Overview - Attributes of the Hybrid Paradigm - Styles of hybrid architectures - Managerial Architectures - Autonomous Robot Architecture (AuRA) - Sensor Fusion Effects (SFX) - State-Hierarchy Architectures - Tiered (3T) - Model-Oriented Architectures - Saphira - Task Control Architecture (TCA) - Evaluation of Hybrid Architectures.

TEXT BOOKS

1. Nicholas Odrey, Mitchell Weiss, Mikell Groover, Roger N Nagel and Ashish Dutta. *Industrial Robotics (SIE): Technology, Programming and Application*, McGraw Hill Education (India) Private Limited, Second Edition, 2012.
2. Robin R Murphy. *Introduction to AI Robotics*, PHI Publication, 2004.

REFERENCES

1. S. R. Deb and Sankha Deb. *Robotics Technology and Flexible Automation*, McGraw Hill Education (India) Private Limited, Second Edition, 2010.
2. Stuart J Russell and Peter Norvig. *Artificial Intelligence: A modern approach*, Pearson, Third Edition, 2009.
3. Roland Siegwart and Illah. R. Nourbakhsh. *Introduction to Autonomous mobile robots*, Massachusetts Institute of Technology, 2004.
4. John J. Craig. *Introduction to Robotics Mechanics and Control*, Third Edition, 2004.

ONLINE MATERIALS

1. <http://nptel.ac.in/courses/112101099/1#video>
2. <http://nptel.ac.in/courses/112101099/31#video>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Classify the elements of the robot system • Derive expression for kinematics and dynamics of robots
Unit II	• Compare different generation of program languages for robotics • Implement the robot control commands for the given scenario
Unit III	• Elaborate AI approaches for robotics • Analyze the hierarchical paradigm architecture
Unit IV	• Infer the advantages and disadvantages of the reactive paradigm • Differentiate hybrid paradigm architectures with other architectures

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Recognize the elements of robot system
- Evaluate the kinematics and dynamics of robot system
- Explain the evolution of robot programming languages
- Develop the program to control robot for the given scenario
- Elaborate AI approached for robot control
- Compare the architectures of hierarchical, reactive and hybrid paradigm with example

L	T	P	C
4	0	0	4

Course Code: CSE418
Semester: VII

BRAIN COMPUTING

Course Objectives

The Course will help the learners to understand functioning of the Brain and Signal Simulation in the Brain, comprehend and implement various Signal Processing and Machine learning techniques and also understand the applications of Brain Computer Interface (BCI).

UNIT - I

15 Periods

Brain Computing Interface: Introduction - Basic Neuroscience - Neurons - Action Potentials or Spikes - Dendrites and Axons - Synapses - Spike Generation - **Adapting the connections:** Synaptic Plasticity - Brain Organization, Anatomy and Function. **Recording and Simulating the Brain:** Recording Signals from the Brain - Simulating the Brain - Simultaneous Recording and Simulation. **Signal Processing:** Spike Sorting - Frequency Domain Analysis - Fourier Analysis - Discrete Fourier Transform - Fast Fourier Transform - Spectral Features - Wavelet Analysis

UNIT - II

15 Periods

Time Domain Analysis: Hjorth Parameters - Fractal Dimension - Autoregressive Modelling - Bayesian Filtering - Kalman Filtering - Particle Filtering. **Spatial Filtering:** Bipolar, Laplacian and Common Average Referencing - Principal Component Analysis - Independent Component Analysis - Common Spatial Patterns. **Artifact Reduction Technique:** Thresholding - Band Stop and Notch Filtering - Linear Modelling. **Classification Techniques:** Binary Classification - Ensemble Classification - Multi-Class Classification - Evaluation of Classification Performance

UNIT - III

15 Periods

Regression: Linear Regression - Neural Networks and Back propagation - Radial Basis Function Networks - Gaussian Process. **Building a BCI:** Major types of BCI - Brain Responses Useful for Building BCI's - Conditioned Responses - Population Activity - Imagined Motor and Cognitive Activity - Stimulus Evoked Activity. **Invasive BCIs:** Invasive BCI in Humans. **Semi - Invasive BCIs:** EcoGs BCI in Humans

UNIT - IV

15 Periods

Non-Invasive BCIs: Slow Cortical Potentials - Movement Related Potentials - Stimulus Evoked Potentials. **Other Non-Invasive BCIs:** fMRI, MEG and fNIR. **BCIs that Simulate:** Motor Restoration: Deep Brain Simulation. **Applications:** Medical Applications. **Ethics:** BCI Security and Privacy - Legal Issues - Moral and Social Justice Issues

TEXTBOOK

1. Rajesh P.N. Rao. *Brain Computing Interfaces: An Introduction*. Cambridge University Press. First Edition, 2013.

REFERENCES

1. Maureen Clerc, Laurent Bougrain and Fabien Lotte. *Brain Computer Interfaces-1 Foundations and Methods*, Wiley India, First Edition, 2016.

2. Maureen Clerc, Laurent Bougrain and Fabien Lotte. *Brain Computer Interfaces-2: Technology and Applications*, Wiley India, First Edition, 2016.
3. Jonathan.R.Wolpaw and Elizabeth Winter Wolpaw. *Brain - Computer Interfaces: Principles and Practice*, Oxford University Press, First Edition, 2011.
4. Andrew Webb. *Statistical Pattern Recognition*. Wiley International, Second Edition, 2002.
5. Arnon Kohen. *Biomedical Signal Processing Vol I and II*, CRC Press Inc., Second Edition, 2000.
6. Bishop C.M. *Neural networks for Pattern Recognition*, Oxford Clarendon Press, First Edition, 1995.

ONLINE MATERIALS

1. https://onlinecourses.nptel.ac.in/noc18_ec02/preview
2. https://onlinecourses.nptel.ac.in/noc19_ee23/preview
3. https://sccn.ucsd.edu/wiki/Introduction_To_Modern_Brain-Computer_Interface_Design

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- Appraise computer as an Interface to the brain - Design and Record Signals in the Brain
Unit II	- Examine brain signals in the format required for specific applications - Implement the various Signal Processing Techniques
Unit III	- Apply Artifact reduction, Spatial Filtering on Signals using various programming Languages - Execute various Machine Learning Techniques over a processed Signal
Unit IV	- Build a BCI by having a subtle understanding on the basics - Develop an understanding on applications in BCI and Ethical Issues in a Brain Computing Interface

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Develop and record signals from the brain
- Analyse pre-processing of a Signal for Signal Enhancement
- Explore required features and classify the Signals
- Apply Machine Learning Techniques on the Signal
- Evaluate and build a Brain Computing Interface with various techniques
- Demonstrate the Implication of BCI on various applications

L	T	P	C
3	1	0	4

Course Code: INT410
Semester: VII

CYBER PHYSICAL SYSTEMS

Course Objectives

This course will help the learner to develop an exposition of the challenges in implementing a cyber-physical system from a computational perspective and aims to expose the students to real world problems in this domain and provide a walk through the design and validation problems for such systems.

UNIT - I

15 Periods

Cyber-Physical Systems: Introduction:- Applications - Motivating Example - Design Process - Modeling- Design - Analysis. **Continuous Dynamics:** Newtonian Mechanics - Actor Models - Properties of Systems - Casual Systems - Memoryless Systems - Linearity and Time Invariance - Stability - Feedback Control. **Hybrid Systems:** Modal Models - Actor Model for State Machines - Continuous Inputs - State Refinements - Classes of Hybrid Systems - Timed Automata - Higher-Order Dynamics - Supervisory Control

UNIT - II

15 Periods

Models of Sensors and Actuators: Linear and Affine Models - Range - Dynamic Range - Quantization - Noise - Sampling - Harmonic Distortion - Signal Conditioning - Common Sensors - Measuring Tilt and Acceleration - Measuring Position and Velocity - Measuring Rotation - Measuring Sound - Actuators - Light-Emitting Diodes - Motor Control. **Embedded Processors:** Types of Processors - Microcontrollers - DSP Processors - Graphics Processors - Parallelism - Parallelism vs Concurrency - Pipelining - Instruction-level Parallelism

UNIT - III

15 Periods

Input and Output: I/O Hardware - Sequential Software in a Concurrent World. **Multitasking:** Imperative Programs - Threads - Creating Threads - Implementing Threads - Mutual Exclusion - Deadlock - Memory Consistency Models - Processes and Message Passing. **Scheduling:** Basics - Rate Monotonic Scheduling - Earliest Deadline First - Scheduling and Mutual Exclusion - Priority Inversion - Priority Inheritance Protocol - Priority Ceiling Protocol - Multiprocessor Scheduling - Scheduling Anomalies

UNIT - IV

15 Periods

Reachability Analysis and Model Checking: Open and Closed Systems - Reachability Analysis - Verifying Gp - Explicit-State Model Checking - Symbolic Model Checking - Abstraction in Model Checking - Model checking Liveness Properties - Properties as Automata - Finding Acceptance Cycles. **Quantitative Analysis:** Problems of Interest - Programs as Graphs - Factors Determining Execution Time - Basics of Execution Time Analysis - Quantitative Analysis Problems. **Security and Privacy:** Cryptographic Primitives - Protocol and Network Security - Software Security - Information Flow - Sensor and Actuator Security -Side-channel Attacks

TEXT BOOKS

1. André Platzer. *Logical Foundations of Cyber-Physical Systems*, Springer, First Edition, 2018.

2. Edward Ashford Lee, Sanjit Arunkumar Seshia. *Introduction to Embedded Systems: A Cyber-Physical Systems Approach*, MIT Press, Second Edition, 2017.

REFERENCE

1. Rajeev Alur. *Principles of Cyber-Physical Systems*, MIT Press, 2015.

ONLINE MATERIALS

1. <https://ptolemy.berkeley.edu/books/leeseshia/>
2. <http://www.ls.cs.cmu.edu/CPSVVIF/slides/Platzer.pdf>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Categorize the essential modeling formalisms of Cyber-Physical Systems (CPS) • Describe the key elements of a CPS such as formal models and hybrid systems
Unit II	• Describe the basic design, architecture and design principles of cyber physical systems • Analyze the functional behavior of Sensors, Actuators and hardware architectures in CPS
Unit III	• Design CPS requirements based on operating system constraints such as Input and Output, Multitasking, and Scheduling • Analyze and verify the correctness of CPS implementations against system requirements
Unit IV	• Analyze CPS implementations through Reachability Analysis and Model Checking, and Quantitative Analysis • Verify the security and privacy constraints in real-time CPS implementations

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Categorize the essential modelling formalisms of Cyber-Physical Systems (CPS)
- Analyze the functional behaviour of Sensors, Actuators and hardware architectures in CPS
- Design CPS requirements based on operating system constraints such as Input and Output, Multitasking, and Scheduling
- Analyze and verify the correctness of CPS implementations against system requirements
- Analyze CPS implementations through Reachability Analysis and Model Checking, and Quantitative Analysis
- Verify the security and privacy constraints in real-time CPS implementations

L	T	P	C
3	1	0	4

Course Code: CSE423

Semester: VII

COGNITIVE COMPUTING

Course Objectives

This course will help the learner to design an intelligent assistive system that can improve the quality of life of elderly, disabled and common public

UNIT - I

15 Periods

Fundamentals of Cognitive science and computing: Approaches to Cognitive Science: Representation and Computation, Connections, Extensions to cognitive science: Brains, Emotions. The Foundations of Cognitive Computing, Design principles for cognitive systems.

UNIT - II

15 Periods

Neuroimaging, Signal processing and Machine Learning: Functional Neuroimaging: Functional MRI, Electrophysiology: EEG and MEG, Simultaneous EEG-fMRI **Electroencephalography Data preprocessing**, EEG feature extraction, statistical learning for BCIs, Adaptive methods in machine learning

UNIT - III

15 Periods

Neural networks and Deep Learning : Convolutional networks: The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data Types, Efficient convolution Algorithms, Random or Unsupervised Features, The Neuroscientific Basis for Convolutional Networks **Neural networks with memory:** Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder, Sequence-to-Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks

UNIT - IV

15 Periods

Applications: Business implications of Cognitive Computing, IBM's Watson as a cognitive system, Building a cognitive health care application, Smarter cities: Cognitive computing in Government, Emerging cognitive computing areas

TEXTBOOKS

1. Hurwitz, Kaufman, and Bowles, *Cognitive Computing and Big Data Analytics*, Wiley, Indianapolis, 2015
2. Maureen Clerc, Laurent Bougrain and Fabien Lotte. *Brain Computer Interfaces-1 Foundations and Methods*, Wiley India, First Edition, 2016

REFERENCES

1. Ian Good Fellow, Yoshua Bengio, Aaron Courville, *Deep Learning*, The MIT Press , 2016
2. Stuart Russell, and Peter Norvig, *Artificial Intelligence - A Modern Approach*, Pearson Education, Prentice Hall of India, 3rd Edition, 2014.

ONLINE MATERIALS

1. <http://nptel.ac.in/courses/106105077/>
2. http://nptel.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Artificial%20intelligence/New_index1.html

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Interpret the principles of cognitive science in terms of computing • Appraise computing and decision making from the perspective of cognitive science.
Unit II	• Represent different forms of neuro imaging and brain signals • Employ traditional machine learning techniques to Neuro imaging and brain signal representations
Unit III	• Apply CNNs to different forms of neuro imaging • Demonstrate neural networks with memory capabilities in the form of Recurrent and Recursive neural networks
Unit IV	• Evaluate a cognitive system developed for a specific application • Build an intelligent cognitive system to improve quality of life

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to:

- Appraise decision making process from the perspective of cognitive science
- Encode different forms of brain signals and neuroimaging
- Illustrate appropriate machine learning technique for a given neuroimaging or brain signal processing application
- Choose a specific neural network architecture for an assistive system
- Demonstrate the potential of neural networks with memory concepts for sequential data modelling applications
- Build an intelligent assistive system to improve the quality of life

L	T	P	C
0	0	24	12

Course Code: CSE400

Semester: VIII

PROJECT & VIVA VOCE

Course Objectives

Project work empowers the learner to

- Understand research articles and technical documents
- Select appropriate scientific and mathematical model based on literature survey to solve the problem
- Develop innovative solutions by applying holistic knowledge and software engineering principles
- Improve Interpersonal skills, including ability to lead and to work cooperatively in a team
- Improve oral and written technical communication skills
- Abide by the legal and ethical standards and catering for social development

Course Description:

The Project work provides the learner an opportunity to apply and extend knowledge acquired throughout the program. Project work can be carried out individually or as a group of two to three persons. It can be carried out either as an in-house project or as an intern at a company or an Institution. This course introduces the dimension of workload management into the program to enable completion of a large, relatively unstructured work

- In the initial stages, the learner is expected to choose a problem, along with the group members. This leads to the formulation of the problem, checking the feasibility of the work to be completed within the given stipulated duration
- The learner analyzes the chosen problem and design computationally efficient optimal algorithms, meeting the requirement specification, so that it leads to building energy efficient Software/Hardware system
- The learner employs advanced tools to implement the methodology by applying project management principles, while keeping in mind the professional ethics, societal and environmental impact
- The learner uses appropriate Software/Hardware testing strategies for the developed system and builds a user friendly interface
- Individual performance of the learner is continuously assessed by periodic reviews of the progress of the project work

COURSE LEARNING OUTCOMES

Upon successful completion of the project, the learner will be able to

- Identify the current societal and technological requirements
- Analyze and discuss the shortfalls of current system by performing literature survey
- Correlate the mathematical, conceptual and computing knowledge with the problem under study, considering uncertainties
- Plan the project development stages and distribute tasks among the team members
- Use technical literature and relevant information sources for designing cost effective solution based on acquired knowledge
- Implement the design by applying computational methods and technical skills using appropriate software tools
- Examine technical uncertainties during system development
- Evaluate the system by using appropriate testing strategies and refine the system through discussions among team members

- Write a report explaining the technical concepts of the developed system
- Apply legal and ethical standards